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ISBN 978-0-309-60133-7 | DOI 10.17226/29292

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SUGGESTED CITATION

National Academies of Sciences, Engineering, and Medicine. 2026. *Frontiers of Statistics in Science and Engineering: 2035 and Beyond*. Washington, DC: The National Academies Press. https://doi.org/10.17226/29292.

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Frontiers of Statistics in Science and Engineering

2035 and Beyond

Committee on the Frontiers of Statistics in Science and Engineering: 2035 and Beyond

Physical Sciences, Systems, and Infrastructure Program Area

Center for Advancing Science and Technology

Consensus Study Report

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This study was supported by a contract and grants between the National Academy of Sciences and the National Institutes of Health (Contract No. HHSN263201800029I/75N98024F00006), the National Science Foundation (Grant No. 2426575), and the National Security Agency (Grant No. H98230-24-1-0019). Any opinions, findings, conclusions, or recommendations expressed in this publication do not necessarily reflect the views of any organization or agency that provided support for the project.

International Standard Book Number-13: 978-0-309-XXXXX-X
Digital Object Identifier: https://doi.org/10.17226/29292

This publication is available from the National Academies Press, 500 Fifth Street, NW, Keck 360, Washington, DC 20001; (800) 624-6242; https://nap.nationalacademies.org.

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Printed in the United States of America.

Suggested citation: National Academies of Sciences, Engineering, and Medicine. 2026. *Frontiers of Statistics in Science and Engineering: 2035 and Beyond*. Washington, DC: National Academies Press. https://doi.org/10.17226/29292.

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COMMITTEE ON THE FRONTIERS OF STATISTICS IN SCIENCE AND ENGINEERING: 2035 AND BEYOND

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LANCE A. WALLER, Emory University, *Vice Chair*
RINA F. BARBER (NAS), The University of Chicago
AMY BRAVERMAN, Jet Propulsion Laboratory, California Institute of Technology
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FRAUKE KREUTER, University of Maryland
XIHONG LIN (NAS/NAM), Harvard University
BRIAN REICH, North Carolina State University
STEPHAN R. SAIN, Jupiter Intelligence
AARTI SINGH, Carnegie Mellon University
DANIELA M. WITTEN, University of Washington
TIAN ZHENG, Columbia University

Study Staff

BRITTANY SEGUNDO, Program Officer, Board on Mathematical Sciences and Analytics (BMSA), *Study Director*
MICHELLE K. SCHWALBE, Senior Program Director, Physical Sciences, Systems, and Infrastructure Program Area
ERIK SVEDBERG, Deputy Director, BMSA and National Materials and Manufacturing Board
SAMANTHA KORETSKY, Research Associate, BMSA
HEATHER LOZOWSKI, Senior Finance Business Partner
MICA PACHECO, Associate Program Officer, BMSA
JOE PALMER, Senior Project Assistant, BMSA

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Reviewers

This Consensus Study Report was reviewed in draft form by individuals chosen for their diverse perspectives and technical expertise. The purpose of this independent review is to provide candid and critical comments that will assist the National Academies of Sciences, Engineering, and Medicine in making each published report as sound as possible and to ensure that it meets the institutional standards for quality, objectivity, evidence, and responsiveness to the study charge. The review comments and draft manuscript remain confidential to protect the integrity of the deliberative process.

We thank the following individuals for their review of this report:

SUDIPTO BANERJEE, University of California, Los Angeles
ANNA BARGAGLIOTTI, Loyola Marymount University
SNIGDHANSU CHATTERJEE, University of Maryland, Baltimore County
IRINA GAYNANOVA, University of Michigan
ANDREW GELMAN, Columbia University
KAREN KAFADAR, University of Virginia
SALLIE KELLER (NAE), U.S. Census Bureau
BO LI, Washington University in St. Louis
MARIANTHI MARKATOU, University at Buffalo
DAVID MATTESON, Cornell University
KNASHAWN MORALES, Hospital of the University of Pennsylvania
ROBERT TIBSHIRANI (NAS), Stanford University
RONALD WASSERSTEIN, American Statistical Association

Although the reviewers listed above provided many constructive comments and suggestions, they were not asked to endorse the conclusions or recommendations of this report nor did they see the final draft before its release. The review of this report was overseen by **ALICIA CARRIQUIRY (NAM),** Iowa State University, and **ELIZABETH STUART (NAM),** Johns Hopkins Bloomberg School of Public Health. They were responsible for making certain that an independent examination of this report was carried out in accordance with the standards of the National Academies and that all review comments were carefully considered. Responsibility for the final content rests entirely with the authoring committee and the National Academies.

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Acknowledgments

We are grateful to the many scholars and leaders who contributed their time and expertise to the committee's information-gathering efforts. Special thanks go to all the speakers who briefed the committee:

Sivan Aldor-Noiman (Banff Advisors), Veera Baladandayuthapani (University of Michigan), Bonnie Berger (Massachusetts Institute of Technology), Catherine Calder (The University of Texas at Austin), Beth Chance (California Polytechnic State University), Wei Chen (Northwestern University), Peter Chien (University of Wisconsin–Madison), Jan Dasgupta (Washington State University), Peng Ding (University of California, Berkeley), Hani Doss (University of Florida), John Duchi (Stanford University), Yulia Gel (Virginia Tech University), Bonnie Ghosh-Dastidar (RAND), Carlo Graziani (Argonne National Laboratory), Trevor Hefley (Kansas State University), Dave Hunter (The Pennsylvania State University), Roshan Joseph (Georgia Institute of Technology), Amarjot Kaur (Merck), Michael Kosorok (University of North Carolina), Donna LaLonde (American Statistical Association), Nicole Alana Lazar (The Pennsylvania State University), Hongzhe Li (University of Pennsylvania), Lexin Li (University of California, Berkeley), Lester Mackey (Microsoft Research), Monnie McGee (Southern Methodist University), Shari Messinger (University of Miami), Jeffrey Morris (University of Pennsylvania), Vijay Nair (Wells Fargo), Dan Nettleton (Iowa State University), Betsy Ogburn (Johns Hopkins University), Eunchun Park (University of Arkansas), Robert Perera (Virginia Commonwealth University), Annie Qu (University of California, Irvine), Ricky Rambharat (Northrop Grumman), Abel Rodriguez (University of Washington), Christopher Saunders (South Dakota State University), David Schimel (Jet Propulsion Laboratory, California Institute of Technology), Lyndsay Shand (Sandia National Laboratories), Dylan Small (University of Pennsylvania), Ralph Smith (North Carolina State University), Michael Stein (Rutgers, The State University of New Jersey), Weijie Su (University of Pennsylvania), Ming Tony Tan (Georgetown University), Anh Tran (Sandia National Laboratories), Lan Wang (University of Miami), Scott Vander Wiel (Los Alamos National Laboratory), Frederi Viens (Rice University),

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Mark Ward (Purdue University), and Linda Young (U.S. Department of Agriculture National Agricultural Statistics Service).

Additionally, numerous members of the broader statistical community participated in a call for perspectives and shared their thoughts on directions for the field. We are grateful for their time. This report was the product of the committee's thoughtful deliberation and dedication on the topic. We extend our gratitude to the broader community for its engagement with this project.

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Preface

The field of statistics is the language of science and innovation. Throughout history, the ability to examine data and information (or lack thereof) critically and provide actionable insight and wisdom grounded in an understanding of the potential biases and uncertainties around these measurements has elevated science and engineering in service to humanity. The field is broad and advances in response to novel ideas and breakthroughs both from within the field itself and its intersections with other disciplines. Statistical advances represent both new statistical thinking and new insights, data, and analytic paradigms from other fields including mathematics and computer science, as well as through collaborations with specific disciplines within science, engineering, medicine, and beyond. This "melting pot" of innovation results in statistics evolving as a dynamic field of inquiry. Statistics also has its own deep roots that grow with new data, assumptions, and computational capabilities, all of which motivate new questions that were unthinkable or considered impossible only a few years earlier. As such, the frontiers of statistics derive from within the field itself and from input into statistics from other fields, and the frontiers provide input from statistics back to these fields.

Because of the close relationship between statistics and other fields in science and engineering, many technological advances today stem from foundational advancements in statistics over the past few decades. Statistics provides the scientific basis for artificial intelligence (AI) and machine learning, which are revolutionizing the world and will bring societal transformations in ways that are difficult to predict. Statistics affords a pathway for transparent, responsible, and trustworthy AI, which is a necessary and important requirement, as these technologies are interwoven with the fabric of society.

Frontiers of Statistics in Science and Engineering: 2035 and Beyond represents a year-long exercise to understand the breadth and depth of the field of statistics and implications for its trajectory over the next 10 years. This intense effort not only reaffirmed the critical importance of the field but also demonstrated its expanded relevance amid the technological advances we all navigate. The life cycle of statistical innovation will remain central to science and engineering

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progress as statisticians work collaboratively, as new questions, new data types, and new methodologies arise.

We would like to thank our sponsors (in alphabetical order) the National Institutes of Health, the National Science Foundation, and the National Security Agency for their support for this study and for outlining such important objectives. The statement of task focuses on science and engineering, which, for the purposes of this study, includes health and medicine. Our report has retained that focus. The field of statistics also has a long history of contributions to and reciprocal development from the social sciences, but this is beyond the scope of the report.

This impactful study was made possible through the dedication and significant contributions of an exceptional committee, including the National Academies of Sciences, Engineering, and Medicine staff. All committee members brought expertise across the breadth of the field. We also recognize the invaluable suggestions and insights from the independent review panel that read a previous draft of the report. The collective and tireless commitment to science, technology, and society was evident throughout the process. We are grateful for the extensive contributions of leaders in our field through various panel discussions, as well as those who provided direct input via our online form.

We want to thank our community and the National Academies for entrusting us with this important role. We hope that by 2035 the study will have paved the way for lasting, transformative impact by bringing the full statistical endeavor to bear.

Katherine B. Ensor, *Chair*
Lance A. Waller, *Vice Chair*
Committee on the Frontiers of Statistics in Science and Engineering: 2035 and Beyond
February 2026

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Acronyms and Abbreviations

ACM	Association for Computing Machinery
AI	artificial intelligence
AISTATS	Artificial Intelligence and Statistics
AoAS	*Annals of Applied Statistics*
AoS	*Annals of Statistics*
AUDITION	Autism Digital Twin Foundations
CDC	Centers for Disease Control and Prevention
CMU	Carnegie Mellon University
CoFES	Center for Computational Finance and Economic Systems
COPSS	Committee of Presidents of Statistical Societies
CSAB	Computing Sciences Accreditation Board
DMS	Division of Mathematical Sciences
DOE	design of experiments
FAIR	findable, accessible, interoperable, and reproducible
GARCH	generalized autoregressive heteroskedastic
GFT	Google Flu Trends
GPU	graphical processing unit
ICSDS	International Conference on Statistics and Data Science
ILI	influenza-like illnesses
IMS	Institute of Mathematical Statistics

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JASA	*Journal of the American Statistical Association*
JPL	Jet Propulsion Laboratory
JPSM	Joint Program in Survey Methodology
JSDM	joint species distribution model
JSM	Joint Statistical Meetings
LLM	large language model
MCMC	Markov Chain Monte Carlo
MD	molecular dynamics
MSR	Microsoft Research
NAIRR	National Artificial Intelligence Research Resource
NASA	National Aeronautics and Space Administration
NIH	National Institutes of Health
NSA	National Security Agency
NSF	National Science Foundation
PCA	principal components analysis
PCS	predictability, computability, and stability
PLoS	*Public Library of Science*
REU	research experience for undergraduate
SBI	simulation-based inference
SIAM	Society for Industrial and Applied Mathematics
SIBS	Summer Institutes for Training in Biostatistics
SLM	small language model
TRIPODS	Transdisciplinary Research in Principles of Data Science

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UC	University of California
VAE	variational autoencoder

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Summary

Statistics is the science through which data become wisdom. Scientific and engineering progress is driven by statistical innovation, and the field continues to evolve through reciprocal interaction with scientific innovation. With the influx of computing capacity, proliferation and interoperability of data, and availability of increasingly complex modeling, the statistical landscape is shifting in ways that will transform its demands and broaden its contributions. This report illustrates the role of statistics as a key element of science and engineering progress. Recommendations in the report highlight critical areas of stewardship, capacity-building, and investment in the statistical sciences to sustain and expand the field's contributions for U.S. competitiveness. This summary presents selected key findings, conclusions, and recommendations that underscore the report's main message.[1]

THE FIELD OF STATISTICS AND ITS FRONTIERS

The field of statistics is broad, with developments occurring wherever data provide information and transforming insight into understanding. Developments by statisticians are enhanced by researchers in other fields who add new questions of interest, data types, and opportunities for data-derived insight. The field of statistics develops both within the discipline and in collaboration with other fields of science, engineering, and medicine. These partnerships are most productive when they are bidirectional, contributing novel developments in statistical theory, methods, and implementation, as well as fresh insights within the collaborative fields. Throughout the report, the committee uses the terms "statistics" and the collective "statistical sciences" broadly to include all these areas of specialization.

[1] Within the body of the report, key findings, conclusions, and recommendations are numbered according to the chapter number and order of appearance. However, selected key findings, conclusions, and recommendations have been included in the Summary in the order that best preserves narrative flow, which may not align with this numbering scheme.

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This report focuses on the frontiers of the field of statistics and, as such, will have a point of view centered on advances in statistical methods, theory, and thinking. However, the frontiers of statistics are not only at the boundaries of the field itself (Chapters 1 and 2) but also, importantly, at the intersection of statistics with new types of data (Chapter 3) and with frontiers of other fields (Chapters 4 and 5). These mutually beneficial interactions between statistics and other fields have profound influence on defining the frontiers of statistical research (Chapter 2), statistical practice (Chapters 4 and 5), and statistical training (Chapter 6) across all disciplines in science, engineering, and mathematics.

Advances in statistics are theoretical, methodological, and applied, or some combination thereof. Statistical theory uses computing and mathematics to study the properties of statistical procedures for estimation and uncertainty quantification to establish if and when the inferences are reliable. While new developments in theory may not have immediate application to problems in science and engineering, they guide statistical practice and provide the statistical foundation for new methodological development (He et al. 2019). These foundations may be under development for decades before statisticians and their collaborators can apply them; theoretical research in statistics defines the basic science underlying its frontier development. New statistical methods, such as estimators or inference approaches, are continuously required to address developments in science and engineering. Applied statistics provides meaningful, quantitatively informed answers to specific scientific questions.

> Key Finding 2-2: New statistical methods afford new insights and new understanding, which often trigger a new generation of motivating questions; thus, the statistical innovation life cycle begins again.

The engine of statistical innovation is often driven by a new scientific question. The scientific question must be translated to a statistical question, which may require robust statistical methodological and/or theoretical development. Finally, these new approaches are used to answer and even refine the original scientific question.

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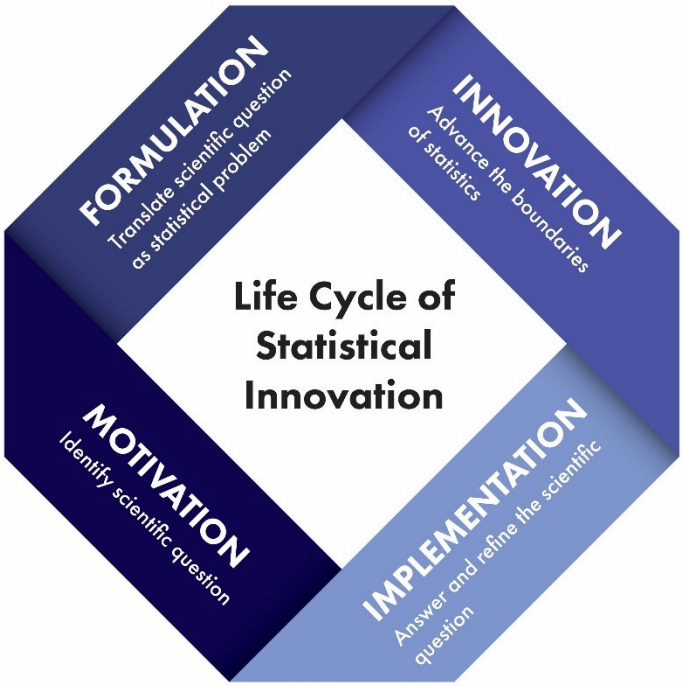

FIGURE S-1 The life cycle of statistical innovation.

Although Figure S-1 illustrates how statistics innovates in response to specific scientific questions, broad science and engineering developments also drive movement in the field. Drivers include (but are not limited to) the evolution of data, increasingly complex systems and modeling, the proliferation of artificial intelligence (AI), and characterization of extremes.

> **Recommendation 7-8: Universities should strengthen support for and champion the contributions of statistical departments, and statistical departments should prioritize hiring, tenure, and promotion practices that reward innovation, deep and ongoing collaboration, and high-impact contributions, including contributions in statistical software, analytic pipelines, and curated data sets. Journals should create environments that encourage and reward high-risk, high-reward research and education initiatives, including novel methodological advances, interdisciplinary applications, and collaborations. Funding agencies and industrial partners should create opportunities for statisticians to serve not only as collaborators but also as**

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leaders in interdisciplinary research teams and education efforts, reinforcing a culture that values their central role in scientific progress.

As described above, the statistical toolset is ever-growing, and a forward-looking approach to statistical training will motivate trainees to think statistically, recognizing and incorporating uncertainty into decision making. A challenge for the field is the timely integration of new inference paradigms and critical advances in statistical thinking into the statistics curriculum. To ensure long-term success, statistics departments (and departments that teach statistical thinking) will have to demonstrate agility with rapid technological change and the growing demands of scientific disciplines for statistical innovation, discussed in Chapter 6.

Historically, statistics has found wide support across the National Science Foundation (NSF), the National Institutes of Health (NIH), and other funding agencies.

Conclusion 2-8: Statistics has been supported broadly across multiple directorates and divisions at NSF, speaking to its critical role throughout science and engineering.

Recommendation 7-1: Prioritized investment in the field's foundations is essential to accelerate trustworthy, data-driven scientific discovery; keep pace with the modern technological evolution; and maximize positive societal impact by enabling future breakthroughs at the frontiers of statistics. The National Science Foundation (NSF) should bolster its ongoing support for statistics, spanning methods, applications, theory, training, and education, and embed statistical science across the agency.

A successful program (or suite of programs) would advance statistical foundations, methods, applications, and training across NSF's scientific and engineering portfolio and include the following:

a. **<u>Foundational Statistical Methods and Applications</u>. Targeted cross-agency funding opportunities should highlight the importance of these translational contributions and the leadership by statisticians.**
    i. **Statistical methods and applications driven by domain scientists with statisticians playing a co-leadership role in advancing the emerging science and technology.**

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ii. **New frameworks and optimality criteria to enable analysis and inference in highly distributed environments, as well as statistical methods for the integrative analysis of observed and model-generated synthetic data.**

iii. **Statistical research explicitly targeting decision making in areas including precision medicine, precision agriculture, resilient infrastructure, robotics, autonomous driving, and digital twins.**[2]

iv. **Methods that integrate quantum computing into scientific inquiry, including statistical algorithms that exploit quantum computing and statistical tools to quantify uncertainty and optimize deployment of quantum sensing devices.**

v. **Statistical research to investigate and improve the quality, scalability, and representativeness of novel data streams and data types, including those that involve human feedback and generative artificial intelligence.**

b. **<u>Foundational Statistical Research</u>. Some practice-informed needs for investment in theoretical research include the following:**

i. **Statistical inference in highly distributed settings, where as much computation as feasible occurs at or as close to the sensor as possible. As data collection becomes embedded in physical and social infrastructure, "edge" computing and federated learning will become requirements.**

ii. **Privacy- and confidentiality-preserving data analysis methods, particularly in distributed settings.**

iii. **Statistical theory to improve inference and understand intrinsic or extrinsic structure in high-dimensional, complex observed, or model-generated synthetic data.**

[2] This report uses the National Academies' definition: "A digital twin is a set of virtual information constructs that mimics the structure, context, and behavior of a natural, engineered, or social system (or system-of-systems), is dynamically updated with data from its physical twin, has a predictive capability, and informs decisions that realize value. The bidirectional interaction between the virtual and the physical is central to the digital twin" (NASEM 2024b, 1).

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iv. **Statistical research to investigate and improve the quality, scalability, and representativeness of all data types.**

v. **Statistical theory to assess validity, accuracy, and reproducibility of models, methods, and data to support decision making under uncertainty in complex systems. These programs should include support for advances in understanding the intersection of heterogeneous data with causal inference methods.**

c. **Foundational Statistical Education. Collaboration with other agencies to establish a nationally coordinated program supporting ongoing evaluation and reform of graduate statistical education, particularly in light of the rapid pace of technological evolution. The coordinated effort should develop strategies to strengthen the domestic pipeline for graduate students in statistics.**

The lists above are not exhaustive but provide several examples of areas with opportunity for rapid and consequential development.

Key Finding 2-4: It is of paramount importance for the statistical community to position itself to take advantage of existing opportunities and emerging application areas by pursuing relevant research priorities and funding calls broadly. Doing so requires strengthening community-level mechanisms and infrastructure that enable statisticians to engage early, collectively, and strategically. Examples include offering leadership and team-science training workshops, sponsoring task forces to develop shared methodological and computational resources, creating grand community research challenges (such as KDD Cup and DREAM challenges), and organizing informal proposers' days in response to large interdisciplinary funding opportunities.

Recommendation 7-2: The National Institutes of Health (NIH) should expand its support for the foundations of statistics in health science across the biomedical sciences, spanning methods, applications, theory, training, and education.

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a. **Continue strengthening NIH support for cutting-edge statistical methodological research and applications, which are essential for accelerating discovery from existing and emerging biomedical data.**
b. **Evaluate and ensure that NIH study sections explicitly recognize statistical methodological research as a priority by assigning proposals to panels with appropriate statistical expertise and by applying review criteria that reflect the nature and impact of methodological contributions.**
c. **Evaluate and ensure that training (T, F) and career development (K) award programs are structured and reviewed in ways that prioritize training of future biostatisticians, apply evaluation criteria appropriate to methodological and interdisciplinary preparation, and involve reviewers with relevant statistical expertise.**

STATISTICS AND DATA

Data are integral to scientific progress, and statisticians specialize in transforming data into insight. Across almost every domain, data are more readily available; increasingly connected to other data; and collected from multiple sources, over multiple time periods, and in multiple locations. This evolution in data types and availability, alongside data-intensive computing, enables new modes of scientific discovery.

Several novel data types have emerged in recent years. These data types include unstructured data, such as images, texts and videos, generated synthetic data, streaming data, and multimodal data. New methods must be scalable to millions of records and amenable to the distributed nature of data.

Frequently, data are collected passively and opportunistically, meaning that there are more data than ever, but they may not be of the right type, capture the metrics of interest, or be structured in the right way. To fully leverage these data, additional work is needed to prepare them for use.

Conclusion 3-1: Statistical research enhances data utility and application. As new data types emerge and as complexity increases, statistics and analytic tools will need to evolve to keep pace with the changing landscape.

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Machine learning and data-driven discovery methods are changing rapidly, often with a goal to automate the search for and linkage of these new data types, especially in exploratory and high-dimensional settings. Data science research encourages the co-development of metadata to provide the information necessary to incorporate uncertainties for decisions and inferences involving fused data (i.e., data that are combined from multiple sources). Generally, the increasing interconnectedness and interoperability of data elements highlight a need for innovations relating to the identification, collection, and fusion of data from multiple heterogeneous sources in ways that allow analysts to model the observation process itself. The analyses of fused data offer great opportunities for theoretical and methodological expansion at multiple levels of complexity.

As a result of increasing data volume and interconnectedness, the costs of storing, finding, linking, managing, and manipulating massive amounts of data are rising. Data may be stored in local repositories supported by cloud computing, but as additional users need access, this paradigm could become prohibitively expensive. Additionally, because different entities collect and store data differently, the full set of data required for a given analysis may not exist in monolithic form. Eventually, a new paradigm for statistical analysis will be required: one that is oriented around peer-to-peer systems that can manage privacy and security concerns but that encourages model and data sharing. All of this suggests that real-time inference on the system itself will emerge as a critical component of statistical inference. Computing will continue to evolve, and decentralized analysis will be required to cope with huge volumes of decentralized data.

> *Conclusion 5-5: As new decentralized technologies—such as the Internet of Things, sensing devices, autonomous vehicles, advanced biotechnology, and blockchain—continue to evolve, the availability and nature of data will continue to change rapidly, catalyzing a new frontier in distributed statistical inference. Traditional methods supporting data privacy also rely on decentralized data collection, storage, and analysis. These interacting motivations and constraints, among others, define an ongoing and evolving challenge to develop statistical methods appropriate for a distributed data setting.*

Statistics must continue to modernize its talent pipeline by equipping a workforce to handle evolving data and to confront the questions that result from this evolution. At the same time, a clear need exists to expand data literacy and data acumen in the professional workforce and in

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the general public (NASEM 2024a; The White House 2025). Statistics is well positioned to lead these efforts and advance statistical thinking as data increasingly inform decision making in science and engineering (Marshman and Dunn 2023), which will require sustained investment from agencies such as NSF, NIH, and the National Security Agency and may be amplified by the work of professional societies.

INNOVATION IN STATISTICAL METHODOLOGY AND APPLICATIONS

The life cycle depicted in Figure S-1 underscores the fact that the field of statistics is always changing; as new scientific and engineering questions arise, new statistical innovations emerge, necessitating new methodological and theoretical developments. The figure also reveals how advances in science and engineering can feed back to synergize statistical development. Often the most exciting developments in statistics surface when new technologies in science or engineering enable scientists to ask questions that were previously impossible. As new scientific and engineering questions arise, the key statistical skills required and the life cycle for statistical innovation displayed in Figure S-1 will remain.

Uncertainty quantification is an essential part of any scientific enterprise and any technological development enabling risk-aware decision making. Uncertainty quantification for increasingly complex models, such as digital twin, hierarchical, multiscale, and hybrid models that also address downstream applications, is an active area of research.

Hybrid statistical–scientific modeling capitalizes on both the insights of scientific principles and mechanistic models and data-driven analytics, such as foundation models, to characterize some phenomenon (NASEM 2025a). Access to high-dimensional, multi-type data; output of physical, mathematical, and predictive models; measures from sensor arrays; and generated synthetic data from predictive models are all increasing. Transdisciplinary collaborations allow hybrid development of data-generative models, analytic models, and predictive models, along with inferential systems for increasing knowledge, supporting decisions, and expanding understanding. However, as noted in the National Academies' (2024b) report on digital twins, more work is needed before hybrid models can scale to complex systems.

> *Conclusion 5-6: Hybrid formulations are often the foundation of uncertainty quantification. Statisticians working at this intersection think carefully about the*

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transparency in assumptions, computational cost, and complexity of implementation—often addressing all frontiers with their contributions. A great opportunity exists to expand support for rigorous collaborations at this interface, with statisticians playing a leading or co-leading scientific role. The benefits span the breadth of science and engineering discovery.

Recommendation 7-4: Funding agencies should collaboratively identify crosscutting, translational issues to inform future programs supporting the emergence of hybrid statistical–scientific models. Such programs should foster collaborative advances in statistical, scientific, and engineering theory, analytic methods, and applications.

Conclusion 5-3: Adaptive and autonomous statistical design—methods designed to change sampling and simulation strategies in real time and in response to insights obtained through sequential learning—will be an important area for cross-agency collaboration. Scientific, engineering, industrial, and other domain-specific substantive problems will provide use cases to guide this research, and ultimately such capabilities can be expected to be implemented through AI agents. Advanced clinical trials and advances in the design of materials provide key examples of statistical innovation in this area.

Recommendation 7-5: In the near term, the National Institutes of Health (NIH) and the National Science Foundation (NSF) should establish an interdisciplinary and translational cross-agency initiative incorporating innovative adaptive and autonomous statistical design, in which statisticians play a critical role in funded teams. Although NIH and NSF may serve as the leaders of this cross-agency initiative, the National Security Agency, Department of Energy, National Aeronautics and Space Administration, national laboratories, U.S. Department of Agriculture, and other federal research agencies are also in a position to benefit directly from U.S. leadership in the space of adaptive and autonomous statistical design.

SYNERGISTIC COLLABORATIONS WITH SCIENCE AND ENGINEERING

The frontiers of statistics intersect with allied fields including mathematics, physics, computer science, engineering, and medicine. These domains employ stochastic models that

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combine disparate data sources and domain expertise, prioritize accurate uncertainty quantification, and strive for scientific knowledge gleaned from observational data and complex experiments.

Synergistic progress is seen across the spectrum of science and engineering and is vital for U.S. competitiveness. For example, the Department of Energy issued a call to automate material design, which benefits from extensive and ongoing statistical collaborations (Department of Energy 2025). A major emphasis includes adaptive experimental design, surrogate modeling, and Bayesian optimization to prioritize candidate materials for further testing. This initiative provides a template for statistical advancement via large-scale open databases of atomic structure and function (NIST 2021).

Expanded interdisciplinary team science, in which statistics plays a key role, is vital to personalized medicine, public health, ecology, aerospace, applied physics, and climate studies, among others. Because statistical thinking underpins trustworthy inference and reproducibility, embedding statisticians in data-driven research across domains can help ensure that statistical rigor and uncertainty quantification are centered early, ultimately leading to more efficient, responsible science.

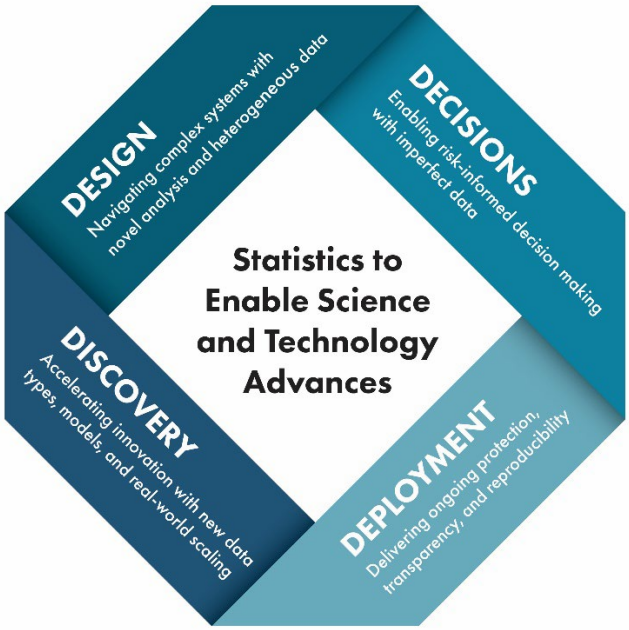

FIGURE S-2 Statistics to enable science and technology advances.

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As an enabler for science and technology, statistics serves a different role in different phases of innovation, from discovery to design to decisions to deployment (see Figure S-2). For the discovery phase, statistical paradigms that use probability models to describe what we may expect in terms of data structure in relation to the unknowns we seek to characterize provide quantifiable probabilistic metrics that balance risk and cost efficiency. Risk-based and transparent decisions from imperfect data and information are at the core of the field. In deployment and sustainment, the role of statistics becomes monitoring to ensure security, trust, reproducibility, and effectiveness in improving some aspect of scientific, commercial, or societal well-being.

Computing infrastructure is undergoing a significant transition. Statistical frameworks often utilize iterative procedures requiring substantial computing resources. Improved computational hardware will enable (1) massively parallelized sampling algorithms, (2) real-time uncertainty quantification for online data collection, (3) real-time robustness and reliability checks, and (4) the ability to democratize experimentation. These advances require close, mutually beneficial collaboration between statisticians and computer scientists.

As computing accelerates, it is important to be cognizant of its environmental impacts. Data centers require enormous amounts of energy and water. In many cases, complex AI models are unnecessary for day-to-day use (e.g., small language models may outperform large language models for many tasks as demonstrated by Belcak et al. [2025] and Sinha et al. [2025]). Researchers need to rely on tools that are fit-for-purpose, opting for those that are less resource-intensive but suitable for the task at hand. Efficient algorithms and deliberate model design, alongside edge computing, could also reduce computational burden.

In today's statistics education, computing is an integral component of curricula (Horton and Hardin 2021). Graduate-level statistics curricula would ideally facilitate collaboration, enable scalability, and encourage reproducibility. The core of these infrastructures requires unified data ecosystems where findable, shared, and version-controlled repositories are kept with standardized preprocessing procedures and in-depth metadata.

Academia and industry play complementary roles within the research ecosystem, and both would benefit from mutual support and increased collaboration. Academic research often focuses on the exploration of new or unproven ideas, while industry often scales up established research. Many technologies and breakthroughs that fuel private sector success are rooted in

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decades of federally funded academic research. For instance, stochastic modeling underpins applications across finance, life sciences, space exploration, complex risk analysis, and weather forecasting.

Public–private partnerships offer a mechanism for academia and industry to support one another. Collaborative models that give academics the freedom to continue foundational research, rather than focusing on research driven by industry needs, are especially valuable because foundational research lays the groundwork for future advances.

Recommendation 7-6: The National Science Foundation, National Institutes of Health, and National Security Agency should invest in undergraduate and graduate training in statistics, biostatistics, and data science. Mechanisms include graduate fellowships, research experiences for undergraduates, and innovative programs that vertically and cross-sectionally integrate students, postdoctoral graduates, faculty, and industry researchers in statistics. Strategic priorities for such support should focus on cross-disciplinary, application-driven training; interuniversity collaboration; partnerships with industry and government; and computational training for the next generation of statisticians capable of leveraging the advanced large-scale computing infrastructure developed in industry today.

Conclusion 4-3: Public–private partnerships increase the vitality of collaborative statistics research for both industry and academia, particularly when academic researchers are given the latitude to focus on foundational research.

Recommendation 7-7: Both the National Science Foundation and National Institutes of Health, as well as other funding agencies, should expand opportunities to advance science through the inclusion of statistics with current and emerging public–private partnerships. Such historical investments have led to breakthroughs in medicine, health care, agriculture, energy, the discovery of new materials, the financial sector, insurance, weather prediction, and resilient infrastructure and communities, and will lead to significant advancements in the future, ensuring sustained U.S. competitiveness.

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STATISTICAL ARTIFICIAL INTELLIGENCE

The statistical sciences undergird AI. Learning algorithms and modern AI grew from advances in random matrix theory. For example, random matrix theory, the study of properties of large matrices with random entries, has been under development for several decades, with many contributions and connections within statistics, mathematics, probability theory, computer science, and other fields. The tools developed in this arena form the foundations of important statistical methodologies, including compressed sensing, covariance estimation, matrix completion, and dimensionality reduction (Ailon and Chazelle 2009; Candès et al. 2006; Chen et al. 2012; Recht 2011). Moreover, random matrix theory plays a crucial role in understanding additional areas of statistics, machine learning, and deep learning, such as the behavior of deep neural networks (Martin and Mahoney 2021). Just as the past 100 years of statistics formed the groundwork for today's AI, the next 20 years of statistics will form the groundwork for the next generation of AI. In many ways, today's AI is simply a vehicle in which machines use statistical theory and reasoning to solve problems at scale in (ideally) a self-supervised manner.

Statistical issues are ubiquitous in data-driven research, and statisticians bring unique expertise to evaluate and enhance statistical rigor and interpretability of AI-based research. Many of the challenges facing today's AI (interpretability, transparency, quantification of uncertainty, etc.) are in fact statistical problems, and the solutions to these challenges will be enabled by contributions from the field of statistics. The use of statistical tools includes a responsibility to quantify uncertainty and assess whether the choice of the model, methodology, or data is appropriate for the task at hand. The rapid development of powerful AI models has enabled many advances in data science, such as using generative models to produce synthetic data or using AI models' predictions for high-stakes decisions such as hiring, and these advances must be paired with a foundational understanding of their statistical properties and potential for errors to ensure safe and fair use.

> *Conclusion 2-4: Investing in research in statistical foundations is critical to ensure a careful and measured use of the tools developed in statistics, machine learning, and AI. Without this investment and support for statistical rigor, these tools could misguide decision making and even cause harm.*

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Statistical development and methods are the foundation of AI and statistics, but the science of statistics, like other technical fields, is also adapting to and exploiting AI. Statistical tasks that are ripe for AI acceleration are those that are manual or human-intensive, including various aspects of data preparation and cleaning, such as looking for outliers, checking for internal consistency, reformatting, regridding, or resampling. Examples of current methodological ingenuity in statistical AI that will remain at the forefront of the field for the next few years include automated hypothesis formulation, neural estimation, data fusion across multiple heterogeneous data sources, dimension reduction using autoencoders for complex data sources, and synthetic data augmentation through generative AI models. Eventually, this list could include fully automated model-building and optimization agents, but this will require the statistics community to reflect on the generic steps in model building and how to structure them in a generalizable way that can be incorporated in the algorithms.

The transformative potential of AI and automation in science and engineering calls for a workforce that develops and deploys ethical and trustworthy AI technology across sectors (NASEM 2025b). As data-based tools become increasingly embedded in everyday life, individuals must be able to engage with such systems thoughtfully and responsibly, treating them as a powerful yet fallible set of tools not fit for every purpose. Instead of passively adopting AI technologies, users need to understand their strengths and limitations, question and evaluate outputs, recognize potential biases, and make informed decisions under uncertainty. Embedding statisticians in AI research teams mitigates risks of algorithmic bias and misinterpretation, helping to ensure that models are not only accurate but also ethically and scientifically sound.

> *Conclusion 6-2: An AI-savvy workforce will not merely adopt these tools but will understand the strengths and limitations of AI, thoughtfully evaluate model outputs, recognize potential biases, and incorporate awareness of uncertainty into its decision making.*

In this context, being AI-savvy does not refer to being able to understand the inner workings of AI but rather to being able to know when and how to use it in decision making. This requires awareness of AI strengths and pitfalls, which need to be supported by training. Many such efforts are underway across K–12, community college, undergraduate, graduate, upskilling,

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and public outreach programs, as exemplified by the U.S. government's executive orders[3] and enabled via investments such as NSF's AI Research Institutes and recent solicitations.[4]

AI education would benefit from expanded training in understanding data, uncertainty, inference, and causality- and prediction-based decision making. The AI revolution affords the community the opportunity to expand the reach of statistics education. The technical performance and proliferation of machine learning algorithms threaten to outpace our understanding and management of their consequences.

Conclusion 6-3: The need for embedding statistical thinking into AI education is both urgent and rapidly increasing, especially as AI tools are adapted and deployed within data-driven workflows.

The U.S. government is committed to advancing AI, as demonstrated by programs discussed in Chapter 2. Statistical rigor is vital to AI research and education, and carefully including statistics in AI-related efforts will maximize effectiveness, efficiency, and safety.

Conclusion 2-9: NSF invests heavily in AI research and education across multiple directorates. Incorporating the depth and breadth of perspectives from the statistics community into these investments increases their value and impact for U.S. competitiveness.

Recommendation 7-3: Federal funding agencies should prioritize funding to expand and accelerate rigorous ongoing statistical thinking in artificial intelligence (AI) research and education initiatives. Successful initiatives would embed statistics into AI programs in the following ways:

a. <u>AI-Based Research</u>. Specifically, federal funding agencies such as the National Science Foundation (NSF), National Institutes of Health (NIH), Department of Defense, Department of Energy, National Aeronautics and Space Administration, and National Security Agency should do the following:

i. Support the identification of crosscutting opportunities to integrate statistical thinking into AI and AI-based research through centers of

[3] See, for example, AI Gov (n.d.) and The White House (2025).

[4] See, for example, Laude Institute (n.d.), NSF (2025), and Office of Communications and Outreach (2025).

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excellence or major collaborations that build on the success of the National AI Research Institutes and broaden statistical research to ensure the accurate, reliable, and reproducible use of AI for improved research results.

ii. **Elevate the role of statistical rigor in data-driven and AI research by incorporating explicit guidance in calls for proposals and proposal review criteria and including reviewers with statistical expertise. Language should include guidance regarding statistical validity of inference and accuracy of uncertainty quantification.**

iii. **Support a comprehensive survey that identifies the footprint of the field of statistics, the statisticians involved, and the foci of their research, throughout the network of National AI Research Institutes.**

b. **<u>AI Education</u>. Future NSF, NIH, and other federal agencies' AI-related training funding, such as to accelerate AI education programs, should include statistical thinking and rigor as a major pillar. Calls should explicitly seek collaboration with the statistics community.**

The frontiers of statistics are defined not only by continued advances in core methodology and theory but also by the field's capacity to engage with new data modalities, emerging scientific questions, and complex societal challenges. As statistics progresses as both a foundational and integrative discipline, its impact will depend on sustained investment in methodological creation, interdisciplinary collaboration, and workforce development. Strengthening these dimensions is essential to national competitiveness by enabling the rapid translation of data into reliable knowledge, technology, and policy and positioning statistical science as a core driver of innovation across science, engineering, medicine, and the data-enabled economy.

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1
Introduction

The frontiers of statistics are dynamic and constantly evolving to meet the shifting needs of science and engineering. Rooted in the pursuit of finding practical solutions to real-world challenges, the discipline brings insight and wisdom through observations and experiments. Throughout history, statistics has been a catalyst for innovation across scientific endeavors. Examples from the 20th century alone include advancement in precision agriculture that increases yields and improves efficiency of land use; pioneering design, implementation, and analysis of human clinical trials for effective and safe medical treatments; groundbreaking analyses of observational data linking the hazards of smoking and the benefits of healthy diet and exercise; and breakthroughs in statistical process control to optimize production and minimize waste in industry.

This report cements the role of statistics as a bedrock of scientific and engineering progress. With the influx of computing capacity, proliferation and interoperability of data (in their many forms), and availability of increasingly complex modeling, the statistical landscape is shifting dramatically in ways that will transform its demands and broaden its contributions. To preserve and defend U.S. global advantage, the nation must focus its statistical research to propel inventiveness and anticipate the future needs of the scientific and engineering community. Given this crucial moment, this report explores the cutting edge of the statistical sciences and envisions its trajectory over the next decade based on the current landscape. The report will outline opportunities for near-term investments, demand for new approaches and focused training, and areas that are ripe for high-risk, high-reward innovation. Throughout, the term “statistics” encompasses biostatistics, and “science and engineering” includes health and medicine, finance and business, geosciences, environmental health, manufacturing, and materials science, among other disciplines.

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STATISTICAL CONTRIBUTIONS

Theoretical advances provide opportunities to expand the reach and impact of statistical methods across all areas of human inquiry. Computational advances bring statistical innovation to the forefront of science and technology by providing access to tools and interpretations. Unlike many siloed areas of research, historically, the field of statistics operates at the intersection of mathematical and probability theory, computational advances, and data-based applications. This intersection is dynamic, evolves with developments in each area, and provides unique insights and opportunities for rapid advancement in probabilistic and mathematical theory, computational implementation, and each area of application (see Gelman and Vehtari [2021] for further discussion of recent themes in statistics and data science research). Going further, statistical innovations arise in

- application-based refinements, extensions, and improvements of existing analytical tools to address new questions, incorporate new data sources and data linkages, characterize complex systems, or build on challenging interactions between data elements;
- novel insights for data interpretation, advances in the types and connections of relevant data, and unfolding concepts of "data" themselves arising from other fields of research; and
- theoretical advances based on mathematical and computational breakthroughs addressing the underlying framework of statistical inference.

Taken together, discoveries in these areas enable expansion of the frontiers of statistical research in various ways, benefitting from new measurement technologies, creative data fusion and harmonization approaches, enhanced computational speed and capacity, and new mathematical concepts and connections. While the categorization of types of statistical frontiers above is helpful, it remains difficult to draw rigid boundaries that clearly delineate innovations in statistical collaboration, statistical innovations born out of other disciplines, and innovations in statistical theory.

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Key Finding 1-1: At any given time, the current frontiers of statistics occur both at the edges of what is possible within thoughtful applications, methodological innovations, and theoretical advances and, perhaps most importantly, at the intersections between them.

STATISTICAL THINKING IN SCIENTIFIC DEVELOPMENTS

Generations of statisticians and statistical practitioners have embedded statistical thinking—a mode of thinking that recognizes and incorporates uncertainty into decision making—and its methodologies throughout science and engineering, an emphasis that is highlighted in the *Guidelines for Assessment and Instruction in Statistics Education* reports from the American Statistical Association (Bargagliotti et al. 2021; Carver et al. 2016). The federation of the field across varied disciplines is a critical accomplishment in support of evidence-based decisions and the scientific method in general. In all disciplines, scientists advance their fields and knowledge through the expert use of statistical methodologies. Many domain-area scientists and engineers also contribute to the statistics itself, driving the growth of fields such as econometrics, psychometrics, statistical ecology, astrostatistics, uncertainty quantification, and data science, to highlight a few. Furthermore, research statisticians collaborate across disciplines to innovate together, as noted above. This cross-fertilization of the field of statistics with so many other disciplines is a landmark driver of breakthroughs in statistics. Addressing the increasing and varied challenges arising from novel data and domain questions from other fields will continue to drive fundamental advances in statistical theory and methodology.

The field of statistics focuses on extracting information, inference, decisions, and wisdom from data, as well as quantifying the uncertainty and randomness of the data. Constraints such as cost, quality, accuracy, reliability, validity, availability, reproducibility, representation, and transparency always factor into data-informed decisions. The move toward open-source statistical software in the 1990s marked a rapid expansion of the field and the democratization of the discipline, bringing basic and sophisticated tools to the fingertips of anyone with a computer and the knowledge to use them (Donoho 2017). At its core, the field of statistics empowers individuals and institutions to make evidence-based, informed decisions. Statistical thinking is a safeguard against poor decisions based on biased or unrepresentative data and methods. For example, without careful statistical oversight, artificial intelligence (AI)-informed decisions such

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as hiring practices may be biased and reflect past hiring patterns that showed preferences for individual characteristics that have nothing to do with a candidate's qualifications (Chen 2023; Rigotti and Fosch-Villaronga 2024; Wilson and Caliskan 2024).

The focus on deriving high-quality information from experiments that were often expensive or data that were difficult to obtain drove much of the discipline in the 20th century. From expertly designed experiments to improve crop production to adaptive clinical trials, many of these methodological advances were derived under a paradigm wherein each observation is expensive and comes with inherent uncertainty that is quantified through expert probability-based statistical methodologies. Methodologies and types of analysis developed during this time period provide a solid framework of distribution-based (or parametric) and distribution-free (nonparametric) methodologies, as well as a combination of the two. The late 20th century also saw the emergence of exploratory data analysis and statistical graphics, which ensure statisticians are answering the right question and not merely the one that is mathematically tractable, as well as tools to assess the accuracy and validity of many of the assumptions upon which statistical methods are based (Donoho 2017; Tukey 1962). These developments provide a rich set of inferential tools going into the future. Areas such as sampling, design of experiments, multilevel modeling, adaptive designs, clinical trials, graphical statistics, and optimality were and remain central to continued statistical innovation.

By the end of the 20th century, advanced modeling methodologies addressing complex dependencies in the scientific problem and the observational data were emerging. These expanded methodologies presented new computational challenges that necessitated simultaneous development of computing solutions and statistical inference methodologies in an active feedback loop. Unique computing approaches offered new directions for statistical methods, and statistical methods afforded insight into the performance and scale of computational analytics. For example, the field of bioinformatics emerged as statisticians and other computational scientists worked with geneticists to advance the genomic boundaries. This forward-looking work resulted in clinical care and biomedical research with revolutionary goals, such as personalized medicine tailored to an individual's genetic profile. The implementation and adoption of these new computational methods were enabled by the statistical community's development and acceptance of the open-source R software environment as well as Python.

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As the field of statistics moved into the 21st century with significant advances in computing; development of high-resolution sensors; detailed images of our globe, our cells, and our brains; and the emergence of social media and broad corpi of integrated text and outcomes—along with many other advances—a rapid influx of massive data emerged from a wide range of sources. With these new data types of varied quality and quantity, the frontiers of statistics continue to evolve rapidly to address these challenges. Even so, it is imperative to remember that not all modern data sets are "large," and a plethora of vital science and engineering questions remains for which high-quality data are carefully obtained, usually at high cost, to answer.

In considering the frontiers of the field, it is essential to recognize that the statistical toolbox is not static; new theory and methods have always been in development to address challenging mathematical problems and access emerging data types enabled by new measurement technologies. Statistical inference is based on probability shaped by assumptions, and methodological statisticians seek to address or modify these assumptions to address current analytic challenges while maintaining accuracy and precision within inferential frameworks.

Statistics plays a critical role in extracting insight and wisdom from data, especially in a rapidly changing, AI-driven world. The frontiers of statistics will simultaneously power AI and evolve with AI. It is more important than ever to maintain the foundational statistical principles that have supported global advancement throughout history. Statisticians understand that more data do not always lead to more information or better decisions. In some sense, the concept of "more data" in the time of big data brings similar challenges as before (e.g., bias, incompatible observations, variation, uncertainty, heterogeneous data quality; see Meng [2018] for more on the "big data paradox"), but it also requires new statistical insight, summaries, and solutions to make best use of the increased access and connectedness of disparate data sets relating to the problem of interest (Mackenzie 2014). At the same time, the "more" also introduces considerations around privacy, bias and legacy patterns, transparency, interpretability, and scalability. Moreover, one cannot ignore the impact of "more" on energy and environmental costs as the massive data centers of today drive much of the information shared across the spectrum of inquiry (NASEM 2025c). Mitigating these realities necessitates a shift toward federated statistical modeling, federated learning, and moving statistical insights between data repositories as a cost- and energy-efficient strategy.

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REPORT DEVELOPMENT AND STRUCTURE

Throughout the information-gathering phase and writing of this report, the committee hosted a series of public panels with statisticians and statistical contributors in different domains and sectors, as described in Appendix B. Additionally, the project included a public call for perspectives that allowed members of the public to offer input to the committee, also found in Appendix B.

To review, describe, and project the frontiers of statistics and respond to the project charge, presented in Box 1-1, this report is organized into five chapters that highlight key dimensions across which statistics is evolving. Chapter 2 articulates the state of the field, maps where it is going, and describes where it must move to support scientific progress in a data-rich world. Chapter 3 addresses the changing nature of data, including the development of novel data types that drive analytic innovation, the need for strong data acumen (Horton et al. 2020), and the continual transformation of data. This chapter explores how statistics and statistical intuition must transform alongside data. Chapter 4 discusses the inter- and cross-disciplinary nature of the field as the committee considers statistical intersections across all aspects of science and engineering. It outlines the importance of the federation of statistics and identifies fields that are using statistical sciences well to define their own frontiers. Chapter 5 illustrates how statistical invention leads to sweeping technological progress, acknowledging that the statistical innovation life cycle is a synergistic and symbiotic relationship between statistics and technological advances. Chapter 6 addresses the necessary changes in statistics education and workforce development required to keep pace with the evolution of the field. The final chapter, Chapter 7, outlines the report's recommendations.

BOX 1-1
Statement of Task

The National Academies will assemble an interdisciplinary ad hoc committee to produce a forward-looking assessment of the current state of the statistical sciences and emerging opportunities for the discipline and its stakeholders as they look ahead to 2035 and beyond. The study may make recommendations to funding agencies on how to adjust and expand their

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portfolios of activities and partnerships to reflect the evolution of the field and strengthen the impact of the discipline.

Specifically, the study will assess:

- **Statistical research.** The state of research in the statistical sciences, examining aspects such as the significance of recent developments and highlighting current and emerging trends, challenges, and directions
- **Statistical research in allied fields.** Interactions between the statistical sciences and the mathematical, computational, engineering, materials science, and related fields such as biostatistics and probability; the statistical foundations that underpin and motivate machine learning, artificial intelligence, the internet, and data science; the growth of data-intensive research arising in fields that incorporate statistical and data science thinking; and the role of statistics as a bridge to translational and use-inspired research;
- **Statistical needs to support scientific and technological advances.** Urgent needs for statistical innovation in areas such as verification, validation, and uncertainty quantification; privacy and security; and working with heterogeneous and complex data, among others, to enable responsible use of nascent scientific and technological developments like large language models; complex and dynamic systems (e.g., digital twins); hybrid models and suites of models that rely on both physics-based and data-driven approaches; broad data gathering and processing efforts; and policies around open data sharing;
- **Importance of statistical sciences and strength of interdisciplinary collaborations.** The role of statistical sciences in key research domains for American competitiveness, including manufacturing, materials science, finance and blockchain, biomedical and biological sciences, public health, medicine, geosciences, environmental health and science, and energy applications; and **statistical education, training, and workforce development.** Approaches toward statistics education and training to enhance workforce development across science and engineering domains (both within statistics and in fields that incorporate statistical thinking, such as artificial intelligence and data science) for a myriad of data-rich applications relevant to industry and technology, innovation and economic competitiveness, health and well-being, national security, and other areas of national interest.

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2
Statistics: A Snapshot of a Dynamic Field

The field of statistics is broad. Advances in the field often fall into one or more of three categories: theoretical, methodological, or applied. Individual research statisticians often seamlessly move between the categories with significant contributions across the spectrum.

THEORETICAL CONTRIBUTIONS

The field of statistics benefits from theoretical advances in mathematics and probability. Statistical theory uses mathematics to study the properties of statistical procedures for estimation and uncertainty quantification to establish if and when the estimates are reliable. Probability theory underlies many methodological and applied developments in statistics. The intersection between statistics and probability theory remains an active area of research. While new developments in statistics and probability theory may not have immediate application to problems in science and engineering, they guide statistical practice and provide the mathematical foundations required for new methodological development (He et al. 2019).

METHODOLOGICAL CONTRIBUTIONS

New developments in science and engineering lead to the emergence of new types of scientific questions and new data settings for which existing statistical methods are unsuitable. These new questions and data demand new statistical methods, such as new estimators or new methods for inference, new graphical displays, and improved strategies for data-informed decision making and communication.

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APPLIED CONTRIBUTIONS

Providing meaningful answers to specific scientific questions is an art unto itself. Even if the statistical method used to answer the question is not new, scientific and statistical insight requires skilled communication to translate the scientific question into a statistical problem and then communicate the statistical result back to a scientific answer. Statistically informed scientific answers to the right questions often lead to new scientific questions and simultaneously advance the boundaries of both the area of application and the science of statistics.

Many—if not most—statisticians contribute to more than one of these three categories throughout their careers and, often, even within a single research paper.

> Key Finding 2-1: Statistics falls into three broad and interrelated categories: theoretical, methodological, and applied. However, a single statistician may contribute to more than one of these categories over the course of a career or even within a single research paper or industry problem. Attributing a development to just one category can be counterproductive and does not represent the reality that these categories are intertwined.

The theoretical underpinnings of the field of statistics are crucial and continuously progressing, often leading the advances in science and engineering. Today's theoretical contributions lay the groundwork for tomorrow's methodological contributions. Furthermore, the theoretical training undergraduate, master's, and PhD students receive today enables the deep methodological understanding required for the applied analysis that they conduct tomorrow.

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THE LIFE CYCLE OF STATISTICAL INNOVATION

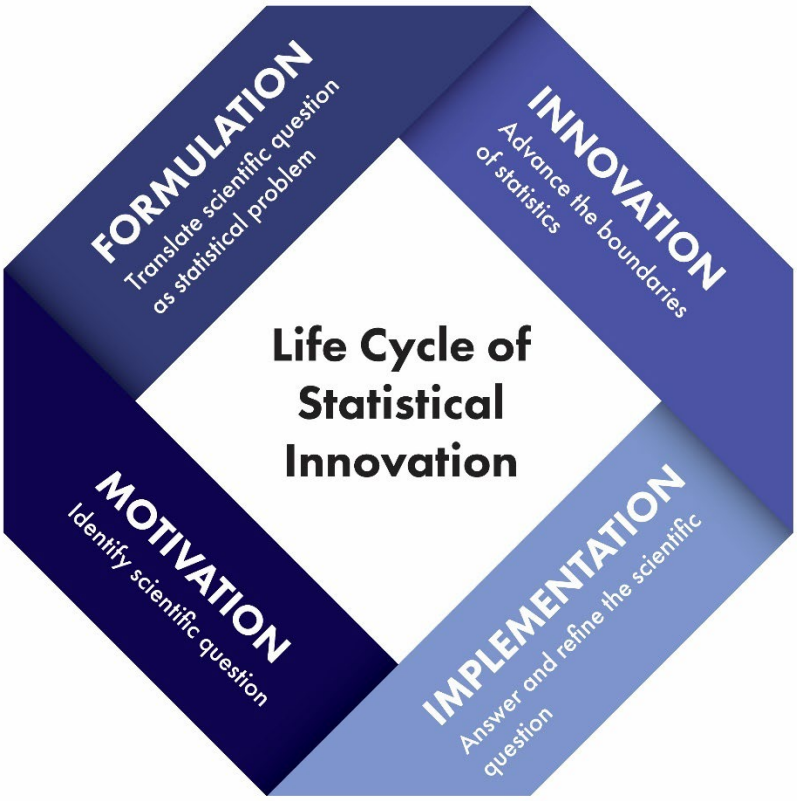

FIGURE 2-1 The life cycle of statistical innovation.

From where does statistical innovation arise? Often, new statistical problems are motivated by a new scientific question. It is the task of the applied statistician to formulate this new scientific question in statistical language. In many cases, even though the scientific question is new, the statistical solution is not; if a satisfactory statistical tool for the scientific question already exists, then no further statistical development is required.

However, the applied statistician often encounters a motivating statistical question that has not been asked before. For instance, the question may arise because of a new data-generating technology. In this case, new statistical methods may be required to provide statistically rigorous answers to scientific questions and perhaps even motivate new statistical theory to fully characterize and improve the performance of these methods.

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While the statistician may provide an initial answer to the scientific question with the best available statistical methodology, additional rigorous answers will follow statistical advances, either in methodologies or theories. Often, methodological and theoretical development reveals that further revision of the scientific question is necessary or helpful. In this case, the cycle of scientific inquiry to statistical methods development continues. In short, innovation in science and engineering motivates innovation in statistics, and, importantly, innovation in statistics motivates innovation in science and engineering.

This cycle—referred to herein as "the life cycle of a statistical innovation"—is shown in Figure 2-1. When statistical discovery arises through this cycle, it can lead to meaningful methodological and theoretical development far beyond the original motivating question, as well as an expansion of the current boundaries of the field. New methods provide new insights and new understanding, which often prompt the next generation of motivating questions and reinitiate the statistical innovation life cycle. Examples of the statistical life cycle are outlined in Box 2-1.

> Key Finding 2-2: New statistical methods afford new insights and new understanding, which often trigger a new generation of motivating questions; thus, the statistical innovation life cycle begins again.

BOX 2-1

Statistical Development and the Life Cycle of a Statistical Innovation

Many of the most important statistical developments of the past 30 years can be viewed through the lens of this life cycle. Some powerful recent examples include shrinkage priors, false discovery rate, conformal prediction, and Bayesian hierarchical modeling and computation.

Shrinkage Priors

The increasing collection of very high-dimensional data—for example, in genomics applications—motivated the need for modeling approaches that can characterize uncertainty in inferences while also handling settings in which there were far too many parameters to apply traditional methods in maximum likelihood estimation or Bayesian inference under non-informative priors. Such problems led to innovations extending beyond these traditional frameworks and resulted in an explosion of interest in shrinkage priors, ranging from early

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spike-and-slab priors (George and McCulloch 1997) to an increasing focus on global–local shrinkage priors that can be expressed as scale mixtures of Gaussians for ease in computation (Polson and Scott 2011). Popular classes of such priors include the horseshoe (Carvalho et al. 2010), Dirichlet-Laplace (Bhattacharya et al. 2015), and generalized double Pareto (Armagan et al. 2013), which have been applied in a vast array of settings ranging from high-dimensional regression to functional data analysis and have motivated algorithm development and theory on posterior contraction rates and other properties. Applications to multivariate data analysis and matrix factorization further motivated a parallel literature on increasing shrinkage priors (Bhattacharya and Dunson 2011), which can automatically infer the appropriate number of factors or rank and are now routinely applied in many settings.

False Discovery Rate

In the mid-1990s, due in part to the development of microarray technology that enabled simultaneous measurement of many features (gene expression values) via a single experiment, awareness of the problem of multiple comparisons in large-scale data increased. While this problem had been studied in statistics for decades (Tukey 1953), previous measures of error were too stringent for practical use in modern problems (namely, the family-wise error rate, which requires a bound on the probability of one or more false positives, regardless of the total number of questions being tested). Consequently, interest focused on the development of a framework for balancing the need for discovery (i.e., rejection of the null hypothesis) with the number of false positives (i.e., Type 1 errors), in a setting where one fully expects to have some true rejection of the null hypothesis (Benjamini and Hochberg 1995). This led to the popularization of the concept of the false discovery rate (which only requires a bound on the proportion of false discoveries), both within and outside the field of statistics, as well as a well-accepted toolkit for false discovery rate control (e.g., the Benjamini-Hochberg procedure; Benjamini and Hochberg 1995; Benjamini et al. 2001).

Conformal Prediction

The success of machine learning and deep learning algorithms has led to their broad adoption across a wide range of prediction tasks, including high-stakes applications such as medical diagnostics or hiring. Motivated by the need to quantify uncertainty in these algorithms' predictions, the field of conformal prediction, whose theoretical development dates back to the 1990s, has seen a surge in popularity since the mid-2010s. Conformal prediction provides a mechanism for calibrating the reliability of predictions in a way that avoids placing assumptions on the data and can be applied to any predictive algorithm (Angelopoulos and Bates 2023; Vovk et al. 2005).

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Bayesian Hierarchical Modeling and Computation

The need to model the complexity of data arising in challenging settings, involving missing data, spatiotemporal dependence, nested data structures, mixed-scale variables, high-dimensional multivariate outcomes, and so on, has motivated an explosion of literature on intricate Bayesian hierarchical models. Some particularly active areas spanning applied-driven innovations, novel probabilistic modeling, foundational theory, and algorithm development include Bayesian nonparametric approaches (e.g., Gaussian and Dirichlet processes; Banerjee et al. 2014; Ghosal and van der Vaart 2017), mixed membership models (e.g., latent Dirichlet allocation; Airoldi et al. 2014), and flexible latent variable models (e.g., Gaussian process latent variable models, variational autoencoders). Challenges in fitting such rich models to complex data while maintaining an accurate approximation to the posterior distribution have spawned massive advances in Markov Chain Monte Carlo algorithms and variational Bayes (Blei et al. 2017) including original algorithms, new theory assessing the performance of these algorithms, and probabilistic programming languages for routine implementation (Winter et al. 2024).

Often, as consensus emerges within the statistics community about the best practical answer to the scientific or engineering question, this answer is communicated to the applied field. For instance, within the field of genomics, the false discovery rate control example in Box 2-1 is now a mainstream tool. Early proposals for false discovery rate control, such as the Benjamini-Hochberg procedure (Benjamini et al. 2001), were followed by plug-in permutation approaches for false discovery rate control and estimation (Storey and Tibshirani 2003), which have since become standard in the field of genomics.

Further statistical development—including software development—and refinement in any given area often continues for years after the initial scientific or engineering question is answered, much to the benefit of both the field of statistics and the scientific domain itself. As statistical tools develop, they improve, adapt to the evolving sources of relevant information in the data, employ advances in theory and computing, and provide more general tools for further application.

In addition, many instances of the statistics innovation life cycle provide meaningful contributions to scientific and engineering domains, even when they do not lead to extensive methodological or theoretical development within the field of statistics. Take, for instance, recent developments in ecology and biodiversity research. It has become common to use DNA

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metabarcoding to improve understanding of which species, such as arthropods and fungi, are present or absent at different sampling sites and times. This leads to large data sets with binary outcomes across many dimensions, rendering popular joint species distribution models in ecology unusable unless all rare species are removed. New Bayesian statistical methods have been motivated by the ongoing, large-scale, international biodiversity project, the Earth BioGenome Project.[5] This statistical progress encourages the advancement of these methodologies by a broader group of statistical scientists.

> Key Finding 2-3: The life cycle of statistical innovation characterized in Figure 2-1 involves the translation of a scientific question into a statistical problem for which new theory and methods are developed and which ultimately leads to an answer for or refinement of the original scientific question.

> *Conclusion 2-1: Statistical scientists drive the statistical innovation life cycle through the adoption and rigorous evaluation of emerging technologies and applications. Sustaining the agility and impact of this innovation cycle requires increased funding and prioritization of research contributions that are practice-informed and application-driven to advance both methodological and applied work across the discipline.*

The Critical Role of Theoretical Statistics

Statistical theory plays a key role in the life-cycle diagram shown in Figure 2-1, providing the fundamental building blocks required for methodological development and inventive applications. Advances in statistical theory are critical to the timely execution of the statistical life cycle. As alluded to above, many of today's rapid advances have deep roots in decades of prior research in theoretical statistics. In many cases, the methodological innovations of the future are predicated on the statistical theory the community prioritizes today.

To demonstrate the power and importance of statistical theory, the following examples illustrate how preexisting statistical theory, developed over the course of decades, formed the fundamental building blocks for breakthroughs in statistical methodology.

[5] The website for the Earth BioGenome Project is https://www.frontiersin.org/journals/science/article-hubs/earth-biogenome-project/explainer, accessed January 16, 2026.

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Martingale Theory to Biostatistics

The innovation around multiple testing and false discovery rate was highlighted above, noting the importance in genomics. Such innovations draw heavily from martingale theory, an area within probability dating back to the 1930s that is used to study the problem of betting on outcomes of stochastic processes. In modern statistics, martingale theory plays a crucial role in *multiple testing methods* (e.g., establishing the validity of the Benjamini-Hochberg procedure and many of its variants; see Lei and Fithian 2018; Storey et al. 2004). Martingale theory is also the key tool behind the *knockoff filter*, a variable selection approach in multiple regression problems, based on generating "knockoff" copies of individual features to detect the false discovery rate of predictions (Barber and Candès 2015; Candès et al. 2018; Sesia et al. 2019; Wu et al. 2007). More recently, martingale theory lies at the heart of the emerging field of safe anytime-valid inference, which provides statistical tools for testing hypotheses over time as data are collected (Ramdas and Wang 2025; Ramdas et al. 2023). In addition, the emerging field of Bayesian predictive inference uses martingale theory to show that coherent Bayesian posterior distributions can be defined through predictive distributions of future observations given past observations without need to explicitly specify a likelihood and prior (Fong et al. 2023).

Matrix Theory to Modern Artificial Intelligence

Learning algorithms and modern artificial intelligence (AI) grew from advances in random matrix theory. Random matrix theory, the study of properties of large matrices with random entries, has been under development for several decades, with many contributions and connections within statistics, mathematics, probability theory, computer science, and other fields. The tools developed in this field form the foundations of many important statistical methodologies, including compressed sensing, covariance estimation, matrix completion, and dimensionality reduction (Ailon and Chazelle 2009; Candès et al. 2006; Chen et al. 2012; Recht 2011). Moreover, random matrix theory plays a crucial role in understanding many additional areas of statistics, machine learning, and deep learning, such as the behavior of deep neural networks (Martin and Mahoney 2021).

Riemannian Manifolds to Characterizing High-Dimensional Systems

Major advances in discovering complex patterns in high-dimensional systems grew out of the theory of Riemannian manifolds. Riemannian manifold theory, developed in the

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mathematical area of differential geometry, explores properties of geometric spaces that generalize the Euclidean spaces used in most statistical settings to a much broader class of locally Euclidean spaces. Parameters and data restricted to non-Euclidean manifolds are common, with some notable examples including the sphere, Stiefel, and Grassmannian manifolds. Riemannian manifold theory is heavily used in developing statistical methods for "object data," which are restricted to lie on a manifold; for algorithms and theory for estimation of parameters restricted to a manifold; and for manifold learning, which forms the basis for a broad class of nonlinear dimensionality reduction algorithms (Goh and Vidal 2008; Zhang and Zha 2004). In a similar vein, the area of topological data analysis provides strong statistical theory to observe patterns and features across a wide variety of systems—for example, in finance or biostatistics (Aguilar and Ensor 2020; Crawford et al. 2020; Rai et al. 2024).

Graph Theory to Social Networks

Understanding social networks and complex dependencies in systems has long been a focus of the statistics community with powerful outcomes (e.g., Holland and Leinhardt 1977; O'Malley et al. 2008; Wasserman and Faust 1994). Graph theory, also heavily studied in mathematics and theoretical computer science, provides a theoretical basis for statistical inference for network data (Sengupta 2025), for graphical modeling of conditional independence structure in data (Geiger and Pearl 1993), and for causal inference based on directed acyclic graphs (Textor et al. 2016; VanderWeele and Robins 2010). In addition, graph-theoretic approaches are increasingly used in machine learning and statistics to improve algorithmic efficiency and provide simpler representations of complex data and systems (e.g., Ucer et al. 2022). This field has expanded in statistics as inference on dynamic graphs provides statistical guarantees and guides our understanding of deep causal mechanisms (Ogburn et al. 2024).

Stochastic Processes to Broad Scientific Innovation

The use of stochastic processes is ubiquitous in science and engineering from implementing adaptive design in materials science, building resilient infrastructure and cities, pricing financial derivatives, to understanding evolving environmental exposures and their downstream consequences (Pavliotis 2014). Stochastic process theory forms the basis for a wide variety of developments in survival analysis and nonparametric Bayes modeling, in which an intricate understanding of properties of Lévy processes, Gaussian processes, and normalized

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random measures is key to model and algorithm development and understanding (Klein et al. 2014; Müller and Mitra 2013).

Robust Statistics and Extreme Value Theory to Predicting Rare Events

Robust statistics improve performance of deep learning methods that capture rare events and handle outliers and missing data, offering reliable inference procedures, alternative loss functions, and divergence measures (Coles 2001; Little and Rubin 2020). The theory of extreme values and rare events has significant implications for addressing future challenges in tracking rare but costly events (De Haan and Ferreira 2006; Galambos 1994). As yet, handling data that are independent in the tails and characterizing multivariate extremes remain open and active areas of research in theory and in practice. This work is desperately needed to support natural disaster prediction and fraud detection, among other low-probability, high-impact events.

The examples above illustrate how theoretical development forms a vital foundation upon which methodological development and original application rely. Not only does theoretical statistical innovation illuminate the mathematical universe, but these expansive (and growing) foundations may have been under development for decades before statisticians and their collaborators could apply them; theoretical research in statistics defines the basic science underlying its frontier development.

> *Conclusion 2-2: Breakthroughs today stem from decades of investments in foundational research within statistics. Theoretical innovations may take years to develop before statisticians and their collaborators can put them into practice. This theoretical research lays the groundwork for the fundamental science that drives advancements at the forefront of the discipline.*

> *Conclusion 2-3: Investments in robust statistics and extreme value theory, for example, have helped experts predict and track natural disasters and thus informed disaster resilience.*

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EVOLVING FRONTIERS

Current Drivers of Statistical Evolution

The life cycle depicted in Figure 2-1 underscores the fact that the field of statistics is always changing; as new scientific and engineering questions arise, new statistical innovation emerges, necessitating new methodological and theoretical developments. The figure also reveals how advances in science and engineering can feed back to synergize statistical development. Often the most exciting developments in statistics surface when new technologies in science or engineering enable scientists to ask questions that were previously impossible or did not capture the interests of the science and engineering communities.

Below is an illustrative but non-exhaustive set of drivers of the statistical frontiers in 2025, which informs the trajectory through 2035. As new scientific and engineering questions arise, and as existing statistical problems are solved, this list will change; however, the key statistical skills required and the life cycle for statistical innovation displayed in Figure 2-1 will remain constant.

AI Changes How Statistics Is Done

According to the World Economic Forum (2024), "Scientists predict that general-purpose AI will transform every part of the scientific discovery process over the next few years. Researchers can draw on past findings to envision new possibilities—AI allows connections to be made and inferences to be drawn that lie beyond the capacity of human minds alone." Machine learning–based AI is itself a technological advancement, and its rapid application to many fields enables scientific and technological progress. The science of statistics, like other technical fields, is adapting to and exploiting AI. Statistical tasks that are ripe for AI acceleration are those that are manual or human-intensive, including various aspects of data preparation and cleaning, such as looking for outliers, checking for internal consistency, reformatting, regridding, or resampling. Many of these types of tasks require understanding how data are collected or produced. To the extent that accurate examples of this process are available on the Internet, it can be much faster to query a chatbot than to search manually. Near-term examples that illustrate how AI changes statistics include the following:

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- *Automated hypothesis formulation*, in which by virtue of the ability of machine learning to find complex patterns in large high-dimensional data sets, AI can reduce the amount of time and effort required to select potential hypotheses worth testing (Lowe 2023).
- *Neural estimation*, which uses machine learning in a simulation-based framework to quickly learn likely parameter values associated with certain types of models (e.g., spatial) from data (Sainsbury-Dale et al. 2023).
- *Data fusion*, where multiple heterogeneous data sources are combined. Such approaches typically lack formal uncertainty quantification so there are great opportunities for statisticians to contribute insight into modeling formal and informal uncertainties (Peterson et al. 2024).
- *Dimension reduction* using autoencoders for complex data sources (Kärkkäinen and Hänninen 2023).
- *Synthetic data generation* and augmentation through generative AI models such as generative adversarial networks and diffusion models can facilitate simulation-based inference (SBI) and estimation of probabilistic models where parametric assumptions may be unreliable (Pezoulas et al. 2024).
- SBI using repeated simulations to explore which scenarios could have plausibly led to the observed data instead of trying to calculate the likelihood of observing the data under various scenarios. SBI enables rigorous statistical inference in settings that were previously beyond reach (Singh et al. 2021).

Eventually this list could include fully automated model building and optimization agents, but this will require the statistics community to reflect on the generic steps in model building and how to structure them in a generalizable way that can be incorporated in the algorithms. The same can be said about intelligent graphical displays. Building on the grammar of graphics philosophy, producing AI-generated automated informative graphical displays of data and information is conceivable.

AI is not only of interest across a variety of science and engineering domains, but it can also automate, expedite, and improve processes that were formerly manual or human-intensive, including large-scale data curation for complicated data analyses.

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Characterizing Accuracy and Uncertainty

Quantifying and communicating uncertainty in parameters, predictions, and decisions of a model is critical to its use. Quantitative measures of uncertainty by themselves are inadequate; understanding the level of confidence in the system, its inputs, and the solutions provided can guide decision makers to make more informed decisions. However, for complex models, such as hierarchical and multiscale models, and machine learning approaches, how to quantify uncertainty both in the model itself and in its downstream applications remains an open problem. As systems like digital twins,[6] for instance, increase in their complexity and incorporate continual or even streaming data, they demand new statistical and computational approaches to fully characterize their uncertainty. Furthermore, innovative efforts such as those at the U.S. Census Bureau result in data products that are modeled with inherent complex uncertainties. Incorporating the uncertainties in modeled information is as important as the estimated mean or predictive distribution.

Conformal prediction, described in Box 2-1, provides a distribution-free framework for quantifying uncertainty associated with predictions of complex models and complex inputs. The classic approaches for accounting for uncertainty in model parameters in prediction intervals break down in high-dimensional, nonlinear machine learning algorithms (e.g., a deep learning classifier). The appeal of conformal prediction is that it uses only prediction errors in a test set to form prediction intervals and requires little understanding of the underlying prediction mechanism. Assuming the prediction observations are exchangeable with the test set, conformal prediction gives finite-sample guarantees of coverage probabilities. Therefore, it replaces reliance on a thorough understanding of the statistical prediction algorithm with a high-level understanding of the data collection procedure, which will be increasingly appealing as models become more complex (Angelopoulos and Bates 2023; Vovk et al. 2005).

Prediction-powered inference, by contrast, considers the setting in which predictions and the corresponding uncertainties from a predictive model will be used to augment gold-standard data available for inference on a parameter of interest (Angelopoulos et al. 2023; Hofer

[6] This report uses the National Academies' definition: "A digital twin is a set of virtual information constructs that mimics the structure, context, and behavior of a natural, engineered, or social system (or system-of-systems), is dynamically updated with data from its physical twin, has a predictive capability, and informs decisions that realize value. The bidirectional interaction between the virtual and the physical is central to the digital twin" (NASEM 2024b, 1).

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et al. 2024). This approach essentially uses the gold-standard data for the task of correcting the outputs of the fixed model, which may require only a small data set if the predictions are approximately correct (Angelopoulos et al. 2023; Zrnic and Candès 2024). Therefore, this approach could exploit machine learning predictions to reduce the sample size required for an experiment if a trustworthy prediction model with quantified uncertainty is available.

Explainable or interpretable machine learning favors models that are more easily understood by humans and characterize relationships between inputs and outputs, where possible. Rather than quantifying uncertainty statistically, explainable machine learning targets the deeper notion of uncertainty about the inner workings of the predictive model. This provides transparency that is critical for building trust in an algorithm and evaluating its fairness (Belle and Papantonis 2021; Rudin 2019). Explainable machine learning is an exciting area at the frontier of statistical research because the task varies widely depending on the setting and the target audience. Solutions range from fitting carefully chosen linear models that lay bare the entire algorithm to be scrutinized by financial regulators, to augmenting machine learning–based diagnoses with human-friendly and user-customized explanations of the driving factors of the prediction.

Methods to detect and address distribution shift are an important concern in real-world settings, as the data to which the model is applied may change over time. If the distribution of the features or outcomes or the relationship between the features and outcomes change over time, then the initial assessments of uncertainty must adapt as well (Hanneke and Kpotufe 2019). For example, a model to detect fraud should be monitored continuously and adjusted as customer and criminal behaviors evolve (Lipton et al. 2018; Park et al. 2020). Predictive models more firmly rooted in causal relationships are less susceptible to distribution shift, but all empirically derived models require ongoing monitoring for model risk (Tibshirani et al. 2019). The frontier of this problem builds on classical statistical tools, such as quality control and population weighting, by incorporating recent developments including transfer learning, particularly Bayesian transfer learning (Suder et al. 2025).

Fairness in machine learning involves characterizing and mitigating the extent to which a model's accuracy or uncertainty varies as a function of covariates or outside variables (e.g., a model's predictions might be more accurate for some demographic or racial groups than for others) (Chouldechova and Roth 2020). Ensuring fairness is critical for high-stakes decisions

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such as hiring, credit scoring, and criminal sentencing (Chouldechova 2017; Romanov et al. 2019). Statisticians play a key role in promoting fairness by identifying potential sources of bias, including non-representative data collection or algorithms, and correcting biases using reweighting or resampling techniques. New methods are being developed to explicitly promote fairness by incorporating fairness measures within the objective function used to train the statistical model. These methods rely on context-specific definitions of fairness and balancing the trade-off between fairness and overall predictive performance.

> *Conclusion 2-4: Investing in research in statistical foundations is critical to ensure a careful and measured use of the tools developed in statistics, machine learning, and AI. Without this investment and support for statistical rigor, these tools could misguide decision making and even cause harm.*

Inference After Data Exploration

Classical statistics considers inference on hypotheses that were selected before looking at the data, often before the data are even collected. However, in contemporary applications, the data are often collected first, either passively or under controlled design conditions, and then hypotheses are selected based on exploration of the data (e.g., feature selection, clustering). Alternatively, data may have been collected for entirely different purposes and under different assumptions. This poses a problem for standard statistical inference techniques; for example, a standard hypothesis test will not control the Type 1 error if the hypothesis was selected from the data, and standard confidence intervals will not attain the nominal coverage if the parameter value was selected based on the preexisting data.

Many applications of promising machine learning methods in science and engineering (e.g., clustering, dimension reduction) suggest hypotheses to test. However, testing those hypotheses using the same data used to generate them is inherently problematic. The scale of the problem increases as machine learning has become more central to many data analysis pipelines.

Post-selection inference, or selective inference, refers to a broad set of strategies for conducting valid inference on a parameter or hypothesis selected from the data.

Conditional selective inference strategies explicitly account for the process of parameter selection in conducting inference. Non-statisticians often generate hypotheses from data and subsequently test those hypotheses as though they were specified in advance of data collection.

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However, this is an inferential mistake, which the statistical community is actively seeking to correct using methods to model the impact of the hypothesis generation directly (e.g., data splitting strategies).

Data splitting strategies split the data into independent (or even dependent) components in order to estimate parameter values using one component and conducting inference on the other.

As science and engineering continue to shift away from problem-specific data collection to test pre-specified hypotheses and toward data collection and integration to both generate and test hypotheses, there is an increasing need for a rigorous and flexible statistical framework that addresses this new paradigm.

Revealing Latent Structure in Complex Data

In analyzing complex data, a common goal is to obtain a lower-dimensional interpretable characterization of variation in the data. These considerations have historically motivated methods such as principal components analysis (PCA), alternative nonlinear data visualization methods such as t-Distributed Stochastic Neighbor Embedding and Uniform Manifold Approximation and Projection, and clustering approaches like k-means. In complex modern settings, these traditional approaches typically encounter practical implementation problems, particularly when applied at scale, motivating ongoing interest in developing better methods for inference on latent structure in data.

As an example, for many diseases, a single treatment is not effective for all patients, and one can obtain greater efficacy by inferring disease subtypes with the goal of targeting treatment to each patient's subtype. Simply applying k-means or classical clustering approaches to patient data consisting of demographic information, clinical variables, and omics measurements tends to produce unreliable results with little information for targeting individual treatments. This motivates improved clustering methods that are better in high dimensions (Chandra et al. 2023; Zhou et al. 2025) and can handle multiview data (Dombowsky and Dunson 2025) while ideally characterizing uncertainty in clustering. Often, simply inferring one set of clusters is overly simplistic, motivating richer alternatives such as inferring multiple binary indicators of patient health or even latent trees, pyramid, or other graphical structures (Gu and Dunson 2023).

In many applications, it is of interest to infer and exploit such continuous latent structures in high-dimensional data. Extrinsic structures include interactions between data elements due to

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known structures and relationships between variables (e.g., physical or chemical dynamics, transmission models of infectious disease in populations, chaotic dynamics in mathematical modeling) embedded within the observed system. The intrinsic structures often represent the hidden associations between variables of primary research interest. Under the manifold hypothesis, high-dimensional data tend to be concentrated near a lower-dimensional subspace. PCA and linear factor models focus on linear subspaces, while manifold learning and nonlinear factor modeling seek to accommodate more general, nonlinear subspaces. There have been rapid advances in recent years in developing linear and nonlinear factor modeling approaches, including consideration of identifiability and inferences on lower-dimensional structure in the data. Many of the successful approaches for generative AI (e.g., variational autoencoders [VAEs]) can be viewed as nonlinear factor modeling methods but with the primary focus being on generating (predicting) new data from a similar distribution to that of the training data and not on conducting statistical inferences on lower-dimensional structure in the data or exploiting such structure for inferences on functionals of the data-generating distribution itself. While there is some work on identifiability for VAEs and related models, in general, highly flexible nonparametric factor models face problems in interpretability and reproducibility and require large training sample sizes for good performance. Recent work develops simple but still flexible alternatives (Xu et al. 2023); for example, such models can even accurately characterize the distribution of bird vocalization spectrograms from small numbers of samples.

This area will continue to grow in importance as researchers from a variety of fields look for statistical models that can capture the complexity of their data while enabling discovery of underlying patterns and structures. To limit over-interpretation and enable reproducibility, it is critical to develop methods that accurately characterize uncertainty in latent structure learning and that are not overly sensitive to violations of modeling assumptions, particularly assumptions that cannot be checked easily against the observed data. It is also of substantial interest to improve the ability of latent structure learning methods to accommodate known structure in the data, such as spatiotemporal dependence and measurement errors. These developments require not only new statistical models but also improved algorithms and corresponding theory. An emerging literature providing theory support for fast posterior approximations using variational inference (Zhong et al. 2025) and pre-training (Mauri and Dunson 2025) supported by a blessing of dimensionality provides promising initial steps.

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Conclusion 2-5: High-dimensional data include both extrinsic and intrinsic geometric structures. Statistical methods and theory are required to enhance discoverability of hidden intrinsic geometric structures while accounting for known extrinsic geometries in the observed or generated data elements. Development of generalized approaches to account for such geometric "lenses" will enhance model accuracy, reliability, generalizability, and analytic efficiency, as well as bridge the gap between AI-driven scientific research and scientific reasoning.

Navigating Trade-offs Under Real-World Constraints

Traditionally, the development of statistical methodology has primarily focused on the aim of accuracy: How can we extract the most accurate possible estimate of the unknown target quantity (say, the correlation between two quantities) using only a limited amount of data? Informally, other constraints also have always come into play; for instance, limited computational capabilities in the early days of the field have shaped the choice of an estimator. In recent years, constraints or aims other than accuracy have been studied broadly in a formal way. Recent research reflects a more nuanced understanding of the trade-offs between statistical accuracy and one or more of the following constraints or aims:

- *Computational constraints*, which place a limit on the computational budget allowed for the estimator;
- *Distributed data or distributed computation*, where the storage of the data used for estimation is decentralized (e.g., different sites, such as individual hospitals, retain their own collected data);
- *Privacy requirements*, where the amount of information about each individual in the data set that can be revealed is limited via the computed estimator or the deployed model;
- *Fairness requirements*, where, for instance, the estimator or predictive model is required to exhibit approximately equal accuracy across different subgroups within the data; and
- *Robustness*, meaning that the estimator must be designed to be insensitive to some particular types of perturbations in the data (e.g., to ensure that the accuracy would not be substantially compromised by a small fraction of corrupted measurements).

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Opportunistic Data Collection and the Federation of Statistical Analysis

Another significant and far-ranging development impacting the frontiers of statistics (discussed further in Chapter 3) is pervasive "datafication" of the physical and social world. This term refers to data collection mechanisms embedded directly in systems that facilitate everyday commercial, scientific, social, engineering, and natural transactions and interactions. For example, supermarket scanners routinely collect data on what shoppers buy and how much they spend; Internet companies collect data on users' likes and dislikes with every click; smart buildings monitor their own states with embedded sensors; patient records are computerized for easy retrieval and analysis; every smartphone continually collects information about its user; and so on. Datafication represents a vast expansion in the opportunistic collection of data, and it has profound implications for the ways we analyze them. These data may be streaming and not stored and would then need to be analyzed accordingly. If they are stored in raw form, they quickly become vast and not easily moved or manipulated. For example, the National Aeronautics and Space Administration's Earth remote sensing data, which are collected opportunistically, are stored in distributed repositories often equipped with cloud computing capabilities so that computation can move to the data rather than moving the data to the analyst's computer. But this model runs the risk of being unsustainable as the number of users and complexity of analyses grow. More generally, because different entities collect and store data differently, the full set of data required for a given analysis may not exist in monolithic form. Eventually, a new paradigm for statistical analysis will be required: one that is oriented around peer-to-peer systems that can manage privacy and security concerns but that encourages model and data sharing. Such a system needs to evolve in time, especially as new data sources become available and old ones disappear.

The implications for the field of statistics are profound. There will be many ways in which any given analysis can be conducted, and optimization will depend on the questions being asked and the topology of the system at the time. In other words, data analysis itself will become opportunistic. Costs associated with data reduction, data and code movement, and computational resources will be dynamic and quantifiable through network and resource monitoring. Transient failures of parts of the system will need to be considered. All of this suggests that real-time inference on the system itself and optimal decision making about how to design data analyses in the face of changing conditions will emerge as critical components of statistical inference.

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Computing itself will continue to mature, with edge computing increasing in importance, and decentralized analysis will be required to cope with huge volumes of decentralized data. Statistical methods and theory that support them will need development to address distributed statistical inference under communication constraints. Existing work in this area includes that of Fan et al. (2021) and Jordan et al. (2018). However, at this time, current work does not address the potential for network topology or other characteristics to change over time.

Federated learning partially addresses the problem of learning from distributed data, but more work is needed to deal with non-independent and identically distributed situations. Z. Lu et al. (2024) provide a survey of current methodology and research, citing communication, model performance, model aggregation, and privacy and security as important classes of challenges. They state, "The profound implication of these lessons is that our challenges extend beyond optimizing models; we must also consider the dynamism of data distribution" (Z. Lu et al. 2024). Clearly, groundbreaking sampling strategies and experimental designs tailored for edge networks, and new types of (approximate) sufficient statistics with well-understood performance guarantees and accurate measures of uncertainty, would help reduce volumes of data exchanged between locations, such as data centers, where inference is conducted.

The coming changes all point to at least three broad but high-impact areas future statistical research should target. First, there is a need for new statistical frameworks for systems where inference is distributed across thousands of autonomous agents. Current theory assumes either centralized or simple distributed settings, but expanded interconnectedness will require understanding statistical inference across hierarchical adaptive networks with varying computational capabilities. Second, this decentralized environment necessitates multiscale statistical modeling and scale-bridging that will operate across different temporal and spatial scales, levels of complexity, and statistical properties. Frameworks to quantify and understand how uncertainty propagates across all these scales and levels are critical. Third, statistical methods are needed to capture and predict network properties, their effects on inference, and their evolution.

Efficient Collection and Use of Data

Across a wide array of applications, data are more readily available than ever through passive, opportunistic collection. However, as noted above, storing large amounts of data quickly becomes computationally prohibitive, and the data that are collected may not be representative.

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At the other extreme, in some large-scale applications, each experiment may be so resource-intensive that data collection is necessarily limited and must be completed strategically. To guide effective data collection and utilization, further work toward strategic design and collection approaches is necessary to best respond to the opportunities provided by current innovations in science and engineering. New techniques are particularly needed as federated data sets, which have privacy and ownership issues, become more prevalent and as data modalities continue to evolve. Some promising directions are described below.

Sampling and experimental design. Starting from factorial design and classical experimental design methods under various optimality criteria (Chernoff 1972; Fedorov 1972), statistical research into sampling methodologies that efficiently leverage the given data budget to collect the most informative samples has continued to grow. While the criteria have existed for a long time and some have efficient solutions or approximations such as D-optimality and Bayesian information design, many criteria result in combinatorial problems, and accurate approximations to some of them have only been found recently (Allen-Zhu et al. 2017; Madan et al. 2019). Beyond passive sampling methods, active methodologies that leverage feedback from prior measurements to sequentially decide what data to collect are also an open area of research. This research ranges from classical statistical work on sequential probability ratio tests (Wald 1945) to modern active learning strategies (Settles 2012), where the focus is on deciding from which samples to collect the labels to train machine learning methods with minimal sample complexity. In particular, conducting effective data collection for training deep neural networks (Ash et al. 2020; Sener and Savarese 2017) and transformer models remains an open problem. Such active sampling methodologies are increasingly being used in applications where data collection carries significant costs such as medical settings, clinical trials, and the design of experiments in science such as materials science, physics, chemistry, and biology.

Self-supervision. Given the increasing needs and costs for collecting labeled data by modern machine learning methods, the AI research community is exploring self-supervised training methods that use unlabeled data in clever ways for robust and accurate representations and reduced reliance on previously labeled data. This research includes training models to fill in parts of an image that are masked (He et al. 2022) or, more recently, sequentially completing sentences in text corpus (Radford et al. 2018). These methodologies enable learning of accurate visual representations and semantic embeddings that require no or minimal labeled data. These

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techniques have fueled the recent rise of powerful AI models such as large language models (LLMs) and vision transformers.

Drawing Causal Conclusions from Observational Data

Causal inference seeks to understand and quantify cause-and-effect relationships beyond associations. It plays a crucial role in a broad range of disciplines such as biology, health, economics, public policy, and social sciences. While randomized experiments remain the gold standard for establishing causality, such approaches are not always possible for both practical and ethical reasons. Many causal inference methods have been developed for observational studies, such as maximum likelihood methods, propensity score matching, G-estimation, instrumental variables, causal mediation analysis, and causal graphical models. As data continue to grow in complexity and volume, particularly in real-world settings where big data are often observational in nature, many challenges and opportunities have emerged for causal inference. These developments call for interpretable, scalable, and actionable causal inference methods that are robust and powerful for handling real-world complexities. Examples of emerging causal inference areas include the following:

- *Integration of machine learning and AI with causal inference*, which enables more flexible and powerful modeling of treatment effects, adjustment for high-dimensional confounding, and improved scalability for complex real-world data (Deek et al. 2024).
- *Causal inference for network data* necessitates that, in the context of infectious disease spread and biological and social networks, individuals are interconnected, and valid causal inference must account for spillover and interference effects (Ding 2024).
- *Personalized and real-time causal inference* focuses on estimating heterogeneous treatment effects across individuals or subgroups, and developing adaptive, real-time interventions and dynamic treatment regimes using methods like reinforcement learning (Wang et al. 2024).
- *Causal inference for multimodal data*, such as causal mediation analysis, offers a powerful framework for uncovering causal pathways by integrating information across diverse data types (Ogburn et al. 2024).

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- *Causal inference and fairness*, where causal reasoning provides valuable tools to assess, understand, and mitigate biases in algorithmic decision making, promoting fairness and trustworthiness.

Test-Time Performance Optimization of AI Models

Conventional approaches to analyzing statistical models often focus on optimizing training procedures to give guarantees on test-time performance, with the inherent assumption that test data represent an independently drawn instance of the training data. Recent advances in AI have demonstrated that one can query the same pre-trained model in different ways and obtain different test performance *without altering any model parameters* at test time. This paradigm, referred to as "test-time optimization," studies how to design test queries (prompting strategies, in the context of LLMs) to get better performance from a fixed model. However, this setting is not well studied, and a theoretical foundation for this paradigm is largely lacking. Some popular efforts for test-time optimization, also known as prompt engineering, include the following:

- *Retrieval-augmented generation*, in which the LLM is prompted to answer input queries by supplementing the information from its training data information with information retrieved from a specified list of documents provided to the LLM at test time (Lewis et al. 2020).
- *In-context learning*, in which the LLM is prompted with a few examples related to the query before it generates a response (Brown et al. 2020).
- *Chain-of-thought reasoning,* in which the LLM is prompted to solve a series of intermediate steps before responding to the input query, thus breaking down the task into easier, manageable reasoning and leading to a final response (Wei et al. 2022).

As the field continues to move toward large foundational models trained on large amounts of data, the utility of these models in downstream tasks rests on being able to adapt for domain-specific prediction and inference. The test-time optimization paradigm plays a central role in enabling this translation and is proving to be a transformative technique that is yielding

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significant improvements. Developing a solid statistical foundation and tools to evaluate performance are crucial steps in unleashing the full potential of test-time optimization.

Characterizing Extremes and Rare Events

Another area of statistical research with deep theoretical roots and current relevance involves the analyses of rare, extreme events such as natural disasters (e.g., flood, hurricanes) or large-scale financial fraud. Many point to the origins of the formal development of extreme value theory to L. Tippet who noticed that, in studying yarn breakage rates, the strength of a thread was related to the strength of its weakest fibers. This work led to collaboration with R.A. Fisher and the establishment of different potential limiting forms for the distribution of maxima and ultimately what is now known as the Fisher–Tippett–Gnedenko theorem or the extreme value theorem (Fisher and Tippett 1928; Gnedenko 1943). This theorem suggests that the (properly normalized) maxima of independent and identically distributed random variables can only converge to one of the Gumbel, Fréchet, or Weibull forms. These distributions were eventually unified into the generalized extreme value distribution that, along with the generalized Pareto distribution, has seen widespread adoption across a variety of scientific fields including engineering, environmental science, technology and communications, and finance and insurance.

More recently, for environmental extremes, interest is evolving from the characterizations of extremes for a single variable to multivariate extremes, including methodological developments for single variable extremes that are correlated in space and/or time, coincident extremes at single or multiple locations (e.g., wind speed and flood depth from tropical storms), antecedent conditions (e.g., prior precipitation influence on flood risk, prior drought conditions on wildfire risk), and multiple stressors on structures or linked systems (e.g., power grids).

Statistical models proposed to explain dependence between extreme events are often too complex for statistical estimation procedures such as maximum likelihood estimation. Development and application of machine learning–based approaches are rapidly growing in extremes, including new paradigms for parameter estimation. SBI, also known as likelihood-free inference, leverages machine learning methods and simulation to estimate model parameters when existing models are too complex for traditional statistical tools like Bayesian inference. The so-called amortized inference approach (Sainsbury-Dale et al. 2023) reframes classical parameter estimation as a supervised learning problem, incorporating neural networks trained on simulated data to predict values of parameter estimates for new data. Such an approach moves

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much of the computational effort from numerical optimization targeted at a single data set to the training of neural networks, which can then be deployed to rapidly predict parameter values for many different data sets.

More far-reaching efforts around machine learning and AI include efforts to improve weather prediction and climate modeling to better characterize extreme events. This application area is changing rapidly on multiple fronts, including a resurgence of weather generators driven by generative AI and AI-powered approaches for weather prediction and climate emulation, as well as digital twins and foundational models.

> *Conclusion 2-6: The field of statistics is always changing in response to the new scientific questions that are asked. The "hot topics" of 2035 may bear little resemblance to those of 2025, but many of the statistical skills forged in 2025 will apply in 2035, and the statistical innovation life cycle displayed in Figure 2-1 will remain constant.*

CATALYZING FRONTIERS ACROSS DOMAINS

The field of statistics underpins a wealth of scientific and engineering advancement, such as the discovery of the Higgs boson (Manole et al. 2024; Van Dyk 2014). Even so, fields like machine learning and applications like particle physics evolve so rapidly that without intentional foresight and strategy, the statistical sciences may end up focused on yesterday's questions from science and engineering instead of tomorrow's. Partnering with other domains of inquiry often requires years of investment and learning, exacerbating the challenge to keep pace with more fluid fields. Statistics adds value to and even catalyzes innovation across these domains, so long as the right incentives exist to foster a fertile environment supporting mutual innovation. Cornerstones of cross-fertilization include collaboration, thoughtful interdisciplinary training, and ongoing and robust professional development.

Collaborative research involving statisticians and domain experts from science and engineering provides numerous mutually beneficial opportunities for expanding frontiers in all fields. Collaboration provides avenues for exposure to new data sets and new questions, which can transform statistical methodology as well as the theory to support that methodology. As new methods result in unique insight, scientists may encounter new questions, and the statistical innovation life cycle begins all over again. This mutual support accelerates the research process

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through sound data collection and management; experience with appropriate methodologies, tools, and algorithms; statistical theoretical prowess to expand methodologies when required; and expertise in the analysis and presentation of results, as well as the development of scientific conclusions based on those results. Finally, collaboration affords statisticians opportunities in scientific leadership, as data—and how data flow through complex and interconnected processes—are often the threads that link the diverse contributions generated across successful multidisciplinary efforts.

Chapters 4 and 5 of this report outline examples of successful collaborations and highlight how these ongoing partnerships benefit both the field of statistics and the fields of collaborators. While these examples are compelling, the committee notes that deep collaboration poses challenges for statisticians and requires skills beyond technical facility with statistical theory and methods alone. High-quality engagements require patience and commitment to building trust. It is important to develop an understanding of the underlying scientific processes, the questions being addressed, and the data being collected and analyzed. Strong communication skills are crucial, including the ability to explain complex statistical concepts and the rationale behind their use, as well as the interpretation of their application. Early involvement in projects improves the quality of collaboration and subsequent research.

A Shared Vision of Statistical Training

As technology and scientific progress undergo rapid change, the long-term vision for statistics education must be one of adaptability, integration, and innovation while still maintaining the discipline's rigor and principled reasoning. Despite how data and computing practices evolve, statistical curricula need to be firmly connected to statistical practice and offer flexible pathways that deepen statistical thinking and learning across a wide range of contexts. The field of statistics needs to continually develop a shared vision of the statistical skills and statistical thinking required for members of tomorrow's workforce, equipping them with technical and collaborative competencies to be rigorous forward-thinkers. This vision requires an educational ecosystem that is agile enough to respond to new tools, challenges, and opportunities, yet grounded enough to continue shaping theory and methods to address emerging issues and future possibilities. By building collaborative infrastructures, fostering partnerships, and embracing progress, the statistics community can lead education efforts that prepare both

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individuals and society to thrive in a data-driven future. Chapter 6 explores opportunities for transforming statistics education and highlights the new statistical insights necessary for rigor and principled reasoning, which, together, will enable the field to sustain and expand its mighty contributions to data-driven innovation for U.S. competitiveness.

Career Opportunities and Professional Development

Statistics graduates and current professionals find strong opportunities across all sectors of society, including academia, government, nongovernmental agencies, and industry. Employment structures are poised to transform with the widespread adoption of AI; the statistics profession is addressing both the opportunities and challenges that arise. Moreover, the landscape of where basic and open-ended research is being conducted is changing. Previously, this was reserved for academic institutions and national laboratories, while partnerships with companies in industry were primarily centered on infrastructure and monetary support to help with scale. However, it is becoming more common for these same companies to invest in groups where they develop tools with themselves as the end "user" in mind. This means that they hire statisticians who might otherwise have been academics to develop theory, build model architectures, and engineer systems for efficient deployment both in-house and to external partners. This dynamic has made large international conglomerates (e.g., Google, Microsoft, Merck, Genentech) major players in terms of scientific inquiry and knowledge production, in addition to their previous role as a reference group.

In the face of this evolving landscape, a thorough understanding of statistical methods offers career agility. First, core theoretical concepts such as variability, uncertainty, and inference will continue to be at the foundation of any new technology (AI and beyond) that will emerge by 2035. Whether a role is in finance, health care, sports, or the geosciences, a strong statistical grounding leads to insight on how to develop information-rich study designs, the capability to build models that take advantage of new types of computing infrastructure, the intuition of when it is appropriate to deploy certain algorithms, and the ability to translate quantitative findings with the clarity needed for general public audiences. Statisticians operate across a wide spectrum, from leading teams in industry to supporting day-to-day business functions and conducting research that advances both the practice and profession of statistics. Academia often links research and education while also emphasizing the advancement of

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pedagogy, supporting the practical application of statistical science, and expanding the development of statistical theory. Education in statistics and data science begins at the earliest stages in the U.S. educational system and progresses as students develop. Doctoral programs in statistics and its related fields produce some of the world's best AI and data science talent, providing a strategic advantage for the nation. Moreover, U.S. doctoral programs in statistics and biostatistics continue to serve as global beacons of intellectual advancement, attracting a substantial influx of international students.

As a field that significantly contributes to the rapid progression of machine learning and AI and also benefits greatly from their applications, leading academic departments in statistics collaborate on these advancements through mechanisms such as the American Statistical Association (ASA) Caucus of Academic Representatives. The caucus recognizes the challenges and appreciates the leadership provided by the field's collective academic research and education departments. In 2022, ASA established the Caucus of Industry Representatives, creating a unique opportunity for industry leaders in statistics, data science, and AI to strengthen and advance U.S. infrastructure. Chapter 6 provides an in-depth perspective on the near-term changes occurring at various levels within the field's academic institutions.

As AI becomes a core driver of innovation across industries, the role of the statistician will shift from being a specialist in inference to being a strategic architect of data-driven systems. This evolution will require statisticians to have expertise in algorithmic thinking, experimental design, and ethics and sociotechnical systems. The rise of generative AI (e.g., LLMs) has already reshaped the job market, especially at the entry level, by automating many routine tasks such as data cleaning, exploratory analysis, and basic model implementation. Consequently, the demand for statisticians who can simply execute predefined analyses is diminishing, while the need for those who can interpret, validate, and integrate AI-generated outputs is growing. Entry-level roles for statisticians are adapting to require fluency in AI-assisted workflows, stronger scientific domain knowledge, and the ability to communicate insights effectively across disciplines. At the same time, new opportunities will continue to emerge in areas like model auditing and red teaming (Feffer et al. 2024; Ganguli et al. 2022), algorithmic fairness, and model-based synthetic data generation—domains where statistical rigor is essential. As noted above, in this new professional landscape, finding strong career opportunities and success on the job market will depend not just on technical skills but also on the ability to collaborate across sectors and guide

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responsible AI development. Statisticians are already at the forefront of many interdisciplinary initiatives where they play central roles in experimental design, algorithm development, and model deployment. Their daily responsibilities often require regular communications with colleagues outside their own field, which naturally positions them as leaders for collaborative work. As such, statisticians will increasingly lead or co-lead teams consisting of domain scientists, engineers, and ethicists. Training will need to place greater emphasis on both oral and written communication skills for interdisciplinary audiences, reproducible research practices, and the implementation of open science tools.

Statisticians have a rich history of contributing to the U.S. patent landscape and technology transfer. While most innovations in statistical methodologies contribute to open science, their use is regularly embedded in specific patents. For example, the 1984 statistical development of the modern machine learning algorithm classification and regression trees appears in an active patent on methods for automatically selecting the trees (Breiman et al. 1984; Steinberg and Cardell 2017). In an AI- and data-driven decision economy, these opportunities will only expand. The collaborative nature of the field supports innovation at the most fundamental levels.

Conferences and Short Courses

Professional societies offer a wide range of development opportunities for working professionals and newcomers to our field. The Joint Statistical Meetings (JSM), held annually in early August, is the largest conference of statisticians in North America, with more than 5,000 participants coming from more than 50 countries. JSM is also one of the broadest meetings bringing together statistical scientists from government, industry, and academia. The meeting typically hosts more than 600 scientific sessions and more than 40 short courses for ongoing professional development. The Institute of Mathematical Statistics (IMS) recently launched the International Conference on Statistics and Data Science (ICSDS), which has gained remarkable traction. ICSDS fosters connections between statistics and modern topics in the field, including data science.

In addition to JSM, each of the member associations in the Committee of Presidents of Statistical Societies (COPSS) hosts a range of professional meetings and training. For example, the Eastern North American Region of the International Biometric Society founded the Fostering Diversity in Biostatistics Workshop, which focuses on connecting underrepresented minority

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students interested in biostatistics with professional (bio)statisticians in academia, government, and industry (Benn et al. 2020). Every year, this workshop provides (1) hands-on computing activities for high school and undergraduate students, (2) roundtable discussions on preparing for graduate degrees and transitioning to the workforce, (3) roundtable discussions for professionals to support and retain students, (4) panels focused on providing career opportunities and mentorship between students and professionals, and (5) a career and graduate school exposition open to students of all levels. As another example, COPSS recently partnered with the National Institute of Statistical Sciences—a consortium of statistical partners providing high-level research in science and public policy[7]—to create a Leadership Webinar Series with the goal of promoting leadership skills for members of statistical societies at any stage in their careers. The series features conversations with leaders throughout the discipline, including individuals from major academic institutions, government agencies, and industry. Invited speakers share their leadership stories and answer questions about their experiences while engaging with online audiences, and each webinar is moderated by a member of the COPSS Emerging Leaders in Statistics program. In short, statisticians have numerous opportunities for advancing their skills and knowledge.

Journals

The leading outlets for statistical research are also expanding. Many statistical journals, such as the *Journal of the American Statistical Association* (*JASA*); *Journal of the Royal Statistical Society Series A, B*, and *C*; *Annals of Statistics* (*AoS*); *Annals of Applied Statistics* (*AoAS*); and *Biometrika*, are highly subscribed—for instance, *AoS* and *AoAS* have more than 700 and 300 institutional subscribers (Qu 2024), respectively, and *JASA* articles have more than 900,000 annual downloads or views (Taylor & Francis n.d.). The research published in these journals focuses on foundational innovations, methodological advancements, and advanced applications, with rigor and reproducibility as key components. This research is increasingly highly cited. An analysis of recent trends reveals that citations are increasingly coming from adjacent applied fields, such as biological or tech applications (Wang et al. 2022). Furthermore, the community regularly publishes in the *Journal of Machine Learning Research*, *NeurIPS*,

[7] The website for the National Institute of Statistical Sciences is https://www.niss.org/about, accessed January 16, 2026.

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Public Library of Science (*PLoS*) *ONE*, *Science*, *Proceedings of the National Academy of Sciences*, and a wide range of public health journals.

In 2024, IMS partnered with the Association for Computing Machinery (ACM) to launch the *ACM/IMS Journal of Data Science*. Furthermore, ASA introduced three new journals: *ASA Discoveries*, *Data Science in Scienc*e, and S*tatistics and Data Science in Imaging*. *ASA Discoveries*, designed to be an additional impactful journal for the statistics profession spanning the spectrum of methodologies and advanced applications, began accepting papers in 2025. *Data Science in Science* was founded in 2022 as an independent journal supported by multiple efforts funded by National Science Foundation (NSF) programs (Matteson 2022). In 2024, the journal was adopted by ASA. *Statistics and Data Science in Imaging* launched in 2024, emphasizing advanced statistical methodologies in the important arena of imaging (Vannucci et al. 2024). The Royal Statistical Society's *Data Science and Artificial Intelligence* published its first volume in January 2025. *Harvard Data Science Review*, founded in 2019, is a leading open-access venue for high-level discussions on data science and statistics.

Over the course of the committee's information gathering, it became clear that greater recognition is needed for interdisciplinary statistical work, particularly outside of biostatistics. The influx of new journals highlighted above will provide top-tier outlets for this critical science. The availability and utility of open-source peer-reviewed publications continue to grow along with requirements for open-access publication and data sharing by many federal sponsors of research. The statistics profession is a strong participant in the ongoing discussion about the evolution of publishing scientific scholarship.

> *Conclusion 2-7: Professional associations for statistics offer robust publishing opportunities for high-quality, rigorous, and reproducible research. New journals supporting statistical science have been established to acknowledge the importance of innovative methods and applications in the field while still upholding the strict review standards maintained by the statistics community.*

The Breadth of Statistical Funding

Historically, statistics has found wide support across NSF. NSF's Division of Mathematical Sciences (DMS) clearly plays a key role in funding statistics research, but because of the way statistics underpins scientific progress across disciplines, funding opportunities can

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also be found throughout the institution. The AUDITION (Autism Digital Twin Foundations) project, for example, is co-led by statistician Huixia Judy Wang and jointly funded by NSF through DMS, the Office of Advanced Cyberinfrastructure Cyberinfrastructure for Sustained Scientific Innovation program, and the Chemical, Bioengineering, Environmental and Transport Systems Engineering of Biomedical Systems program. In the *Amstat News* series that features Q&A with NSF program officers and awardees, awardees cite various divisions outside DMS whose programs have funded their work. One example is the Innovation and Technology Ecosystems Division within the Technology, Innovation, and Partnerships Directorate; program director Jemin George notes that "statistics PIs [principal investigators] are crucial to the division's mission due to their expertise in data analysis, statistical modeling, and quantitative methodologies" and that statisticians "are not only eligible but are integral to the success of Innovation and Technology Ecosystems–funded projects" (*Amstat News* 2024). Other issues in the series highlight statistics research funded by other areas of NSF including the Directorates for Computer and Information Science and Engineering; Biological Sciences; and STEM Education. Of course, DMS remains critical to funding statistics research, but statisticians are also encouraged to explore funding opportunities in other areas of NSF. The ASA conference "NSF@75: Advancing Statistical Science for a Data-Driven World" offered a similar look at the broad reaches of statistics: it showcased 80 papers presenting work funded by NSF, from "Statistical Analysis of Complex Image Data: Applications in Medicine, Biology, and Beyond"; to "Understanding Terrorism via Advanced Spatio-temporal Point Process Models"; to "Generalized Fiducial Inference with Applications to Forensic Science."

Conclusion 2-8: Statistics has been supported broadly across multiple directorates and divisions at NSF, speaking to its critical role throughout science and engineering.

The National AI Initiative is a coordinated federal approach by the U.S. government that aims to accelerate AI research and the integration of AI systems across society. Numerous federal agencies, including NSF, the National Institutes of Health, and the Department of Energy, are involved. NSF in particular has broadly supported AI since the early 1960s as one of the largest non-defense federal funders of AI research (NSF n.d.b). The National Artificial Intelligence Research Resource (NAIRR) is one approach that is supported by numerous federal agencies—including NSF—as well as private and nonprofit organizations. NAIRR connects U.S. researchers and educators to AI resources in order to strengthen the U.S. AI research ecosystem

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and accelerate AI and AI-powered discovery and innovation (NSF n.d.c). An ongoing development, it demonstrates another way the United States is moving to advance AI.

These approaches demonstrate a healthy push toward advancing AI from across NSF and the wider U.S. government, but statistical rigor is vital to supporting AI research that cannot be forgotten. Many of the challenges facing today's AI, including interpretability and uncertainty quantification, are, in fact, statistical problems, and the solutions to these challenges will be enabled by contributions from the field of statistics. New statistical methods are needed to address challenges of modern AI systems including interpretability and accountability, hallucinations, quantification of uncertainty and exploration for foundation models, and causation for decision making. NSF's Transdisciplinary Research in Principles of Data Science (TRIPODS) is one program that is careful to highlight statistics, with the aim of bringing together the statistics, mathematics, and theoretical computer science communities to develop the theoretical foundations of data science. Under TRIPODS, new large institutes comprised of several universities across the United States have been created to study the foundations of data science. The Texas A&M Research Institute for Foundations of Interdisciplinary Data Science, for example, was funded by a TRIPODS grant and is led by statistician Bani Mallick and a cross-disciplinary team with expertise spanning computer science, mathematics, and engineering (Hutchins 2019). The committee encourages continued focus on foundational work that recognizes the importance of statistics, and it emphasizes the urgency of supporting statistical rigor as AI research advances.

> *Conclusion 2-9: NSF invests heavily in AI research and education across multiple directorates. Incorporating the depth and breadth of perspectives from the statistics community into these investments increases their value and impact for U.S. competitiveness.*

> Key Finding 2-4: It is of paramount importance for the statistical community to position itself to take advantage of existing opportunities and emerging application areas by pursuing relevant research priorities and funding calls broadly. Doing so requires strengthening community-level mechanisms and infrastructure that enable statisticians to engage early, collectively, and strategically. Examples include offering leadership and team-science training workshops, sponsoring task forces to develop shared methodological

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and computational resources, creating grand community research challenges (such as KDD Cup and DREAM challenges), and organizing informal proposers' days in response to large interdisciplinary funding opportunities.

Conclusion 2-10: NSF would benefit from an overarching survey of statistical progress within the National AI Research Institutes. Increasing the footprint of the field of statistics in the NSF-supported AI space and providing coordination across statisticians working at this interface are paramount. Professional societies may be well positioned to support this effort.

Key Finding 2-5: The foundations of statistics consist of theory, methodology, applications, training, and education, with the practice of statistics serving as a central driver of its development and evolution. Statistical ideas emerge, advance, and mature through use, interaction with real-world complexity, and iterative refinement informed by both theoretical insights and application.

Conclusion 2-11: The statistical innovation life cycle operates as a dynamic continuum between scientific inquiry and innovation in statistical theory, methods, and applications. The broader statistical community would benefit from recognizing the many ways in which researchers and practitioners advance the field throughout this life cycle. The community could support statistical innovation through transdisciplinary collaboration by recognizing high-risk, high-reward foundational research and enabling statisticians to contribute as leaders and collaborators across disciplines. Accelerating this cycle is critical for addressing today's data-driven challenges, and existing infrastructure and incentives for statistical research and education need to balance recognition across the full spectrum of research.

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3
The Changing Nature of Data

Data are foundational to science and engineering. The concepts and models relating to data are fundamental to the dynamic intersection of theory and application that drives innovation, refines methodologies, and provides ongoing and bidirectional links between statistics and science and engineering. The very nature of data, both in concept and in practice, is constantly evolving due to changes in measurement technology, interconnectivity, and computing power, and these changes drive the frontiers of the field of statistics.

Modern science arose from careful observations of natural phenomena in search of replicable patterns. Early development frequently involved observational data—that is, data observed without experimental control (e.g., Florence Nightingale's observations of seasonal patterns of mortality among British soldiers during the Crimean War in the mid-1800s; McDonald 2014). In the 20th century, further advances in scientific observations motivated statistical developments in experimental design, randomization, sampling design, formal testing, and statistical estimation. Statistical thinking allowed investigators to design data collection and analyses to improve precision, gain efficiency, and focus conclusions while minimizing uncertainty and bias. In settings where randomized experiments were infeasible or unethical, observational data from non-experimental settings drove the development of new inferential models and laid the groundwork for modern causal inference, increasing understanding of the settings and assumptions under which causal conclusions can be derived from observational studies.

In recent decades, data relevant to fields of scientific inquiry are increasingly interconnected or interoperable and are collected by multiple sources, over multiple time periods, in multiple locations. The rapid increase in ongoing data generation, aggregation, and collection from sensors, administrative data systems, citizen-generated sources, personal devices, and social media—coupled with decreased costs of computational processing and storage—drives

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developments in data science, machine learning, and artificial intelligence (AI). The availability of multiple sources of data of differing types, differing resolutions, and differing levels of accuracy motivated concepts, theory, and methods of data fusion (e.g., for multisensor networks [Mitchell 2012] and for broader applications linking mass amounts of simulation-generated synthetic data and observations from multiple sources [Banerjee et al. 2014; Hünermund and Bareinboim 2025; NRC 2013; Szalay 2011; Wikle et al. 2023]). Data-intensive computing supports new modes of scientific discovery but also raises new challenges. For example, data fusion from multiple sources fuels machine learning prediction models, but fusing current predictions into the same predictive model often can degrade prediction performance and lead to model collapse (Shumailov et al. 2024).

Many frontiers of the field of statistics are driven by the need to understand data and their causal relationships. The ongoing motivation in scientific and engineering data analyses is to provide the most accurate, reliable, and reproducible inference possible while minimizing errors, costs, losses, and risks. The frontiers of statistics are motivated, enabled, and evaluated by how the term "data" is applied and interpreted.

EVOLVING CONCEPTUAL FOUNDATIONS OF DATA

Data, both in concept and in practice, continue to evolve rapidly. As new data types arise, and as data become more immediate, interrelated and granular, the frontiers of statistics must and will continue to adapt to these changes.

Historically, statistical methods were developed under a core set of assumptions, including assumptions that data were costly to obtain, gold-standard data were best generated through carefully designed experiments, and data sets were collected and fully controlled by the team of researchers conducting the analysis. Under these assumptions, analysts sought to gain the most precise insights from the smallest number of highest-quality data observations available. Analysts sought to maximize information under constraints of time and cost. Under these settings, statistical research both in theory and in practice pioneered the fields of experimental design, hypothesis testing, and statistical inference throughout the 20th century.

Statisticians are often familiar with structured data. Several novel data types have emerged in recent years as part of the AI revolution, including unstructured data, such as images, texts and videos, model-generated synthetic data, streaming data, and multimodal data. Analysis

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of these data types presents substantial challenges and opportunities. Furthermore, statisticians are increasingly working with synthetic data—that is, values generated from underlying mathematical, simulation, or machine learning models—and this presents additional challenges as they are used in place of or to augment traditional data. Take, for instance, blockchain. Blockchain represents another new format of data consisting of anonymized data streams providing an efficient, immutable, and transparent record of all transactions among a network of peers. Originally developed for cryptocurrency, blockchain offers application in other key areas involving transactions including health care and supply chain management. Because all transaction records are available to all members of the network, the data produced are massive and ripe for statistical analysis to understand behaviors and interactions within the complex systems being monitored. Currently, the most common machine learning methods applied to blockchain data focus on anomaly detection and price forecasting. New statistical tools and innovations are needed to model, extract, and understand more complex latent structures embedded within the peer network in order to monitor network evolution, identify drivers, and predict future behaviors. New methods must be scalable to millions of records and amenable to the distributed nature of blockchain data.

> *Conclusion 3-1: Statistical research enhances data utility and application. As new data types emerge and as complexity increases, statistics and analytic tools will need to evolve to keep pace with the changing landscape.*

Data used in statistical research are often an amalgamation of both designed data collection and data generated by others. While in many areas of science, engineering, and statistical research, the experimental setting still thrives and drives its own innovations in the design of data collection—clinical trials, complex survey samples (e.g., the American Community Survey), neuroscience, and wastewater epidemiology, for instance—increasingly, the data used for statistical inference are often "found," "curated," or "fused" rather than designed. Data are frequently byproducts of interconnected technological systems, often with ongoing measurements shaped more by engineering constraints and user behaviors than by statistical sampling principles. This suggests a shift from using designed data sets to combining and processing data products, whether as the outcome of ongoing data collection and extraction or by the release of official government or administrative data sets (complete with metadata, or

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data about data) that can be combined with other data sources, rather than assuming raw data will only be collected for one study and fit to existing statistical models.

Machine learning and data-driven discovery methods are evolving rapidly, often with a goal to automate the search for and linkage of these new data types, especially in exploratory and high-dimensional settings. Detailed information regarding completeness, accuracy, and uncertainty quantification is lacking at times, but data science research encourages the co-development of metadata to provide the information necessary to incorporate uncertainties for decisions and inferences involving fused data (i.e., data that are combined from multiple sources). That said, successful integration of such data types into research paradigms requires more than simple adjustments to existing methods; it provides exciting opportunities to rethink foundations of inference beyond a closed experimental system to embrace the modeling of imperfect or biased data that contain related but heterogeneous information regarding the research questions of interest, in order to design creative inferential methods that are theoretically sound and strategically robust in complex, dynamic data settings.

To illustrate an example of such heterogeneous but informative data products, consider the following data and research setting from the field of ecology outlined in Box 3-1.

BOX 3-1
Data and Ecology

Modern ecology and biodiversity monitoring has been transitioning in recent years to incorporate automated data capture methodologies to identify which species are present at particular locations and times. For example, the European Research Council–funded Lifeplan project is currently finishing collection of a data set containing all of these types of data for sites around the world (Hardwick et al. 2024). In addition, interest in leveraging massive data produced by citizen science has emerged, with a particular emphasis on data from users of bird identification apps such as BirdNET (Kahl et al. 2021). The raw data produced by these efforts take the form of vast quantities of data including (but not limited to) audio recordings, images, metabarcoding, and nucleotide sequences. It is a non-trivial problem to infer which species are present based on audio segments and images collected under non-ideal field conditions by instruments ranging from sophisticated audio censors to cell phones. It is similarly challenging to accurately classify which species are present in a pooled sample of arthropods or fungi based on sequencing of nucleotides in the coefficient of inbreeding region of focus for metabarcoding.

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The current state of the art in the field is to combine statistical joint species distribution models (JSDMs) with AI and bioinformatics pipelines to infer how species communities are structured and depend on covariates, such as habitat conditions, human interference, and climate. For example, there are carefully developed deep neural network classifiers that produce well-calibrated predictions of presence or absence of each bird species in each audio segment along with a probabilistic measure of uncertainty. These algorithms are even being deployed within the recording devices using edge computing to bypass the expensive storage and transmission of the raw audio in a new National Science Foundation–Academy of Finland project (Kennedy 2019). DNA metabarcoding requires a delicate and complex series of steps to classify which species, or more accurately operational taxonomic units, are present in each sample.

In this data landscape, statisticians are faced with numerous substantial hurdles requiring fundamentally new statistical methods. Quite recently ecologists focused largely on single species distribution models, but the above data streams call for JSDMs that can handle massive numbers of potential species, many of which are extremely rare. In statistical terms, the response is a huge dimensional and highly sparse binary (or count) vector for each sampling site and time. The JSDM must account for covariate effects and statistical dependence in co-occurrence across species as well as space and time. Handling the extreme sparsity and dimension while crucially obtaining accurate uncertainty quantification in predictions and parameter inferences may be challenge enough (Mauri and Dunson 2025; Ovaskainen et al. 2025). However, this challenge is exacerbated by the fact that the inferred species are not measured perfectly but are inferred from noisy images and audio or DNA. In addition, clear bias exists in which species are recorded or captured in the trap, which should be addressed via some appropriate measurement error or “occupancy model.” Accurately characterizing the entire process from latent populations to samples is potentially necessary to make truly unbiased inferences and predictions.

This example highlights an increasing challenge for statisticians moving forward—to adapt rapidly to a massive and complex data landscape, often involving multiple types of non-Euclidean or non-standard data objects that are difficult to analyze directly, while maintaining the interpretability, validity, and reproducibility of statistical inferences. Addressing this challenge often requires combining state-of-the-art AI and other algorithmic tools with statistical modeling approaches.

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As the example in Box 3-1 illustrates, there is great potential in combining, exploring, and analyzing such a wide variety of data types and sources, but successfully doing so requires data analysts to thoughtfully engage statistical skills and leverage interoperability while also understanding the data-generating processes involved and their associated complexities, uncertainties, biases, and imperfections. Particularly as data continue to proliferate, it is of paramount importance that research continues to accept, describe, and model data flaws. Fields including ecology, climate science, population studies, and system sciences often make use of a hierarchical structure based on the process model (a model of the underlying dynamic process of interest) and the data model (conditional on the process model, a model of how observations derive from the underlying process) (Berliner 1996; Wikle 2003; Wikle et al. 2003). The data model provides an important conceptual element in the statistical analysis of "found data" as a mechanism for prioritizing how to learn from imperfectly observed data. This modeling of the act of observation, what it preserves and what it may obscure, frames a conceptual way to consider data from other sources deriving from the same complicated underlying process of interest.

In addition to modeling how one observes data, the evolving nature of data requires expansion of both the fundamental concepts of data analysis and the assumptions associated with them, as illustrated by a few brief conceptual examples. First, the highly distributed nature of data also requires extensions of fundamental concepts in statistical theory such as sufficiency, the definition of how data summarize information regarding unknown parameters (Lee et al. 2017) wherein information regarding model parameters is considered within and between distributed data sets. Second, the multiple sources of data call into question familiar assumptions of independent and identically distributed observations; the related, but weaker, assumption of exchangeability often applies more broadly. The concepts of independence and exchangeability are related (e.g., de Finetti's famous theorem) but subtly different, offering expansion of the impact of information flow through data linkages, particularly within a Bayesian inferential framework. Third, as previously mentioned in Chapter 2, consider how rethinking multiple testing in light of then-newly available sequence data provided the foundation for the false discovery rate (Benjamini and Hochberg 1995). This changed the question of interest in the testing of multiple hypotheses. Traditionally, in applications comparing each of a handful of independent tests to a null hypothesis, the assumption would be that the null hypothesis was true

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for all tests (e.g., Bonferroni corrections). In contrast, the false discovery rate tested millions of alleles but fully expected some fraction of null associations to be rejected (i.e., to be true discoveries), and the goal changed from controlling the rate of any rejection of the null hypothesis to the goal of controlling the proportion of rejections that were false. Effective merging of uncertainty into coherent models of variation, information, and bias requires new and more expansive thinking to create inferential tools that fully draw from the information contained in the rich data resources available. Finally, consider the linkage of heterogeneous data sources and types, which requires careful consideration and modeling of uncertainty through probabilistic measures of bias and variance or deterministic surrogates for uncertainty quantification (NRC 2012).

These examples and others appearing throughout this report illustrate how familiar concepts in statistical thinking and theory emerge in response to new assumptions, new availability of broad classes of data, new linkages between separate data sets, and new questions of interest enabled by new data. Such strategic use of evolving concepts and availability relating to measurable data goes well beyond simply adding robustness to analytic methods and instead prioritizes strategic thinking to broader classes of model specification, domain adaptation, and out-of-domain generalization of results, all fertile areas for ongoing expansion of the frontiers of statistics.

EXPANDING MODELS OF DATA FUSION

Concepts of data collection and design are well developed for randomized experiments, and such methods are sufficient in that setting. However, the increasing interconnectedness and interoperability of data elements highlight a need for innovations relating to the identification, collection, and fusing of data from multiple heterogeneous sources, in ways that allow analysts to model the observation process itself (i.e., a formal or informal data model as defined above). Such modeling incorporates specific models of uncertainty (or uncertainties), biases, and influencing factors driving observed values of data. Relevant data sets may include official statistics or administrative records, and such data releases become statistical products (data infused with documented information regarding empirical or modeled quantifications of uncertainties) rather than simply data sets or collections of observations (Molfino et al. 2024; Shipp et al. 2024). The field of statistics will increasingly include principled guidelines for

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planning, collecting, and documenting the fusion of data from official statistical products with data collected or abstracted across multiple data sources. Indeed, customized data collection toolkits mature in parallel to expanding and interoperable data sources, providing tools for finding and linking data. However, these toolkits should always inform the data models mentioned above; effective data collection toolkits gather, summarize, and document the information required to model the complicated, multilevel, multistage data collection and observations process(es) for data fusion and its associated uncertainties.

The analyses of fused data offer great opportunities for theoretical and methodological development across multiple levels of complexity. Machine learning models for data discovery offer exciting frontiers for exploration and knowledge building across data sources and types. Detailed survey sampling protocols, advanced dimension reduction methods, and detailed Bayesian hierarchical models all offer frameworks for extension into probabilistic inference for complicated, interconnected data and process models. The interactions of these fields are especially attractive for enhancing interpretability of machine learning and extending data fusion across traditional statistical methods. Indeed, many investigators are heavily engaged in this interface (Donoho 2017), offering ample new opportunities for exploring the interconnections between statistical concepts and data access and availability (Bradley et al. 2021; Keiding and Louis 2016, 2018).

The rapid expansion of generative AI for data discovery, data exploration, and exploratory inference represents active and open areas of research (Ruta et al. 2025). Generative tools also raise the possibility of model-based synthetic data generation, an ongoing evolution of simulation-based concepts in statistical and mathematical modeling (e.g., Monte Carlo simulation, bootstrap, jackknife, multiple imputation, and agent-based modeling) to generate individual survey responses (e.g., to preserve individual privacy, to enhance coverage of small, underrepresented subgroups, or to fill in responses to sensitive questions) (Bisbee et al. 2024; Mathur et al. 2024; Shrestha et al. 2025). Exploring inferential accuracy and reliability across synthesized data types and linkages offers another frontier of statistical practice and theory to better understand the potential and boundaries of accuracy for such inferences.

Consider health studies, which have historically been underpowered for women and other underrepresented populations due to limited subgroup sample sizes and the biases from directly applying the results of the large sample-sized population to the underrepresented populations.

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Emerging approaches such as transfer learning offer an attractive path forward by borrowing strength from large, well-characterized data sets of a source population while adapting models, such as distribution shift models, to capture population-specific effects, thereby improving efficiency and stability of estimates for small, underrepresented cohorts. In parallel, generative AI–enabled synthetic data methods can create realistic privacy-preserving data sets that reflect underrepresented populations. Robust statistical methods can be developed by integrating large synthetic data with observed data of underrepresented populations to empower the analysis of the underrepresented populations when real-world data are limited (McCaw et al. 2024). When paired with rigorous and robust statistical methods, validation, and bias assessment, these approaches have the potential to reduce historical inequities and strengthen discovery in underrepresented populations in health research.

In practice, the evolving nature of data requires guidelines and performance assessments for how to integrate and fuse information from multiple data sources, as well as specifications for data reporting to facilitate effective and efficient fusion of a given data set into potential future analyses. The FAIR Principles framework (keeping data findable, accessible, interoperable, and reproducible [Wilkinson et al. 2016]) can lead to improved data management and stewardship. Working with and toward shared data provides exciting new collaborative opportunities but necessitates creating unconventional methods for understanding data quality, identifying whether data should be fused, and considering how to fuse disparate shared data. Special care will need to be taken in the case of machine learning. Kapoor and Narayanan (2023), for instance, use data leakage as a way to explore reproducibility issues surrounding machine learning, and they provide a model information template for documenting various data preparation steps in a machine learning workflow.

OPPORTUNITIES AND CHALLENGES

The ongoing transformation and increased interconnectedness and interoperability of data sets and statistical products offer ample opportunities for growth at the statistical frontiers; however, these changes also raise important and novel challenges that require rethinking standard approaches.

First, collecting and analyzing emerging types of (non-probability) survey data; unstructured data, such as images, texts and videos, and generated synthetic data (e.g., predicted

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observations from mathematical, simulation, or machine learning models); and multimodal data will require tailored collection, sampling, learning, and analytical methods to ensure accurate, unbiased, and reliable inference. In some projections, synthetic data from generative AI and statistical algorithmic methods will occupy a significant portion of emerging data in the near future (Gauthier et al. 2025), much of it derived from images, texts, and videos collected without random sampling. Expanded concepts of inference and accuracy are needed to structure statistical techniques and analyses for these novel data types, again to ensure and quantify accuracy and reliability of conclusions. Research in science and engineering must continue to advance proactive plans for data sharing and improve standardization and use of data documentation and citation practices to maintain robustness and improve reproducibility of results and conclusions (Brown 2021; Buneman et al. 2020; Silvello 2018).

Second, the increased use of fused data sets also requires advances in measurement, assessment, and accounting of data quality and uncertainty across heterogeneous data sources and data types. A probabilistic measure of uncertainty offers a common language for understanding how and when to use data of varied types. Third, while the rapid expansion of cloud storage has enabled tools for finding and fusing data sets relating to particular research questions, many limitations remain. Two limitations include the following: (1) potential barriers to access due to proprietary or privacy concerns, and (2) the inability of cloud platforms to store or process all data simultaneously, requiring dynamic optimization of approved access. Relying only on available data may result in biased results and limit generalizability of scientific conclusions and engineering solutions. For instance, data may suffer from selection bias, and using these data in subsequent decision making could compound any discrepancies (Wiens et al. 2024).

> Key Finding 3-1: As new data types emerge and evolve, statisticians will need new methods for structuring, sampling, modeling, and analyzing these data.

Several specific challenges offer opportunities for expanded statistical frontiers, including the linkage of privacy-preserving data sources; detection and mitigation of paper-mill publications enabled by data sharing; and the detection, prevention, and remediation of malicious data generation and insertion. These examples involve maintaining inferential integrity in the presence of injected noise (for privacy preservation) or injected signal (for disruption or influence of results).

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The extension of formal disclosure avoidance algorithms (e.g., differential privacy) to data fused from multiple sources offers opportunities to implement and ensure secure data integration across distributed networks. In addition, focused research is needed into the extension of formal protocols, algorithms, and strategies (e.g., privacy-preserving algorithms and synthetic data; Ji et al. 2025; Raghunathan 2021) for broader settings of fused data, inference, and decision making in integrated data settings. How best to fuse confidential data remains an open question—that is, to what extent can privacy and confidentiality be ensured in a connected data world (Rocher et al. 2019)?

While the enhanced accessibility of publicly available data sets offers great potential as described in the sections above, the general availability of large data sets also tempts so-called paper-mill types of publications based on scraping data for any sort of effect, as well as generating publications framing any significant finding within standard reporting styles for rapid publication (Byrne and Stender 2025; Suchak et al. 2025). Such semi-automated publication extends concerns of p-hacking or fishing for significant results into a cottage industry promising publication to junior scientists seeking to pad their professional credentials (Elliott et al. 2022; Head et al. 2015; Stefan and Schönbrodt 2023). Such practices are clearly on the rise and require nontraditional approaches to identify and mitigate their impact on science and engineering, future generative AI summaries and reviews, and the erosion of trust by the public in scientific and engineering research.

Finally, the majority of analyses based on interconnected data resources implicitly assumes benevolent intent from data providers. However, in addition to unintended biases of benevolent actors, malicious data generation and insertion of data elements have the potential to mislead, misdirect, or otherwise influence outcomes of analyses based on fused data from different sources. Statistical research into reliable and accurate empirical methods to recognize, describe, and mitigate deceptive data among fused data elements, as well as improved methods to detect and prevent data manipulation, builds upon a foundation of statistical forensics and fraud detection efforts in finance, tax reviews, and image analysis. The dual-use nature of data to enlighten and obscure patterns and conclusions sets up an ongoing competition wherein so-called red teaming may be used to ensure data integrity and protect against misleading analyses and conclusions (Bezuidenhout 2013; Teichmann and Boticiu 2023).

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4
Statistics Across the Spectrum

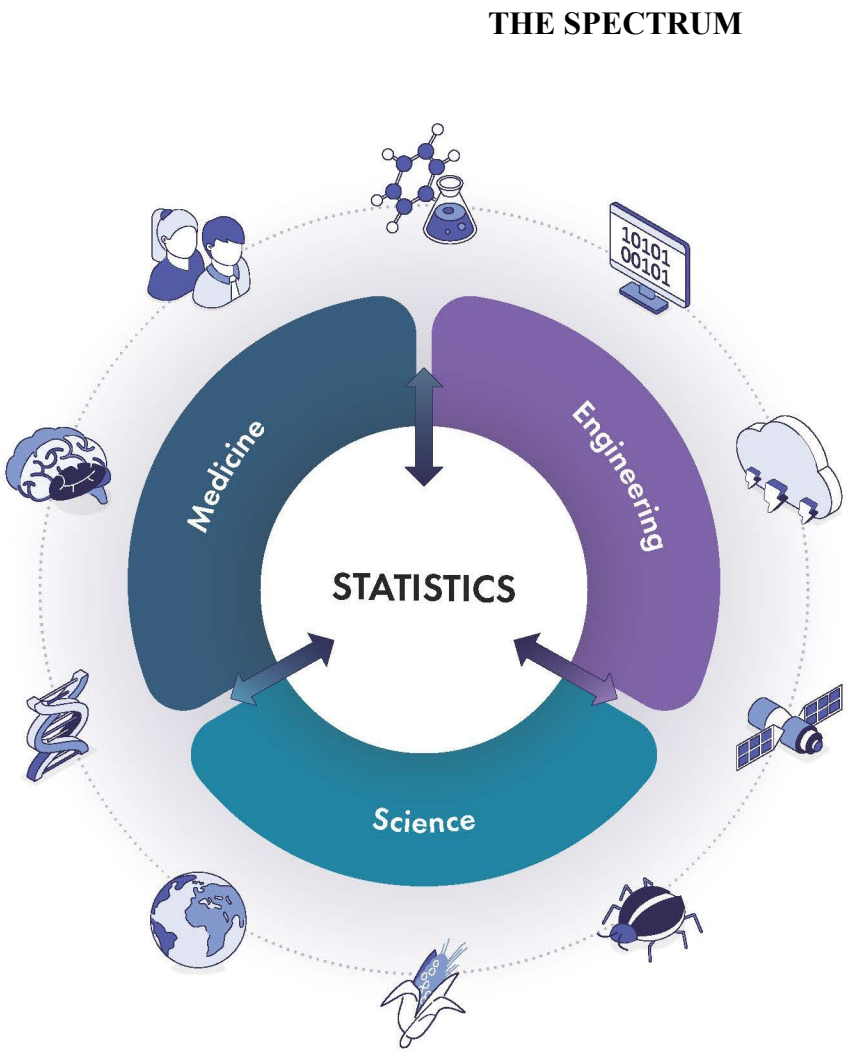

FIGURE 4-1 Statistics: The core of data-driven scientific progress.

Most research areas in science and engineering are data-driven, frequently producing novel data types and structures and requiring creative statistical thinking and advanced methodology. Conversely, methodological ideas or theoretical insights can often bridge

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seemingly unrelated data-driven applications. This dynamic is illustrated in Figure 4-1. Collaborative research has become the norm, blending complementary skills to tackle complex problems well beyond what any single researcher could achieve. Statisticians who work at the intersection of theory, methods, and application can be natural leaders of multidisciplinary teams. Particularly effective statisticians are those who demonstrate a willingness to engage, listen, and learn across disciplines. Such talented individuals are drawn to statistics from a variety of backgrounds because the field offers a rigorous foundation for distilling data into information and serves as a bridge to a wide range of scientific and technological areas of application. These skills and traits remain critical components in the educational programs for statisticians.

Whether via experimental design or complex data collection, feature creation, predictive modeling, causal understanding, or uncertainty quantification, statisticians offer tremendous benefits in collaborative research by helping to drive scientific innovation, speed the scientific process, use resources efficiently, and improve outcomes. Furthermore, a statistician's involvement in applications and collaborations drives statistical innovation, leading to statistical methodology and theoretical development. As noted in Chapter 1, synergistic advances occur across the spectrum from applications to methodological developments to theoretical advances in the field of statistics.

Supporting the contributions of this talent and its central role in team science through recognition, mentorship, and leadership development is essential. Professional societies, such as the American Statistical Association (ASA), have made important strides in this direction, and programs like the ASA Leadership Institute (ASA 2018) and the annual Committee of Presidents of Statistical Societies Emerging Leader Award (COPSS n.d.) represent promising efforts. At the same time, statisticians who wish to lead multidisciplinary initiatives must think beyond disciplinary boundaries and foster the skills needed to collaborate, communicate, and innovate across diverse fields.

COLLABORATION WITH ALLIED FIELDS

As statistical concepts spread through the scientific and engineering community, the frontiers of statistics intersect with and complement the frontiers of allied fields including mathematics, physics, computer science, engineering, and medicine. While statisticians offer

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unique perspectives, each of these fields builds stochastic models that combine disparate data sources and domain expertise, prioritizes accurate uncertainty quantification, and strives for scientific knowledge gleaned from observational data and complex experiments. The remainder of this section is devoted to exploring examples of successful collaborations between statistics and various applied domains.

Weather Forecasting

GenCast (Price et al. 2025) is a probabilistic machine learning weather forecasting model trained on decades of reanalysis data that shows improved speed and medium-term prediction performance compared to a purely physical approach. Conceptually, GenCast is grounded in statistical principles. It essentially models the conditional probability distribution of future weather states based on the current and previous weather states. A forecast trajectory is predicted by factoring the joint distribution over successive states, each of which is sampled autoregressively. In practice, GenCast is implemented as a conditional diffusion model, which is a generative machine learning technique to estimate probability distributions of complex data and generate new (unseen) samples through a process of iterative refinement (e.g., Karras et al. 2022; Sohl-Dickstein et al. 2015; Song et al. 2021).

GenCast owes its success to the abundant reanalysis data produced by assimilating observational data with physics-based models, thus linking and building on domain science and statistical analyses to devise and evaluate data-assimilation methods. This successful application generalizes more broadly to physics-informed machine learning, which blends statistical theory and machine learning algorithmic scale with the constraints of physics-based models to fill scientific knowledge gaps and improve computation while avoiding potential solutions falling outside of known physical laws and associations. Statisticians are well positioned to contribute to this frontier in the physical sciences, given their expertise in interdisciplinary collaboration and data fusion through methodologies such as hierarchical modeling. The novel elements of physics-informed machine learning will also push the frontiers of statistical methodology, as statisticians will seek to focus the fidelity of their statistical models to underlying processes (see the process model discussion in Chapter 2), thereby developing more robust predictive and explanatory models and increasingly relevant statistical inference.

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Experimental Physics

Another important example is the 2012 discovery of the Higgs boson. This breakthrough drew on advancements in statistical theory and methodology, including a robust debate about the foundations of hypothesis testing to devise an objective approach that simultaneously protects against false discoveries and maintains sensitivity for specific alternatives (Van Dyk 2014). Ongoing research in this area continues to provide statistical innovation. Manole et al. (2024) propose a new strategy for establishing the background rate of the event that signals the presence of the Higgs boson, a key to verifying its existence. Because many physical processes can lead to this event, the rate must be estimated empirically. Manole et al. (2024) combine data for auxiliary events with similar properties to the Higgs boson with optimal transport maps to correct for distributional shifts between the auxiliary and background distributions. This is an excellent example of statisticians working in tandem with theoretical physicists to craft a solution that requires expertise in both fields.

Biomedical Sciences

Medical science and neuroscience have witnessed dramatic advances in the past two decades, and statisticians are central to much of this work. As noted in Chapter 3, the emergence of high-dimensional data, with thousands of genes or voxels, sparked a new multiple-testing paradigm to control experiment-wide error rates and promote reproducibility. Functional spatiotemporal modeling has advanced synergistically with neuroscience as statisticians seek optimal ways to understand the complexities of our brains. Subsequent statistical advances in methodologies for imaging led the ASA to launch a new open-source journal, *Statistics and Data Science in Imaging* (Vannucci et al. 2024). Rapid evolution of imaging technologies ensures that collaboration-motivated statistical contributions to imaging will continue through the next decade and beyond. In neuroscience, the emergence of imaging coupled with causal and population-level methods brings further understanding of human behavior and medical advances. Educational programs that specialize in biostatistics remain essential in their emphasis on interactions between the time of assessment and disease manifestation and outcomes, as well as on the measurement technologies that generated the related data.

Moving forward, as artificial intelligence (AI) is poised to change biomedical research, biostatisticians are in a critical position for furthering this research. A recent summit organized

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by the editors of the *Journal of the American Medical Association* addressed pressing topics at the intersection of health care and AI (Angus et al. 2025). The summit raised a few critical questions relating to the field of biostatistics: (1) How is effectiveness established in the context of AI as it applies to biomedical research? (2) Given the rapid pace of development of AI technologies, how can effectiveness be established more quickly than the conventional years that randomized clinical trials take to be accomplished? (3) What are appropriate frameworks for clinical trials in the area of AI? Thoughtful answers to these questions require the involvement of biostatisticians and provide fertile ground for further foundational research.

Conclusion 4-1: The foundations of statistics—across theory, methods, and applications—have provided the scientific infrastructure for major advances in medicine, public health, neuroscience, genomics, and, more broadly, the biomedical sciences. Historically, the National Institutes of Health (NIH) has provided critical support for groundbreaking statistical development supporting these advances. Continued NIH investment in foundational statistics research and training is needed to sustain and to advance the frontiers of both biomedical science and the field of biostatistics itself. Such joint investment in biomedical science and biostatistics is essential for continued development of accurate, reproducible, and actionable insights from modern, complex data sources. Expansion of this support structure would encourage ongoing innovations in the foundations of statistical research focused on health and medicine.

Conclusion 4-2: Areas of investment that NIH might prioritize include statistical and machine learning methods for the analysis of multimodal genomic data, biobanks, and electronic health records; causal inference for complex biomedical data; adaptive designs and real-time intervention; federated and privacy-preserving statistical learning from health care data; and integrative statistics and AI methods for biomedical science and public health.

Materials Science

The field of materials science explores the atomic structure and function of materials. For example, understanding the semiconducting properties of silicon under different environmental conditions allows advances leading to more powerful microchips. The Materials Genome

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Initiative[8] is a multi-agency project launched in 2011 with the aim of streamlining the development of such new materials. A major emphasis of this initiative is to combine physical knowledge and mathematical modeling of material performance with advanced statistical tools including adaptive experimental design, surrogate modeling, and Bayesian optimization to prioritize candidate materials for further testing. This initiative has been a huge success and provides a template for statistical synergistic advancement via large-scale open databases of atomic structure and function (NIST 2021). There is a call to automate material design that directly benefits from rigorous and ongoing statistical collaborations (Department of Energy 2025). As a specific example, the 2025 Institute for Mathematical and Statistical Innovation Uncertainty Quantification for Material Science and Engineering workshop demonstrated the power of adaptive design and the emerging integration of Gaussian process modeling and deep learning in this space (IMSI 2025). New science in this arena, representing the full integration of material scientists and statisticians, is expected to advance the discovery of new materials rapidly.

Meteorology and Climate Science

Both meteorology and climate science involve the study of weather but focus on different timescales and objectives. Meteorology is generally concerned with weather prediction, while climate science is focused on characterizing the distribution of weather over longer time periods. Both areas are data-driven and have been and continue to be heavily influenced by statistics and statistical thinking (Nychka 2000; von Storch and Zwiers 1999). The climate system encompasses Earth's atmosphere, oceans, cryosphere, and other interacting systems that collectively influence Earth's weather. The relevant data are diverse and involve direct observation (e.g., weather stations), remote sensing (via, e.g., satellites, drones), proxies (e.g., ice cores, tree rings), and reconstructions and reanalysis, as mentioned above, as well as output from numerical weather prediction systems and climate models. Statistical methods have been used in and developed for a variety of situations (Zwiers and Von Storch 2004). Examples include collecting and analyzing observational data over time in a nonstationary environment that includes changes in observational technology; understanding the dynamics of the climate system connecting unobservable states to both direct observations and indirect measurements (proxies);

[8] The website for the Materials Genome Initiative is https://www.mgi.gov, accessed January 16, 2026.

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weather forecasting, including data assimilation, ensemble prediction, and understanding and correcting for forecast error and bias (e.g., model output statistics); designing and analyzing climate model ensembles, including the assessments of uncertainty and methods for integrating the ensemble members into a unified analysis or characterizing past climate through paleoclimate reconstructions (hockey stick report); and characterizing weather extremes and their impacts including applications to public health, agriculture, power generation, and other things.

Such research generated methodological improvements in a variety of areas, including spatial statistics, Bayesian methods, methods of multivariate extremes, and uncertainty quantification, to name a few. The work remains critically important today and will continue to garner strong statistical insight over the next decade, particularly as our banks and investment vehicles, as well as the insurance industry, prepare for impacts of climate change.

AI and Computer Science

Advances in deep learning have impacted science and engineering and the development of statistical methodology. A recent example is the re-emergence of simulation-based inference (SBI) (Cranmer et al. 2020). Statistical methods are often limited to using closed-form distributions; however, there are examples, such as multivariate extreme value analysis, or heavy-tailed processes, where theoretically motivated distributions cannot be easily computed (Schlather 2002; Vankov et al. 2019). SBI unlocks these distributions for use in statistical methods by approximating them (or the resulting posterior distribution) using simulated data sets (refer to Chapter 3's discussion of model-generated synthetic data) to estimate parameters and train a deep learning model to emulate the distribution function (Banerjee et al. 2014; Wikle et al. 2023). Statisticians will continue to realize advancements in statistical methodology by incorporating computational and conceptual advances in allied fields.

Box 4-1 offers a deep dive into how statistics and finance furthered one another.

BOX 4-1
The Impact of Statistics in Finance

Statistics as a discipline fundamentally changed banking and global financial markets; simultaneously, questions asked from finance advanced the boundaries of the statistical sciences (Ensor 2024, Zhang et al., 2005).[a] For example, it is critically important to understand

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the volatility in market returns, which led directly to the creation of the class of time series models known as a generalized autoregressive heteroskedastic (GARCH) models to study volatility (Henderson n.d.). The 2003 Nobel Prize in Economics was awarded to Robert Engle for bringing forward these ideas. GARCH models and their Bayesian hierarchical counterparts, stochastic volatility models, are commonplace in the toolkit of today's financial econometrician and statistician. The statistics innovation life cycle is in full force at the intersection of statistics, econometrics, probability, applied mathematics, and computing in the important area of finance.

The study of financial markets has always relied on strong statistical modeling, inference, and visualization. As financial markets are temporal in structure, the subfields of time series and stochastic processes are critical for this industry. Today, the use of statistics has shifted from drawing inferences from patterns in historical data to quantifying risk, predicting market movements, and automating decision making and action in real time. Statistical principles, from basic concepts like standard deviation and beta to complex models like the stochastic Black-Scholes-Merton option pricing model, the time series GARCH and the Bayesian complement of stochastic volatility models to understand market volatility, multivariate copula approaches to complex dependence structure, and large-scale Monte Carlo simulation to quantify systemic risk-based scenarios, form the core of every major financial market's operations and thinking. Progress in statistics and AI transfers immediately to this sector of the global economy. The application of advanced statistical methodologies, whether in the context of portfolio optimization, options pricing, or high-frequency trading, allows professionals to navigate uncertainty and operate with a level of precision and speed previously unimaginable.

A fundamental issue in the professional's quantitative toolkit is model risk and a lack of understanding of the underlying principles on which the conclusions and actions are based. The global financial crisis of 2008 is often attributed to the failure of complex risk structures packaged as credit default swaps and priced using Gaussian copula technology (MacKenzie and Spears 2014; Salmon 2009). The Gaussian copula model (and indeed any model) provides a tool that must be used with care, making sure to account for ways in which the model may not reliably describe the data; excessive trust in this imperfect model led to a severe and detrimental global impact as investors over-leveraged their positions based on faulty understanding (Arnade 2013). The role of leverage is often also not well understood from a probabilistic perspective (Ginley et al. 2015).

[a]This sentence was changed after the release of the report to include an additional reference.

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EVOLUTION OF SYNERGISTIC COLLABORATION

Synergistic advances will continue and expand across science and engineering due to the changing nature and connectedness of data (described in Chapter 3). Expanded interdisciplinary team science wherein the statistical field plays a key role continues to occur in personalized medicine, neuroscience, adaptive experimental design and clinical trials, materials science, meteorology, and climate studies, among many others. For instance, deep Gaussian processes drive advances in cosmology. Personalized medicine and neuroscience are moving to a stage where inference will be explanatory and causal, progressing well beyond predictive models. These advances require essential transitions in statistical thinking and careful understanding of information, including the subtleties of time-to-event constructs and population representation. An appreciation of these nuances is critical for meaningful scientific findings and recommendations. In general, every field of human inquiry benefits from synergistic progress in statistical methodologies. This section outlines a few specific opportunities for targeted support and enhancement.

Advances in Computing

The rapid evolution of computing technologies is key to the advancement of AI and machine learning. Its impact on fundamental statistical methodology and theory is equally profound. For example, as graphical processing units (GPUs) and other chips become faster and more memory-efficient, researchers will have increased capabilities to explore more complex models, larger data sets, and computationally intensive inference techniques that were previously computationally impractical. The committee anticipates continued dramatic increases in processing speeds and memory capacity, as well as task-specific optimal hardware.

Computing Enables Large-Scale and Real-Time Statistical Methods

Statistical frameworks, such as those involving Bayesian methods, nonparametric processes, and high-dimensional optimization, often utilize iterative procedures requiring substantial computing resources. Improved computational hardware will enable (1) massively parallelized sampling algorithms (e.g., Hamiltonian Monte Carlo, variational inference, and full integration of machine learning to provide relevant Bayesian posterior distributions); (2) real-time uncertainty quantification in emerging online data collection settings (e.g., continuous

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scraping of text from the Internet); (3) real-time robustness and reliability checks; and (4) the ability to democratize experimentation such that statisticians can rapidly prototype pioneering architectures, tune hyperparameters at scale, and automate model selection using ensemble and meta-learning techniques, including Bayesian transfer learning (Suder et al. 2025), Bayesian stacking (Yao et al. 2022), and consensus Monte Carlo methods (Scott et al. 2016). Optimizing these advances requires close, mutually beneficial collaboration between statisticians and computer scientists.

Computing Powers Collaboration

New computational resources create opportunities for statisticians to contribute to various emerging application areas and scientific domains. For example, one of the most compelling intersections of AI and statistical modeling with quantum chemistry and biophysics is in molecular dynamics (MD) simulations, where the scientific aim is to model the physical movements of atoms and molecules over time. These simulations are foundational in drug discovery, materials science, and protein folding studies. However, traditional MD simulations are computationally intensive, often requiring weeks of runtime on conventional central processing units. With the advent of AI accelerators—such as NVIDIA's tensor cores, Google's tensor processing units, and custom application-specific integrated circuits—researchers will be able to integrate well-calibrated statistical approaches (e.g., bootstrapping, model averaging) in MD workflows across multiple simulation trajectories, enhancing the robustness of downstream scientific conclusions. The newly established National Artificial Intelligence Research Resource (NAIRR) aims to expand researchers' access to these kinds of AI resources, including computational resources.

Quantum Computing and Statistics

The fundamental stochastic nature of quantum physics means that quantum computers do not provide deterministic results but rather probabilistic outcomes that require novel statistical frameworks for accurate analysis and interpretation. The extreme fragility of qubits leads to quantum decoherence and noise, requiring sophisticated statistical tools and techniques for effective quantum error correction and mitigation without collapsing the quantum states. Designing quantum algorithms requires a fundamentally different approach than classical programming. Statisticians are needed to help develop and optimize algorithms that can leverage

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quantum mechanics for specific problems like large-scale optimization, Monte Carlo simulations, and machine learning—areas where quantum computers promise significant speedups. New statistical methods are required to interpret results from quantum experiments and simulations and to assess the validity and reliability of these new computing paradigms. Intense foundational statistical research is required to bring fault-tolerant, useful quantum computers to fruition.

Quantum computing is also set to provide a scientific paradigm shift. While it is in its early stages of development, the theoretical advantages are difficult to overstate. Fundamentally new statistical methods and algorithms will be needed to exploit this new technology (Layden et al. 2023). Standard operations such as exactly copying a variable are impossible in quantum computing, and there are currently non-trivial error rates for basic operations. Despite these challenges, emerging test cases show great promise. To illustrate the potential, consider Markov Chain Monte Carlo (MCMC) sampling. MCMC is the longstanding workhorse of Bayesian statistics but can be prohibitively slow for large problems. This approach is conducive to quantum computing because the algorithms are stochastic and high-dimensional. Lopatnikova et al. (2024) show that incorporating quantum computing results in a cubic speedup compared to classic algorithms. For statisticians to play a role in developing new quantum algorithms, substantial time and training investment is required by researchers to understand the underlying physics, and interdisciplinary teams are likely the best path forward. Investments in such team science will pay off with unprecedented analytical power focused on state-of-the-art models when quantum computation becomes widely available.

Constrained Resources and Environmental Impacts

As computing continues to accelerate, it is increasingly important to be cognizant of its environmental impacts. Data centers require enormous amounts of energy and water, making it necessary to consider sustainability needs (NASEM 2025c). In many cases, complex AI models are unnecessary for certain day-to-day uses. It is critical that researchers use tools that are fit-for-purpose, opting for those that are less resource-intensive but work as needed.

Consider the case of large language models (LLMs) versus small language models (SLMs). Belcak et al. (2025) argue that SLMs are "sufficiently powerful, inherently more suitable, and necessarily more economical for many invocations in agentic systems" than LLMs, and research is ongoing to explore cases where SLMs outperform LLMs and how these models

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can be selected appropriately (Sinha et al. 2025). The edge computing paradigm—in which computing is increasingly moved from the cloud to the "edge," closer to the local device on which data are created—offers another promising direction for improving sustainability of computing (Carvalho et al. 2021). In general, efficient algorithms and deliberate model design can reduce computational burden.

Novel Data Architectures

Emerging data infrastructures are enabled by statistics and also impact how data are stored and preserved. One relatively new and widely known development in data architecture is blockchain. Blockchain is a distributed, tamper-resistant ledger that records transactions across a network without relying on a central authority. While initially developed for digital currencies such as Bitcoin and Ethereum, blockchain applications now extend to health care (Treiblmaier et al. 2024), commerce, agriculture (Colorado Department of Agriculture n.d.), art, government, and financial markets. This expansion has generated vast new data streams and complex challenges. To ensure these technologies are trustworthy, transparent, and beneficial to society, statistical science is crucial. Statistics offers the methods to evaluate performance, detect fraud, assess risks, and promote responsible adoption.

Statistical approaches are particularly valuable in areas with direct societal and policy relevance. For example, fraud detection methods strengthen the integrity of cryptocurrency markets, while portfolio analysis in decentralized finance improves security and efficiency. In health care, statistical methodologies are crucial for balancing interoperability of patient data with privacy protection. Similarly, as real estate markets explore tokenization, statistical tools can be used to evaluate risks and design effective safeguards (Saari et al. 2022; Swinkels 2023). These applications demonstrate how statistics can both address pressing challenges and support innovation in emerging blockchain systems.

Meeting these demands requires advancing statistical methodologies tailored to blockchain data. New models for dynamic, multilayer networks can capture the complexity of transactions, while topological and geometric tools can reveal hidden patterns of illegal actions in financial markets. Uncertainty quantification, through statistical methodologies, ensures that AI applied to blockchain data produces reliable insights. Scalable computational methods make analysis feasible at the volumes of data blockchains generate. By embedding statistics into

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blockchain development, researchers, policy makers, and industry leaders will foster systems that are resilient, equitable, and trusted, thereby ensuring blockchain technologies contribute to stronger markets, better governance, and measurable benefits for society.

There is a plethora of public blockchain data (AWS n.d.; Shamsi et al. 2022; Victor et al. 2021). An overview for statisticians is provided by Akcora et al. (2022). Furthermore, the *Journal of the Royal Statistical Society Series A* plans to publish a special issue on statistics and blockchain in early 2026 (Gel and Rambharat 2024) that will cover a wide range of statistical applications in the blockchain ecosystem.

BUILDING A COLLABORATION ECOSYSTEM

Statistics often stitches diverse domains together, providing not only technical expertise but also a thorough understanding of data, analytical tools, and domain-specific motivations across science and engineering. To foster meaningful collaborations, an accessible infrastructure and thoughtful cross-sector partnerships will be necessary.

Benefits of Statistical Collaboration

For all data-driven research, including statisticians on teams—and placing statisticians in leadership and co-leadership roles where appropriate—is necessary to ensure that statistical rigor, methodological innovation, and uncertainty quantification are embedded from the outset rather than retrofitted after critical decisions have been made. This early inclusion is essential for advancing science responsibly and efficiently. Furthermore, leadership roles empower statisticians to shape research agendas, training programs, and ethical standards. Statistical leadership can ensure foundational principles guide applied work, preventing misuse of algorithms or misinterpretation of results. This guidance is especially critical in fields like public health and biomedicine, where decisions have significant real-world consequences.

The considerable opportunities offered by incorporating statistical expertise within team science come with some key challenges. In the future, the many modes of collaboration for statisticians within large research teams will continue to move beyond traditional academic consulting centers and government-supported mathematical institutes. This growth will be driven partly by (1) the need for statisticians to be more than data consumers and take leadership roles in project design, implementation, and evaluation and (2) the speed and flexibility with which

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machine learning and AI significantly increase the pressure on researchers to expedite technological development and produce tangible products.

Tools to Facilitate Collaboration

The continued integration of statistics with other scientific domains will depend on the development of robust infrastructures that facilitate collaboration, enable scalability, and encourage reproducibility. The core of these infrastructures requires unified data ecosystems where findable, shared, and version-controlled repositories are kept with standardized preprocessing procedures and in-depth metadata. UK Biobank[9] offers an example of such a unified data ecosystem; it has become a global reference model for large-scale biomedical research. It combines standardized data collection and storage, rigorous preprocessing, and rich metadata to create a findable, accessible, and version-controlled resource. UK Biobank has also put in place comprehensive training, policies, and protocols to ensure ethical practices, particularly around data privacy, participant consent, and responsible and collaborative research conduct. This framework not only underpins trust in the resource but also fosters collaboration across government, academia, and technology partners. UK Biobank demonstrates how robust governance and shared infrastructure can accelerate innovation while maintaining the highest ethical standards.

Similarly, in online data collection environments (e.g., adaptive clinical trials, real-time monitoring systems, or digital phenotyping platforms), model version control will become an equally critical component. To that end, scalable computing hardware will be essential as models will need to be retrained, fine-tuned, or recalibrated often in real time. Here, shared coding platforms and integrated documentation tools will foster and expand transparency and consistent interactions across disciplines. Governance frameworks need to evolve in order to support reproducibility and accountability including audit trails, workflow managers, and ethical review processes (Committee on Professional Ethics of the American Statistical Association 2022). Of growing concern is the enforcement of transparency requirements for publications and products from for-profit institutions. As AI architectures become larger and GPUs become increasingly coveted resources, industry-based groups may be advantaged to conduct research at the frontier. The landscape becomes even more opaque when one considers privately funded

[9] The website for the UK Biobank is https://www.ukbiobank.ac.uk, accessed January 16, 2026.

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institutes that sit in between academia and industry. Journals could hold authors from outside the academy to the same standards and disclosures as those from academic institutions. One way to enforce statistical rigor and transparency for proprietary models meant for commercial purposes is from within the organizations that create them (i.e., self-regulation). For example, Google and Microsoft release AI transparency reports (Google 2025; Microsoft 2025) each year on the practices that they take internally for their products and research. The real socio-ethical challenge here is that this system requires trust within these companies (e.g., honesty about the way user information is used or not used for AI models). A more extreme version of this is the regulation (O'Brien 2024) that any AI model trained on 1,026 floating-point operations per second must be reported to the U.S. government. One approach may include appointing a third-party task force whose role is to ensure statistical rigor and uphold some notion of a transparency and reproducibility standard before any AI model is moved to commercial markets.

Public–Private Partnerships

Academia and industry play complementary roles within the knowledge and innovation ecosystem, and both would benefit from more mutual support and increased collaboration. Academic research often focuses on the exploration of new or unproven ideas, while industry often serves to scale up established research. Due to this nature, academia plays a foundational role in driving industry innovation. Many technologies and breakthroughs that fuel private sector success are rooted in decades of federally funded academic research. Google is a classic example; it was borne of the National Science Foundation's (NSF's) multi-agency Digital Library Initiative, built by Stanford University students (NSF 2004). In statistics specifically, stochastic modeling benefits from a long history and today underpins applications in numerous fields, from finance to the life sciences to weather forecasting. Plenty of the progress in stochastic processes is a direct result of NSF funding, through researchers like Monroe Donsker, Srinivasa Varadhan, and Daniel Stroock (NSF n.d.a).

As a result, statisticians, industry researchers, and members of allied fields will benefit from co-located spaces, creative organizational structures, and other deliberate mechanisms that can help address some of the issues that arise in academic and industry collaborations. These challenges include differing goals and objectives concerning the artifacts or products of the collaboration, the ownership of intellectual property, differing timeline pressures, and use of

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industry and academic resources (e.g., data, software, algorithms, compute). Some recent examples of scientific collaborations between academic institutions and sectors outside of the academy include (1) pharmaceutical (e.g., Eli Lilly and North Carolina State University), (2) technology (e.g., Microsoft, the Broad Institute of the Massachusetts Institute of Technology, and Harvard University), (3) national laboratories (e.g., Jet Propulsion Laboratory; the University of California [UC], Los Angeles; and the California Institute of Technology), and (4) private institutes (e.g., Arc Institute with Stanford University, UC San Francisco, and UC Berkeley). Co-located spaces foster real-time dialogue and a better understanding of the scientific context behind any data being generated. They also generate deeper literacy in statistics for scientists from allied fields.

Public–private partnerships offer a mechanism for academia and industry to support one another. Collaborative models that give academics the freedom to continue foundational research, rather than focusing on research driven by industry needs, are especially valuable. Foundational research lays the groundwork for future advances. Thus, public–private partnerships are most valuable when they not only fund research but also safeguard foundational principles, ensuring that rapid deployment does not compromise statistical validity.

NAIRR provides an example of a public–private partnership, supported by 14 federal agencies and 28 private sector and nonprofit partners. It aims to connect U.S. researchers and educators to AI resources with the ultimate goal of accelerating AI and AI-powered discovery and innovation (NSF n.d.c).

Conclusion 4-3: Public–private partnerships increase the vitality of collaborative statistics research for both industry and academia, particularly when academic researchers are given the latitude to focus on foundational research.

Conclusion 4-4: Such historical statistical investments have led to breakthroughs in medicine, health care, agriculture, energy, the discovery of new materials, the financial sector, insurance, weather prediction, and resilient infrastructure and communities, and will lead to significant advancements in the future, ensuring sustained U.S. competitiveness.

Looking ahead, Chapter 5 discusses the training needs for the next generation of statistical scientists in a future of interconnected teams and interconnected data. The committee

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echoes some key points here to set the stage for expanded discussion in Chapter 5. As previously noted, statisticians bring many valuable skills to interdisciplinary collaborations. This includes methodological expertise and familiarity with different aspects of data analysis, from collecting data to choosing the right methods (or developing new ones if necessary) to interpreting analysis results to presenting those results for broader audiences. Sound statistical thinking is imperative to help guide this process and bring together the many different threads being worked on by the team, which typically involve data and how data move between different parts of the process.

Training Toward Collaboration

To improve outcomes arising from collaboration, statisticians will need expertise in the domains in which they apply their knowledge. Intentionally incorporating cross-training and interdisciplinary projects into postsecondary statistical education is paramount, as discussed further in Chapter 6. Statisticians must also possess strong communication skills, which involve the ability to explain complex statistical concepts in the development and application of methods; to learn technical concepts from other fields quickly; and to link the results of statistical analysis back to the motivating scientific or engineering objectives. Explaining statistical concepts to collaborators includes explaining the relative merit of different tools, such as neural networks or LLM embeddings, whose names have become buzzwords but whose actual workings and power may not be well understood by non-statisticians.

A significant need exists for sustained and effective two-way communication while collaborating. In this context, communication entails being a good listener and asking appropriate questions to develop an understanding of the scientific and engineering problems and goals of domain experts, as well as translating those into statistical frameworks. With the widespread use of statistical methodologies throughout science and engineering, it is essential to meet the collaborative team members where they are on their statistical journey. Implementing effective collaborative workflow processes can enhance the quality of collaboration.

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5
Statistical Innovations That Enable New Scientific and Technological Advances

This chapter examines how statistical sciences facilitate scientific and technological innovation and how emerging technologies are reshaping the landscape of statistics. Chapter 2 described the life cycle of statistical innovation. Chapter 3 explored how data and data collection are changing. Chapter 4 highlighted successful examples of collaboration between statistics and domains within science and engineering. This chapter connects and expands these points by proposing that these changes in (1) the nature of data, (2) how data are collected, and (3) how researchers collaborate to understand these data are both drivers and products of technological advancement. This self-reinforcing cycle also drives and, crucially, is driven by statistical advancements. The collection of the many ways statisticians explore, model, visualize, interpret, and understand data represents an inherent and crucial part of technological advancement. Such "data-driven innovation" is explicitly addressed in the Organisation for Economic Co-operation and Development's report *Data-Driven Innovation: Big Data for Growth and Well-Being* (OECD 2015) and in Sharp et al. (2024).

The next decade will bring important changes to the ways science and engineering are conducted and the way technology is developed and deployed. Much of this will be driven by new hardware technologies and continuing developments in artificial intelligence (AI). On the hardware front, there are two domains of particular relevance: new technologies for computing and new technologies for sensing and data collection (NASA Ames Quantum Sensing Technical Meeting Organizing Committee 2024). In the National Academies of Sciences, Engineering, and Medicine's 2019 report *Quantum Computing: Progress and Prospects*, Chapter 1 provides a concise history of progress in computing capabilities. A finding from the report states, "Moore's law for integrated circuits resulted from a virtuous cycle, where improved technology generated exponentially increasing revenue, enabling reinvestment in R&D [research and development]

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and attracting new talent and industries to help innovate and scale the technology to the next level." But the report also notes that the scaling benefits of Moore's Law are slowing, and the virtuous cycle for hardware will not likely resume until a commercially viable quantum computer becomes available (NASEM 2019b). Conversely, some, such as Jensen Huang, argue that while transistor-level scaling has indeed slowed, system-level innovation is driving rapid gains in computing. For example, Huang has argued that advances in architecture, accelerators, packaging, and software (sometimes called Huang's Law) continue to accelerate performance and reduce effective costs. This means that new algorithm design and statistical research can and should adapt to leverage these evolving computing paradigms, without waiting for quantum computing.

On the data collection front, sensing technologies such as satellite remote sensing and medical imaging continue to advance, and by 2035 quantum sensing is expected to bring revolutionary changes. According to the National Aeronautics and Space Administration's Earth Science Technology Office, "As a future capability, quantum sensing has the potential to be revolutionary in advancing the accuracy and precision of science measurements, of Earth and beyond. By utilizing various properties of quantum mechanics—entanglement, superposition, interference, squeezing, etc.—quantum sensing technologies may offer significant advantages over traditional sensing methods" (NASA ESTO n.d.). In the biomedical realm, "Quantum sensors enable new possibilities in biomedical applications due to their high sensitivity…. The atomic length scale of quantum sensors and their coherence properties enable unprecedented spatial resolution and sensitivity" (Aslam et al. 2023). At the same time, quantum sensing will introduce new types of statistical noise, owing to quantum mechanical fundamentals, for which statisticians should prepare. This will require a deep dive into quantum sensing technologies and presents ample opportunities for incorporating statistical components as the technology develops and comes online.

In addition to quantum sensing, networks of smaller satellites, with instrumentation such as synthetic aperture radar, should also come online soon. This technology could have a dramatic impact on environmental applications with faster timescales for monitoring rapid change events such as floods or wildfires. With increased spatial coverage and much faster timescales, new statistical methodologies for dealing with high-resolution, rapidly generated data will need to be developed, allowing similarly rapid development and deployment in such emergency events.

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Unmanned aircraft systems will also contribute data and information, further increasing the demand for accurate, reliable, and timely statistical methods to fully gain the value offered by these technologies in emergency applications.

More broadly, there are many development efforts for improved, expanded, or entirely new Earth- and ocean-observing systems (e.g., in-situ, remote sensing, and models). Such technologies are individually impressive but when coupled will feed the next generation of weather and climate models, increasingly including those that are AI-driven (Calewaert et al. 2024). This confluence of rapidly obtained and interconnected data generation will require statistics to handle the diversity and volume of these new data and the speed at which they are obtained. Machine learning and AI will play a crucial role in handling the data and in the applications leveraging the data. Weatherhead et al. (2018) discuss the need for design and evaluation criteria for these new systems.

The field of statistics is positioned to capitalize on many of these areas of rapid development. Agentic AI, for instance, goes beyond current AI capabilities by taking action to achieve specific goals, plan, learn, and interact with the real world autonomously. This highlights the importance of equipping AI agents with the ability to characterize the biases and uncertainties in inferences and decisions that drive their actions. Interestingly, consequences of at least some of those actions may be known in real time; thus, it may be possible to learn how to reduce biases and uncertainties on the fly. In the future, statistical inference may become embedded in autonomous systems making consequential decisions. This elevates statistics to a real-time, mission-critical component of intelligent systems. Realistically, additional constraints imposed upon AI are environmental, and they present an opportunity for the development of resource-efficient AI. Focused AI solutions, such as small language models, are proving highly beneficial as resource constraints, including environment, time, and cost, are incorporated into the widespread use of AI (Ornes 2025).

Broderick et al. (2023) provide an insightful look at issues of trust in machine learning, and many of them carry over to the more general concepts of AI including statistical methods. In particular, the authors distinguish between development and deployment phases of methodological innovation, acknowledging that the trust requirements are different. The committee builds on this framework to illustrate the life cycle of scientific and technological advancement and how the field of statistics links to each step in Table 5-1. The framework in

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Table 5-1 posits that the development phase discussed in Broderick et al. (2023) can be further broken down into discovery, design, and decision making. Deployment is augmented to include sustainment activities required to keep the technology supported and updated as it may continue to progress. Accordingly, the columns of Table 5-1 represent main phases of the life cycle. The rows detail the enablers that allow for success in each of the phases. The cells in the body represent broad categories of ways in which the enabling areas contribute to achievements in discovery, design, decision, and deployment (see Figure 5-1). In what follows, the committee devotes a section to each phase, and subsections will discuss contributions of the enablers and relevant examples. Statistical methods are pervasive throughout but always intertwined with other crucial scientific and technological domains. Moreover, statistical thinking provides a holistic, rigorous, uncertainty-aware paradigm for scientific and technological achievement that naturally integrates diverse knowledge, maximizes the value of massive amounts of routinely collected data, and provides a framework for optimal decision making (Chance 2002; Wild and Pfannkuch 1999). Each cell in Table 5-1 below presents non-exhaustive examples of how the enablers intersect with the stages of development.

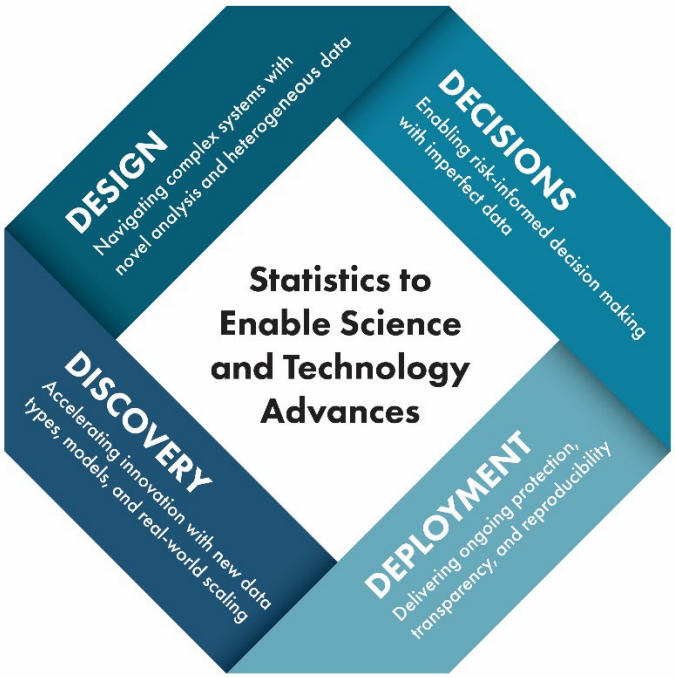

FIGURE 5-1 Statistics to enable science and technology advances.

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TABLE 5-1 Stages of Scientific and Technological Advancement

	Stages of Scientific and Technological Advancement			
Enablers	**Discovery (1)**	**Design (2)**	**Decision (3)**	**Deployment and Sustainment (4)**
New technologies for data acquisition, including sensors	New data types reveal new scientific insights	New analysis methods in response to new data collection	Advances in scientific understanding or societal benefit	Security considerations
Data, domain knowledge, and models	Data-driven science; experimental design	Fusion of heterogeneous, complex data; data assimilation	Uncertainty quantification and risk analysis; digital twins; computer experiment	Privacy, open data, data sharing
Computation	Scale for real-world relevance	Optimization; complex dynamical systems	Simulation-based Inference	Reproducibility; statistical algorithms for reduced computational burden
AI and machine learning and statistical methods	Accelerating the development cycle	Quality of approximate solution versus cost	Accuracy and autonomy	Trustworthiness, policy, and protection against unintended consequences

The following sections discuss the ways in which emerging technologies are reshaping the landscape of statistics, and how statistical sciences facilitate scientific and technological innovation at each of the four stages of scientific and technological advancement.

STAGE 1: DISCOVERY

New discoveries result from new ideas inspired by efforts to explain the real world, as seen in the example outlined in Box 5-1. Sometimes this is because current theories do not adequately account for what is observed, and sometimes it is because new information suggesting alternative explanations presents itself. In the former case, current theories are often encoded in models; the first problem is whether the models' predictions agree with observations, after which the models may be updated and the question asked again. A rigorous statistical framework for such hypothesis testing has existed for a very long time. The latter case might be described as *hypothesis discovery*, and it has been supercharged by the avalanche of opportunistically collected data and by AI and machine learning methods trained on these data.

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Finding previously unknown patterns and relationships is what machine learning does very well. Current work includes studying how AI can be leveraged for knowledge discovery in this way (e.g., J. Lu et al. 2024). Statistical theory and methods are crucial for determining whether these relationships are real or not in the sense that they may result from randomness rather than structural sources. Moreover, AI models are not inherently explanatory. They are often so complex that it is exceptionally difficult to do more than exploit the relationship through prediction—and only so long as the underlying structural relationships do not change. Hence, testing in the presence of uncertainty is needed (Dunson 2018).

Over the next 10 years the committee expects the avalanche of data to accelerate and AI to become agentic. To harness these developments for scientific and social good, statistics must evolve to play its part in discovery (and testing) using heterogeneous, highly distributed, nontraditional data types that do not necessarily conform to existing mathematical formalisms. A logical progression exists from (1) the advent of new data types delivered by new ways of collecting data; to (2) increased importance of experimental design principles for interrogating these data (even if no actual "experiments" are conducted); to (3) the need for highly scalable statistical methods, and the theory that underlies them, for on-the-fly testing; and finally to (4) development of AI agents that can autonomously execute analyses in these areas and make decisions as part of the data pipeline. These steps form a virtuous cycle of their own. What starts with discovering new relationships ultimately leads to new questions, and this in turn informs new data collection strategies or reinterpretation of existing data.

BOX 5-1

Data-Driven Science and Genomics

A key example of this cycle is the rapid development of statistical methodology in conjunction with the growth of the field of human genomics over the past several decades. A crucial step in the growth of this field was the 2003 completion of the Human Genome Project, which sequenced the human genome over a span of multiple years; since then, technology has advanced to the point where it is possible to sequence an individual's genome in hours (Cara 2023). The public availability of human genome data has enabled progress on a vast range of research questions, from population genomics (e.g., studying migration patterns of ancient populations) to applications in medical diagnostics and public health (via genome-wide

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association studies). More recently, a different technology has emerged to give a more fine-grained view of genetics: single-cell transcriptomics, which is the process of measuring gene expression levels in individual cells—this was awarded the *Science* 2018 Breakthrough of the Year (in nonhuman cells, at the time, but this technology has been applied to questions in human research as well) (*Science News* Staff 2018).

These developments in genomics and genetics exhibit close connections to accompanying innovations in statistical methodology. One example is visualization and dimensionality reduction techniques for data exploration and interpretation. For instance, t-Distributed Stochastic Neighbor Embedding (Hinton and Roweis 2002) and Uniform Manifold Approximation and Projection (McInnes et al. 2018) are two techniques that provide low-dimensional embeddings for high-dimensional data, enabling the identification of important and interpretable clusters or subpopulations in the data. At the same time, drawing conclusions from clusters that have been identified via visualization of the data can lead to false discoveries, and more recent methodology leverages tools from the field of selective inference to correct for this issue (Gao et al. 2024). A second example is the role of hidden Markov models for modeling the human genome. This classical statistical model has existed for more than 50 years and over the past several decades has found an influential application in modeling the strong dependencies between nearby locations on the genome (e.g., Hobolth et al. 2007; Li and Stephens 2003), reflecting ancestral population structure. This model offers solutions to questions such as identifying population structures, imputing missing genotype data (Browning et al. 2018; Li and Stephens 2003), correcting for dependence when testing hypotheses to determine which genetic variants are associated with a disease (Sesia et al. 2019), and many more.

The statistical community has spent the past decade exploring how to minimize computational complexity. Scalable computational solutions are achieved by investing heavily in the initial computation to train the inferential procedure and then reuse this pre-trained inference model in a forward fashion. Related approaches include variational inference, integrated nested Laplace approximations and Laplace approximations, expectation propagation, and approximate Bayesian computation. In probabilistic machine learning, these ideas are encapsulated in amortized inference. Multivariate spatiotemporal extreme value methodologies capitalize on this hierarchical framework.

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Amortized inference methods reframe classical parameter estimation as a supervised learning problem, incorporating neural networks (or some other machine learning algorithm) trained on simulated data to predict values of parameter estimates for new data. The fundamental idea is to create a mapping from a data set to the parameters of statistical models that represent the data-generating mechanism. Not only are these methods showing error performance comparable to traditional approaches, but they are also changing the conversation about where to apply computational effort. Many areas benefit from such approaches, and generalizing these ideas for more complex models is an open area of work. There are opportunities to improve amortized inference methods as well as to generate a theoretical framework to better understand the performance and limitations of this new paradigm.

Key Finding 5-1: Statistics provides a mechanism for understanding the behavior, robustness, and reliability of AI and machine learning models.

Key Finding 5-2: Statistics could benefit from new research on computationally efficient hierarchical modeling and amortized inference using AI and machine learning to make inferences on vastly larger scales than is currently possible.

STAGE 2: DESIGN

Increasingly we are in an era in which many possible types of related data are obtainable, but these data differ in quality, volume, and the degree to which they relate directly to questions of interest. As discussed in Chapter 3, it is of paramount importance to develop statistical methods for combining these data to synthesize the current state of knowledge, and to design subsequent data collection efforts to address knowledge gaps in a manner targeted to improve our ability to address the questions of interest accurately. For example, when the inferential goal is to obtain daily spatial maps of where different species of birds occur, one may combine prior information on migration patterns for different species, opportunistic citizen science data from apps (e.g., BirdNET)[10] containing observations of which species were present at particular locations and times, and line transect surveys at different locations conducted by bird experts within a Bayesian spatiotemporal model (Coxen et al. 2017; Ogawa et al. 2024). The resulting

[10] The website for BirdNET is https://birdnet.cornell.edu, accessed January 16, 2026.

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posterior can then be used to design the locations of future line transect surveys by choosing the locations in which new data would lead to the greatest reduction in posterior variance.

While on the surface this resembles classical Bayesian design, such as in the context of choosing computer model inputs relying on Gaussian process emulators, a problem increasingly faced in emerging applications is the presence of many different sources of data. For example, in the above case, various data include prior information elicited from bird experts on migration, cheap app data that are plentiful but of varying quality and potential sampling bias, and high-quality surveys conducted by scientists. An ongoing challenge is how to consider the complex multiview nature of the data in designing new data collection efforts. Statistical models that can link the different data sources while considering the lower-quality nature of certain sources provide a key approach for tackling such problems.

> Key Finding 5-3: An open area of work is capturing the complex nature of data during data collection.

The design phase of a new scientific or technological advancement comes after a discovery has been made and at the point when the discovery has viable potential to improve scientific understanding or yield engineering, financial, or societal benefits. The importance of statistical design for precision agriculture, for instance, is outlined in Box 5-2. In other words, design means entering a prototyping phase. In this phase, statistical rigor is crucial in quantifying the performance of the new technology under different design choices and potential scenarios. A coherent quantitative formalism is needed to compare the impacts of these choices on the new technology and to make informed decisions about them. Moreover, the new technology may itself include new statistical methods required to fully exploit the new data types that the new technology enables. For example, hyperspectral imaging in remote sensing yields data in hundreds or even thousands of spectral channels with complex nonlinear dependencies and demands new dimension reduction strategies to build pipelines for processing these data at scale. Visualization can play a role in informing and guiding data analysis and identifying appropriate models during the design phase.

The highly distributed and interconnected data environment promised by the Internet of Things will also require practical “data fusion” methods for combining heterogeneous data types and sources in a streaming fashion. Only then will new technologies designed to exploit these data sources be able to do so. For instance, AI agents operating in real time will need to be

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situationally aware, acquiring and integrating multiple kinds of data generated by multiple kinds of sensors at multiple scales. In studying disease progression, data exist at the molecular, cellular, organ, person, and population levels, and these data evolve over different timescales (Alber et al. 2019). In climate and weather systems, it is well known that phenomena at different spatial and temporal scales all impact both long- and short-term behavior (Frank et al. 2024).

Any time AI is deployed, it is critical to ground its performance against known data. The technical performance and proliferation of machine learning algorithms threaten to outpace our understanding and management of their consequences. It is particularly critical, then, to validate AI technologies in the health domain, where decisions could make a difference between life and death. Responsible use dictates that black box models undergo experiments with gold-standard data to determine whether the answers provided fall within a reasonable range. However, validating AI models is still an open area of work (NASEM 2024b).

> Key Finding 5-4: Further research is needed to rigorously ground-truth and validate AI models.

More generally, as long as emerging technologies enable new ways to measure data, statistics will play a critical role. In many ways, today's AI is simply a vehicle in which machines use statistical theory and reasoning to solve problems at scale in (ideally) a self-supervised manner. A good example of this is within biology, where new data modalities such as spatial transcriptomics have required novel statistical frameworks to deal with phenomena like technical dropout due to limits such as inefficient mRNA capture, limited sequencing depth, and sample preparation biases. This causes a data analytic challenge, because it is difficult to distinguish true non-expressed gene counts from those that are zero simply due to technological inefficiency. Statistics offers the ability to deal with varying rates of zero-inflation and noise in these data, while the promise of AI foundation models is to utilize these principles to carry out downstream tasks (e.g., predicting which cell types make up a "spot" in a histopathology image). New statistical methods are needed to address challenges of modern AI systems including interpretability and accountability, hallucinations, quantification of uncertainty and exploration for foundation models, and causation for decision making. Until AI systems can reason over statistical principles to deploy new theory for unseen data modalities, this dependency on statisticians to push fields and technology forward will never go away.

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Conclusion 5-1: The rapid advancement and deployment of AI technologies call for continued and increasing contributions from the field of statistics. Immediate need and opportunity exist to invest in development and deployment of innovative statistical theory and methods to ensure accurate, reliable, and reproducible conclusions drawing from and using AI technologies, as well as to accelerate trustworthy scientific discovery by integrating AI and rigorous statistical principles. Statisticians bring unique expertise to evaluate and enhance the rigor, interpretability, and trustworthiness of AI models and AI-based research and application.

Particularly as models increase in complexity, such as in the case of hierarchical modeling or adaptive design, ways to measure robustness are lacking. Adaptive statistical design—a term that encompasses methodologies using both online and offline data to make sequential decisions for optimizing data collection for discovery, insight, and decision, sometimes in an autonomous way—is crucial across many scientific and engineering disciplines where data are expensive to collect, such as robotics, materials science and engineering, transportation, precision agriculture, resilient infrastructure, and advanced clinical trials. Various fields are developing frameworks toward addressing these types of questions, both in theory and in practice (e.g., sequential hypothesis testing, off-policy evaluation, bandits, and reinforcement learning). Funding for automated cloud-based laboratories to accelerate scientific and engineering discovery is increasing.

Conclusion 5-2: Interdisciplinary effort is greatly needed to build a more comprehensive understanding of principled methodologies for adaptive statistical design problems and their statistical performance at scale, with statisticians playing a critical role in these endeavors.

Conclusion 5-3: Adaptive and autonomous statistical design—methods designed to change sampling and simulation strategies in real time and in response to insights obtained through sequential learning—will be an important area for cross-agency collaboration. Scientific, engineering, industrial, and other domain-specific substantive problems will provide use cases to guide this research, and ultimately such capabilities can be expected to be implemented through AI agents. Advanced clinical trials and advances in the design of materials provide key examples of statistical innovation in this area.

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Multiple federal research agencies and organizations are poised to benefit from U.S. leadership in the space of adaptive and autonomous statistical design. The U.S. Department of Agriculture's breeding programs, for instance, will benefit from cross-disciplinary initiatives that provide adaptive statistical design and intervention. Plants grown in more productive areas of a field will tend to produce higher yield. Here, adaptive weather and geospatial models that quantify uncertainty of "growth" over seasons can help farmers make better breeding decisions (e.g., when to invest in certain types of soil, which hybrid lines to cross). As another example, the Department of Energy's national laboratories aim to identify new molecules and materials that (1) have less toxicity for human beings or (2) present less direct harm for the environment. Adaptive and autonomous statistical designs would increase the long-term efficacy of discoveries that national laboratories make as well as illuminate when these discoveries may need to adapt in response to changes in the surrounding conditions.

Another data source in the collaborative ecosystem is model output, either actual or virtual. Models encode knowledge and express hypotheses. Model output is actual if it is stored and virtual if the output is produced as it is needed. Either way, new technologies often rely in some way on models, especially on complex dynamical systems. For example, spacecraft navigation, aircraft control systems (autopilot), chemical process control, power-grid management, and biomedical devices all rely on dynamical systems models to detect, understand, and react to state changes in real time. They are crucial in the design of new technologies because they provide mechanisms to answer "what-if" questions as scientists and technologists explore different scenarios.

Hybrid statistical–scientific modeling capitalizes on the insights of scientific principles and mechanistic models in conjunction with data-driven analytics such as foundation models, taking advantage of all available information to further elucidate some phenomenon (NASEM 2025a). The interface between science, engineering, and statistics is expanding. Simultaneously, access to high-dimensional, multi-type data; output of physical, mathematical, and predictive models; measures from sensor arrays; and model-generated synthetic data from predictive models is increasing. Transdisciplinary collaborations allow hybrid development of data-generative models, analytic models, and predictive models, along with inferential systems for increasing knowledge, supporting decisions, and expanding understanding. However, as noted in the 2024 National Academies' report on digital twins, "[H]ybrid modeling approaches that

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combine data-driven mechanistic modeling approaches are a productive path forward for meeting the modeling needs of [complex systems such as] digital twins, but their effectiveness and practical use are limited by key gaps in theory and methods" (NASEM 2024b, 49).

> *Conclusion 5-4: Transdisciplinary collaborations allow hybrid development of data-generative models, analytic models, and predictive models, along with inferential systems for increasing knowledge, supporting decisions, and expanding understanding. Hybrid statistical–scientific models leverage available and synthetic data and known scientific principles to better understand complex phenomena.*

Finally, given that these data sources and capabilities will be placed in a highly distributed, decentralized environment, the design problem inherently includes questions related to trade-offs between cost and quality. Execution of code and movement of both data and code have costs in dollars and time. The quality of the results of a computation is a function of its uncertainty and accuracy. Better results often come at higher cost, but the cost of a unit improvement in quality is typically unknown. This gives rise to a host of important statistical problems from different vantage points. First, in an existing distributed system, what is the best way to deploy computational and data assets to minimize cost subject to a constraint on quality (or vice-versa)? Second, what is the best way to design a distributed system given an analysis objective? Third, given that a distributed computational environment will certainly develop organically and without central planning, how should the design of a scientific or technological advancement come to understand the characteristics and topology of that environment and utilize them to the maximum extent? Fourth, how should existing statistical and AI methods be modified to account for the distributed nature of data and computation?

> *Conclusion 5-5: As new decentralized technologies—such as the Internet of Things, sensing devices, autonomous vehicles, advanced biotechnology, and blockchain—continue to evolve, the availability and nature of data will continue to change rapidly, catalyzing a new frontier in distributed statistical inference. Traditional methods supporting data privacy also rely on decentralized data collection, storage, and analysis. These interacting motivations and constraints, among others, define an ongoing and evolving challenge to develop statistical methods appropriate for a distributed data setting.*

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Advances in computational modeling techniques such as digital twins are changing the landscape of technology design, enabling in-silico design and testing of prototype complex, dynamical systems. Statistical models as well as statistical optimization methods are increasingly the enablers of such techniques ranging from design and analysis of Monte Carlo simulations to sampling methods for diffusion models and Bayesian deep nets. New research in dimension reduction and optimization methods that can leverage the structural constraints and physics of digital models of complex systems and enable robust distributional analysis is needed to enable surrogate modeling and emulators of complex systems at scale.

BOX 5-2

Design of Experiments in Precision Agriculture

In the field of sustainable agriculture, statistical design plays an important part in identifying effective inputs and farming practices that may increase crop yields. Many statisticians explore how much nitrogen fertilizer should be applied to improve production. With too little nitrogen, the plants may not be as hardy. Too much nitrogen can lead to runoff in waterways and harm the local ecosystem. Traditional design of experiments (DOE) has been applied in agricultural settings since the early 20th century (NRC 1995; Telford 2007), but researchers also use statistical methods such as Bayesian hierarchical modeling to exploit observational data in their analysis.

DOE can be limited in agricultural settings. Farmers may be unwilling or unable to allocate multiple sites for blocking, and the outcomes do not necessarily generalize to broader settings or regions. Data-driven approaches rely on large-scale data sets such as crop insurance rating data, farmer survey responses, or satellite images, which capture observations on larger scales. However, because these data are not generated through a structured DOE framework, they are not comprehensive and are likely biased. For instance, crop insurance rating data may suffer from selection bias, since it will only capture data from farmers who opt in to the insurance programs.

Researchers are pursuing hybrid approaches that combine the well-identified effects of DOE with the grander scale of data-driven analysis and prediction. Data fusion methods might be used and weighted with the ground truth of structured experiments. The causal estimates developed through DOE may be applied to the observational data. In a panel discussion with the committee (NASEM 2025f), participants emphasized that a successful approach to

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experimental design in sustainable agriculture is only feasible with intentional interdisciplinary collaboration.

STAGE 3: DECISIONS

The field of statistics has long been associated with the optimization of decision making in the presence of uncertain observations, estimated parameters, and differing costs of mistakes. The classical distinction between a Type I error (rejecting a null hypothesis when it is actually true) and a Type II error (failing to reject a null hypothesis when it is not true) becomes starkly relevant when considering false positive and false negative tests in medical diagnostic screening (Waller and Levi 2020) and AI algorithms for automated résumé review and hiring recommendations (An et al. 2025; Caliskan et al. 2017). More specifically, the decision phase of a new scientific or technological advancement comes after a prototype or initial implementation is in place and explores, in depth, how the new knowledge or technology will be used by others. In a scientific context, the decision might be whether or to what extent a new theory is adopted or accepted by individual researchers all the way up to a whole community. In a technological context, the decision might be to certify that a new machine, AI agent, drug, or early warning system performs as claimed to within verifiable uncertainty. It is at this stage that questions around robustness and uncertainty quantification arise. How robust are decisions made under various assumptions and models to changes in those assumptions and models? Do the assumptions and models hold across different potential applications? How will a user know?

Digital twins provide a controlled laboratory environment in which to study interactions between models, data, and decisions. They allow users to isolate specific parts of that ecosystem for intensive study and to perform "what if" experiments to explore different scenarios under alternative scientific and technological designs. However, bringing digital twins at scale to fruition will require new theoretical and methodological developments beyond algorithmic advancements on existing methods. For instance, digital twins will require unique approaches to hybrid modeling of both data-driven and mechanistic modeling, resolving components of disparate scales, advanced sampling strategies for systems with large spatiotemporal variability, frameworks for data (and model) collaborations across disparate systems, and verification,

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validation, and uncertainty quantification that are embedded in a continual setting and dynamically adapt to changes across the digital twin components (NASEM 2024b).

Uncertainty quantification is an essential part of any scientific enterprise and any technological development enabling risk-aware decision making. The uncertainty quantification moniker has traditionally meant something specific in the applied mathematics community: the act of "accounting for all sources of uncertainty and quantifying the contributions of specific sources to the overall uncertainty" (NRC 2012). Of course, statistical models inherently carry uncertainty through the use of random variables. In 2012, the Society for Industrial and Applied Mathematics and the American Statistical Association jointly launched the *Journal of Uncertainty Quantification*, recognizing the important contributions from both communities. The communities have moved toward each other considerably in recent years as the advent of AI, machine learning, and data science is of intense interest to both. This has led to a more unified approach in which statistical and mathematical tools are extensively intertwined. Simulation-based inference (SBI) is a good example in which models (of any kind) are used to simulate data that stand in for complex likelihoods.

Hybrid formulations are often the foundation of uncertainty quantification. Statisticians working at this intersection think carefully about the transparency in assumptions, computational cost, and complexity of implementation, often addressing all frontiers with their contributions. Examples include (1) combining physical models on weather and scientific knowledge of fire expansion with real-time observations in a Bayesian hierarchical framework that supports real-time forecasting of the expansion of the edge of a fire, and (2) integrating generative AI foundation models within a statistics framework to provide sound statistical inference for questions in neuroscience. More hybrid methods will be needed as the underlying systems increase in complexity.

> *Conclusion 5-6: Hybrid formulations are often the foundation of uncertainty quantification. Statisticians working at this intersection think carefully about the transparency in assumptions, computational cost, and complexity of implementation—often addressing all frontiers with their contributions. A great opportunity exists to expand support for rigorous collaborations at this interface, with statisticians playing a leading or co-leading scientific role. The benefits span the breadth of science and engineering discovery.*

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From personalized medicine to recommendation policies to maps of natural-hazard risk, statistical output is embedded in societal decision making. Generative AI, such as large language models, offers an opportunity to advance the qualitative sciences by enabling systematic extraction, representation, and quantification of information from texts that capture human experiences. Realizing this potential requires the development of generative AI–enabled statistical methods that integrate statistical inference, uncertainty quantification, and bias assessment with AI-driven text representations. Continuing to translate research into real-world impact will require vigorous verification of new technologies to ensure they are robust, reliable, and enable new and beneficial discovery in the social sciences. Specific examples discussed in Chapter 2 include interpretable machine learning and algorithmic fairness.

A frontier in developing tools that will improve decision making is to leverage the social sciences to account for human dimensions in the deployment and vetting of new technologies (NASEM 2019a). One example is using our understanding of how humans learn to trust a technology, to improve the technology and its validation. Another example is moving from developing algorithms to optimize the performance of a task (e.g., diagnosing a disease) to developing algorithms that best support a human performing a task. This will require a framework for human learning and decision making, including identifying knowledge gaps and optimal modes of presentation.

As previously noted, uncertainty quantification, risk analysis, and SBI provide formal mechanisms for enabling informed, robust decision making. Traditional statistical procedures, including maximum likelihood and Bayesian estimation, prescribe functions that map data to parameter estimates. The functions typically involve numerical optimization (for the maximum likelihood estimation) or integration (for the posterior mean) that can be tedious to compute. Great effort has been made to improve computation with clever approximations and numerical algorithms. Rather than perform difficult computation on a given data set of interest, amortized inference (Ganguly et al. 2023) is a different paradigm that approximates the map from data to parameters offline, often using machine learning, that can then be applied virtually immediately when a data set is present. As discussed previously, SBI is another emerging example in which a neural Bayesian estimator approximates the posterior mean estimator by sampling from prior distribution and a data-generating model and then regresses the simulated parameters onto the simulated data using deep learning. Amortized inference fills important gaps in the classical

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statistical toolbox. Since most of the heavy computation is done offline, it is conducive to real-time analysis or distributed computing environments.

> Key Finding 5-5: Statistics could benefit from new research in amortized inference using AI and machine learning to conduct inference on larger scales and across distributed environments.

> Key Finding 5-6: SBI uses models to provide training data for statistical inference and machine learning predictions. Statistics and discipline-specific model understanding need to be used together to properly interpret the results of SBI.

Communication is paramount to incorporating statistics into decision making. Translating quantitative results into digestible next steps and policies is a skill statisticians need to cultivate. A key part of communication is visualization, an active area of research that is crucial to the evolution and utility of statistics. For example, *The Grammar of Graphics* (Wilkinson 2005) revolutionized the concept of the theory of information display, building off the influential books by Tufte (*Visual Display of Quantitative Information*) and the exploratory data analysis paradigm advanced by Tukey (1977), and motivated extended theory and practices for data visualization by statistical and data scientists. As an example, the field of spatial statistics benefited greatly from advances in displaying and interpreting information geographically.

STAGE 4: DEPLOYMENT AND SUSTAINMENT

The final phase in the development of a new scientific or technological advancement is deployment and sustainment. This is the point at which the prototype has been determined to be suitable for use by others, having proven itself to be a reliable tool for decision making. A scientific advancement may now be built upon by others, or a technological advancement may be offered commercially. The role of statistics becomes monitoring to ensure security, trust, reproducibility, and effectiveness in improving some aspect of scientific, commercial, or societal well-being. We can expect these new technologies to monitor themselves by producing and analyzing data on their own performance, and perhaps even taking action when a breach or other undesirable behavior occurs. The symbiotic relationship between statistical and scientific or technological advancement is on full display here, as the highly interconnected nature of these new systems leads to new data types and the need for new methods to handle them.

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Statistics plays a major role in both preventing and thwarting attacks after a technological advancement is deployed. Statistical analysis not only can monitor data logs for pattern matching with known attacks but also can be used for early detection of anomalies as well as for proactive study of the effect of new types of attacks and vulnerabilities. For example, for deployed AI models, distribution matching tests can alert about drifts and anomalies in data distributions, check robustness of model and explanations to various perturbations, as well as help quantify uncertainty and robustness of predictions under various metrics such as studying jailbreaking and hallucination in foundation models.

Reproducibility

Reproducibility is fundamental to science. The lack of reproducibility increasingly hampers scientific progress and serves to reduce trust in research in general and statistical conclusions in particular. While there are many causes for the lack of reproducibility in many studies, a key factor is the pressure to publish only statistically significant results, which often leads researchers to check for significance before deciding what to publish. Although many negative effects of this practice can be reduced through strict adherence to experimental and analysis protocols, a movement is underway to develop new theories and methods that recognize the motivation to publish significant findings while ensuring valid statistical inference at the process level (Ramdas et al. 2023).

As the field of statistics continues to evolve, having robust and reproducible frameworks ensures the highest form of scientific integrity and also enables transparent collaboration across sectors. Statisticians are leading efforts to formalize reproducibility standards. For example, in 2016, the *Journal of the American Statistical Association* (*JASA*) introduced an initiative that requires authors to include their code, data, and workflow details when submitting their work for publication. The journal also created a new editorial role, the associate editor of reproducibility, to implement these standards. Importantly, *JASA* has committed to upholding the same reproducibility criteria for all papers, regardless of whether they are written by authors in academia or industry—signifying that scientific publishing requires a certain level of transparency for the public. This effort reflects a growing recognition that reproducibility is not just a technical concern but a cultural shift in how research needs to be conducted, reported, and shared. Other scientific disciplines are attempting to overcome reproducibility challenges,

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especially with journals like *Nature* attempting to support better practices by implementing their own version of reporting standards. Even the National Institutes of Health's (NIH's) data sharing policy (NIH Grants and Funding n.d.) and the Generalist Repository Ecosystem Initiative are shaping the infrastructure for reproducible science. This situation creates a natural opportunity for statisticians to further unify reproducibility practices across sectors. Journals are certainly one medium to help establish robust change both within the profession and in the broader scientific community. Expanding groups like the associate editors of reproducibility at *JASA* or the Statistical Review Committee at the *Proceedings of the National Academy of Sciences of the United States of America* could be one avenue for such leadership.

As data repositories continue to grow, reproducibility will become more dependent on data versioning, which is a common practice already for code and models. Resources that are routinely updated, such as federal databases at the National Oceanic and Atmospheric Administration, Environmental Protection Agency, and U.S. Census Bureau or industry databases including CELLxGENE from the Chan Zuckerberg Initiative, the UK Biobank, and UniProt, highlight the importance of data versioning.

In parallel, the rise of AI models that are continuously trained in online settings also underscores the importance of robust model versioning. Here, each checkpoint of a model training run along with a description of the training data used, the hyperparameter values, and evaluation metrics deployed are also documented. Platforms such as Weights & Biases[11] and Hugging Face[12] will be increasingly important in enabling reproducible pipelines and model development. At the same time, reproducibility must be approached pragmatically. Interdisciplinary collaborations often span academia, industry, and government, each with distinct incentives and constraints. Open data and code can conflict with proprietary interests and privacy concerns. Flexible frameworks that balance openness with ethical and commercial realities are needed.

To support this shift, the statistical community might consider how it incentivizes reproducible work. Awards like the *JASA* Reproducibility Award are a step in the right direction, but broader recognition is needed. This is especially important for junior researchers contributing code, data curation, and infrastructure that may not result in traditional publications.

[11] The website for Weights & Biases is https://wandb.ai/site, accessed January 16, 2026.
[12] The website for Hugging Face is https://huggingface.co, accessed January 16, 2026.

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Finally, reproducibility is a vital part of statistics education. Future training ideally would include best practices such as thorough documentation, clean and informative READMEs, version control workflows, and strategies for working with restricted-access data. These skills will empower the next generation of statisticians to lead reproducible, impactful research in increasingly complex and collaborative environments. Without a robust pipeline of statisticians trained in uncertainty quantification and reproducible workflows, the United States risks falling behind in areas such as AI and biomedical innovation.

> *Conclusion 5-7: Research to assess the validity, accuracy, and reproducibility of models leads to higher-quality results, reduces bias, and decreases the risk of misinterpretation, and thus is critical for responsible decision making. Topics of particular interest include the development and evaluation of improved and robust measures of risk, fairness, trust, transparency, and accountability.*

> *Conclusion 5-8: The statistics community needs to recognize, support, and coordinate efforts focused on reproducibility and high-caliber statistical research. Recognition for these efforts includes publications, promotions, tenure, and scholarly awards and fellowships. The U.S. statistics community has led efforts to promote rigor and reproducibility in research and peer-reviewed publications, which become increasingly imperative with the proliferation of AI.* JASA *implemented a formal reproducibility review of all accepted submissions in 2016, guided by a dedicated reproducibility editorial team (Wrobel et al. 2024). Reproducibility has become a standard expectation across the spectrum of professional statistics journals. Leading scientific journals have adopted similar initiatives; for example,* Nature *and* Science *each have statistical review boards, and rigor and reproducibility are central to all NIH-funded research. Statisticians contribute to the broad corpus of research on advancing and using AI, while maintaining the fundamental, non-negotiable constructs of sound statistical practices.*

THE RISK TO SCIENCE AND TECHNOLOGY WITHOUT STATISTICS

This chapter illustrates the varied ways in which the statistical sciences undergird scientific and engineering innovation. Without the intentional structure the field affords statistical, machine learning, and AI tools, outcomes may be ineffective at best and harmful at

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worst (see Box 5-3). As outlined throughout the report, the rapid development of powerful AI models has enabled many advances in data science, such as using generative models to produce synthetic data or using AI models' predictions for high-stakes decisions such as hiring, and these advances must be paired with a foundational understanding of their statistical properties and potential for errors to ensure safe and fair use. Multiplicity and false positives in genomics are inflated in the absence of theoretically grounded understandings of inference and multiple testing. Such uncontrolled false positives can lead to misunderstood disease mechanisms or patient risk of a given condition. Extreme and rare events, such as might occur in weather or in financial markets, are, by definition, challenging to predict, and a rigorous statistical approach that can account for such extremes is critical. Wrongly predicting a catastrophic event can lead to wasted resources, and failing to accurately characterize a natural disaster may result in a lack of preparation or even the loss of human life. Heuristic notions of privacy and anonymization can be vulnerable to attack, and principled formulations of privacy (such as differential privacy) are critical for ensuring reliable guarantees to individuals' privacy when their data are used for research. Unregulated black box models have been used to predict criminal recidivism, which may inform pre-trial sentencing. Rigorous statistical thinking is an imperative for the deployment of real-world tools, and support for the statistical sciences will need to mature in tandem with technology.

BOX 5-3

Google Flu Trends

Google Flu Trends (GFT) went offline in 2015 after high-profile failures to characterize the prevalence of influenza-like illnesses (ILI) accurately. The idea behind the product was that Google would leverage its vast amounts of search query data to nowcast real-time influenza cases—bridging the 2-week lag of Centers for Disease Control and Prevention (CDC) data—at a more granular level than was currently available, an approach predicated on the notion that search frequency on specific terms (including ILI-related terms) is correlated with the proportion of visits to clinicians for ILI.

As a tool with the potential to guide public health decisions, GFT was markedly unsuccessful. GFT missed the 2009 A-H1N1 pandemic, a virus that resulted in 12,469 deaths, according to CDC estimates (CDC and NCIRD 2009). One post-mortem assessment suggested

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that, at the time, the algorithm was "part flu detector, part winter detector," thus ill-equipped to capture the nonseasonal virus (Lazer et al. 2014b). In a 108-week window from 2011 to 2013, the GFT numbers were higher than those of the CDC for 100 of them (Lazer et al. 2014a). At one point, the Google influenza estimates were almost double the official numbers (Butler 2013).

The GFT algorithms relied on opportunistic data not fit for purpose. Furthermore, flu-like symptoms are not exclusive to ILI, and early assessments suggested that media coverage of the flu influenced search frequency. The models were not recalibrated against updated CDC data over time and thus did not capture temporal shifts. Moreover, midway through GFT's life cycle, Google's search algorithms underwent changes that impacted the queries; these changes were never introduced in the flu trends models, underscoring the need for investment in continuous data validation. GFT development and deployment failed to follow statistics' best practices and validation. At a minimum, these sorts of unchecked approaches toward analytics undermine the trust of the public and decision makers in data-informed decision making.

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6
Statistics Education, Training, and Workforce Development

As statistics enters an era shaped by the rise of artificial intelligence (AI) and progressively interconnected or interoperable data products, statisticians must critically view, evaluate, and expand the frontiers of statistics education and workforce development to support and enhance the dynamic theoretical, methodological, and collaborative frontiers detailed in the preceding chapters. Statistics must continue to modernize its talent pipeline by equipping a workforce to handle evolving data and to confront the questions that result from this evolution. At the same time, a clear need exists to expand data literacy and data acumen in the professional workforce and in the general public.[13] Statistics is well positioned to lead these efforts and advance statistical thinking as data increasingly inform decision making in science and engineering. Such statistical thinking helps people accurately interpret, trust, and effectively employ autonomous tools based on data, algorithms, and machine learning.

Statistics education reform will have to leverage existing and emerging interdisciplinary connections to remain responsive to accelerating technological advancements and emerging needs of data-driven innovations. Building on a legacy of success and strong educational infrastructure (e.g., Bargagliotti et al. 2021), statistics education will need to capture the field's adaption to data availability and structure and the rapid expansion of foundational methodologies and theories. Emerging directions in statistics education provide opportunities for the community to contribute meaningfully through leadership, collaboration, and creation.

This consensus study highlights the ongoing interplay between statistical theory, methods, applications, and interdisciplinary collaborations, bringing forward a continuum perspective of the progression of the field of statistics. This continuum is reflected in advances in the statistics education spectrum. As new methodologies emerge to solve new problems, a challenge for the field is the timely integration of new inference paradigms and critical advances

[13] See, for example, NASEM (2024a) and The White House (2025).

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in statistical thinking into the statistics curriculum. The committee views curricular advances in the same light, extending and including the translation of new statistical paradigms, with associated broad applications, into the education pipeline. Earlier examples of statistical breakthroughs including Bayesian hierarchical modeling, adaptive experimental and clinical design, false discovery rate control, and causal inference are all finding places in the statistical curricula. These important statistical inference methodologies provide a wide range of statistical solutions with links to compelling but challenging collaborative applications, making them suitable to be included in advanced undergraduate curricula and folded into core knowledge at the graduate level. The continual evolution of statistics curriculum is essential to meet, support, and enhance the expanding needs of science and engineering.

This chapter outlines key elements of a vision for statistics education across the pipeline in the next decade—elements based on the significant research innovations and ever-changing trends and dynamism reported in earlier chapters. While statistics education has been grounded in rigor and deep reasoning about data and experiments, it can be difficult for its traditional structures and curricula to effectively and efficiently adapt in a timely manner to grow with the needs of learners and the rapidly evolving discipline (Gibbs and Taback 2021). Potential challenges to continued success and growth include the temptations for an overly expansive core curricula with numerous prerequisite courses required in advance of training in emerging, high-demand methods of current interest; limited integration of the computing (Horton and Hardin 2021) and real-world applications driving the frontiers (Raman et al. 2023; Wright et al. 2024); limited collaboration with industry; and fragmented statistical training across disciplines (Alzen et al. 2024).

Key to motivating the frontiers of statistical training is the notion that the field of statistics is dynamic. In other words, the statistical toolbox is not a finite set of tools; rather, it is a fluid set that allows core statistical concepts of uncertainty, estimation, inference, and modeling to grow to meet challenges presented at the frontiers of science and engineering. The toolset is ever-growing, and a forward-looking approach to statistical training will involve motivating trainees to think statistically, not training them how to “do statistics” (e.g., DeLuca et al. 2025). An important aspect of such education is teaching learners how to identify problems worth solving. Statistical education of the future might cultivate an engineering mindset to gain intuition about pressing problems that matter and balance elegant solutions with approaches that

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scale to real-world implementation (NASEM 2025d). The American Statistical Association's (ASA's) *Statistical Education of Teachers* reports (*SET* and the in-progress *SET II*) offer guidance for the development of teachers of statistics (Franklin et al. 2015).

> *Conclusion 6-1: A forward-looking approach to statistical training would incorporate core statistical concepts and teach students how to think statistically, as well as learn which problems are worth solving, rather than rely on a finite and dated set of tools. Training goes beyond black box implementation and supports the deep understanding of the evolving tools in the toolset.*

The sections below outline several frontiers in statistics education and training aimed at modernizing statistical training by enhancing statistical thinking, embedding computing and ethics into all levels of learning, and strengthening nationwide academic ties with industry, government, and nongovernmental agencies. This chapter highlights priorities for coordinated efforts among professional societies, funding agencies, and academic departments that would advance an evolving educational ecosystem and cement the discipline's central role in shaping a future where data and AI are used effectively and responsibly to enhance U.S. competitiveness in science and engineering.

STATISTICS EDUCATION: OPPORTUNITIES AND SIGNIFICANCE FOR 2035

Statistics education focuses on equipping a specialized workforce with analytical reasoning and a strong foundation in theory and methods for solving problems that involve the collection and analysis of data—skills essential for extracting insights for evidence-based decision making. Increasingly, these skills are shaped by the changing nature of data, AI and machine learning, and other technological breakthroughs, which is cause for excitement but also concern over the potential for unintended consequences (Bargagliotti et al. 2020).

Data, AI, and Statistical Thinking

As discussed in Chapter 3, the proliferation, interconnection, and evolution of data; the rapid development of computation and machine learning models; and the integration of AI into daily life are transforming nearly every discipline and sector of society. The transformative potential of AI and automation in science and engineering calls for a workforce that not only

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uses such tools but also develops and deploys ethical and trustworthy AI technology across sectors. Furthermore, as data-based AI tools become increasingly embedded in everyday life, individuals must be able to engage with such systems thoughtfully and responsibly, treating them as a powerful yet fallible set of tools not fit for every purpose. Instead of passively adopting AI technologies, users need to understand their strengths and limitations, question and evaluate outputs, recognize potential biases, and make informed decisions in light of uncertainty (NIST AI 2023).

> *Conclusion 6-2: An AI-savvy workforce will not merely adopt these tools but will understand the strengths and limitations of AI, thoughtfully evaluate model outputs, recognize potential biases, and incorporate awareness of uncertainty into its decision making.*

In this context, being AI-savvy does not refer to being able to understand the inner workings of AI but rather to being able to know when and how to use AI in decision making. National conversations call for systematic integration of education in AI concepts and tools across all levels of learning (The White House 2025). Important considerations include preparing the population for an increasingly data-driven world (NASEM 2018), equipping AI-users in data-driven environments (NIST AI 2023), enabling AI-ready leaders and a workforce across all sectors (NASEM 2025b), and accelerating AI research for solving grand challenges.

Statistics Education Supports the Needs for Interpretable, Reliable, and Transparent Solutions

Statistical thinking—a mode of thinking that recognizes and incorporates uncertainty into decision making—is indispensable at every stage of today's data-driven technological innovation pipelines: discovery, design, decision, deployment, and sustainment (Marshman and Dunn 2023). The needs for statistical principles, including questioning of core assumptions, and methods vary across each of these stages, as outlined in Chapter 5. Statistical approaches guide study design and data collection and processing in the discovery stage. During the design phase, statistics connects disparate data, extracts information from the data, and enhances computing. Statistics helps set and question modeling assumptions, structure uncertainty quantification, and mitigate biases as part of decision making. End users for such data-driven solutions must be able to understand the values and limitations of these tools, interpret the results, and make informed decisions. At the deployment stage, statistical principles help guide evaluation of a model's

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privacy guarantees, trustworthiness, security, and reproducibility. Furthermore, statisticians bring an important skill set regarding how best to use data to ground-truth the decision process behind AI-supported recommendations. The final layer is arguably the most important. Without it, it is difficult to ensure the rigor and accountability of the earlier stages.

As a result, statistical thinking is requisite throughout the development and use of data and AI. AI education will benefit by increasing its focus on equipping the public with statistical thinking as a foundation for working with AI, specifically, by expanding training in understanding data, uncertainty, inference, and causality-based and prediction-based decision making (Friedrich et al. 2022). This moment presents great opportunities for the statistical community to lead and to thoughtfully expand the reach of statistics education in response to the growing role of data and AI across society.

Statistics offers formal theory, methodological frameworks, and a set of guiding principles for identifying and correcting biases in data and algorithms, quantifying (structured and often complex) uncertainty, and supporting sound knowledge discovery and inference in the presence of uncertainty. Statistics education has a critical role to play in meeting the urgent need for training that supports the ethical, interpretable, reliable development and use of AI and data-driven solutions. The advancement of the technical performance of machine learning algorithms threatens to outpace our understanding and management of their consequences. These concerns are grounded in real risks and trade-offs requiring that we understand and quantify bias and uncertainty. The statistics discipline can expand its role in defining and leading the mission of disseminating principled and trust-building approaches across the AI innovation ecosystem through education and collaboration.

The statistics education community faces both opportunities and challenges. On one hand, advancement in science requires the statistics community to broaden its education and engagement (Sanchez 2025). On the other hand, such efforts must be tailored to different audiences, scalable, and grounded in core statistical principles.

Conclusion 6-3: The need for embedding statistical thinking into AI education is both urgent and rapidly increasing, especially as AI tools are adapted and deployed within data-driven workflows.

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Current Statistical Curricula: Gaps, Challenges, and Opportunities

Current statistical curricula have strong foundations. However, the current structure must continue to evolve to address the changing nature of data, include learners from different backgrounds, better bridge and expand statistical training for individuals in other disciplines, and continue to expand engagements with industry and non-academic entities.

Curricula, Pedagogy, and Structure

A key strength of statistics education lies in its commitment to a formal and principled framework. The field of statistics, grounded in mathematical reasoning and probabilistic thinking, shapes a pedagogical tradition that emphasizes clarity, rigor, and conceptual understanding. Across institutions and levels, effective statistical training emphasizes critical thinking about data, over mechanical application of formulas or definitions, and provides opportunities for engagement with real-world data problems. As stated above, it is more important to be able to think statistically than to "do statistics."

The most effective statistical pedagogy considers how statistical curricula support the needs of learners (Franklin et al. 2005; NASEM 2018). For instance, if an introductory course, while well organized and logically constructed, mirrors more advanced courses in design, differing only in the level of mathematical sophistication expected, the field will likely fail to appeal to students seeking more of an appreciation for and literacy in data and statistics. Students may find these courses overly abstract or technical, missing an opportunity to see statistical ideas as meaningful concepts to address real-world questions as well as opportunities for necessary discussions of data quality, contextual decision making in the face of uncertainty, the iterative nature of applied work, and the importance of mutually beneficial interdisciplinary collaboration.

Integrating Statistics Education Across Fields

The excitement about data science and machine learning across all scientific disciplines and professional sectors motivates ongoing investments in data science curricular development (NASEM 2018, 2020, 2023). ABET, a global engineering accreditation entity, began accrediting undergraduate data science academic programs in the 2022–2023 cycle through CSAB (formerly titled the Computing Sciences Accreditation Board), an ABET member society representing the Association for Computing Machinery (ACM), ASA, the Institute for Operations Research and Management Science, and the Institute of Electrical and Electronics Engineers Computer

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Society). Training in statistics, machine learning, and data science has become an essential component of professional education in science and engineering (CSAB 2021). In fact, there are numerous free opportunities for self-guided learning in this space—for example, the Massachusetts Institute of Technology's Open Learning offerings (MIT Open Learning 2024). However, the structure and ownership of such training vary significantly, and the opportunities for essential statistical thinking connecting concepts and applications may be limited.

In many fields, statistical content is embedded within domain-specific curricula, often taught with field-specific language and tailored to disciplinary norms. While this contextualization makes the training relevant to downstream work within a field, it can also create silos; methodological researchers may find it difficult to make connections with those in science and engineering fields, and the current workforce in a particular discipline may struggle to adopt the latest statistical and machine learning advances to address emerging data-based challenges. Such fragmentation of statistics education and separation between fields with strong potential for collaboration can limit translation between fields and pose inefficient use of training resources, missed opportunities for collaborations, delayed innovation, and potential disconnects across both academia and industry. Conversely, interdisciplinary courses that are developed around fundamental concepts—such as information, uncertainty, and robustness—and are team-taught can present different points of view, and these types of courses may have the potential to attract a variety of students from different disciplines and cultivate collaboration.

Statistics education, in collaboration with domain disciplines, can help move the training of applied scientists and engineers beyond the current focus on simply using statistical methods and machine learning tools toward understanding what answers and insights these tools provide. Training should be grounded in the active intersections between statistical thinking and domain theory: how to assess uncertainty, how to integrate domain knowledge, how to understand and evaluate limitations, and how to interpret results in context.

Conclusion 6-4: Recent advances by ABET and CSAB, of which ASA is a member, established accreditation criteria for undergraduate data science programs, which went into practice in the 2022–2023 cycle. These criteria recognize the importance of contributions across computing, statistics, and applied mathematics. ABET—through CSAB—is also pursuing criteria for master's programs in data science as well as a breadth

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of AI-focused programs. The criteria for these programs clearly articulate the important role of statistics and statistical thinking.

Experiential Learning and Collaboration Across Industry, Government, and Nongovernmental Agencies

Statistics curricula are strengthened by expanded collaboration with motivating challenges from industry, government, and nongovernmental agencies, fostering a culture and infrastructure of entrepreneurship and advanced problem-solving, as well as synergizing translation of research to practice (i.e., catalyzing links between the frontiers of statistics with the frontiers of excellence in science and engineering, the importance of which is emphasized in Chapter 4: Statistics Across the Spectrum). Within statistical training environments, while individual faculty may engage with industry or the public sector through consulting or research collaborations, these efforts would benefit from becoming more formal components of systematic departmental strategies. Many industries lead innovations in AI and data science and apply statistical methods at scale and in novel ways.

As they have for some time, current graduates in statistics pursue exciting and rewarding research careers at the frontiers of the technology, finance, and pharmaceutical industries or government agencies, in addition to career paths to academia. However, the transition from training to careers at the frontiers of science and engineering could be eased through better alignment with career pipeline needs and career flexibility. Students seek statistics curricula that are connected to the skills necessary for a modern data science and machine learning career across all sectors. In response, statistics education benefits from the inclusion of deliberate efforts and structured mechanisms to systematically engage with industry, national laboratories, government agencies, and other non-academic entities.

It is important to note that there is no “one-size-fits-all” approach to student programs promoting interactions beyond academia. Successful university programs identify partnerships that best meet the needs of their students (Gutierrez et al. 2025), align with available educational resources, and leverage local partnerships and impact as well as national connections. Statistics departments across the country employ several useful mechanisms by which their students at all levels benefit from enhanced non-academic ties. One mechanism is to have a subset of faculty with active external ties, such as Amazon Scholars (Amazon Science n.d.) or laboratory–university partnerships such as the Department of Energy Laboratory Directed Research and

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Development program (NASEM 2025b). These faculty then engage students in external research projects and target their teaching and mentoring to better prepare students to address non-academic expectations and challenges while simultaneously educating colleagues. Another mechanism is to encourage non-academic partners to provide case studies for students. Case studies are incorporated in consulting courses and in dedicated summer programs such as Research Experiences for Undergraduate (REUs) students. These examples demonstrate that students can gain relevant experience and strengthen the pipelines from academic departments to industry internships and jobs.

Faculty may also engage in collaborative research with industry partners. Such efforts are similar to academic collaborations, but they can also bring substantial logistical challenges (Rossoni et al. 2024). Challenges include different timelines to reach objectives as well as a need for clear negotiated agreements regarding data access, data use, and intellectual property in order to facilitate access to and use of industry infrastructure, code, data, and algorithms. Addressing the challenges requires careful thought and implementation. Successful examples exist where institutions offer joint advising with industry professionals for graduate students. Again, this mechanism is similar to faculty collaborating to support doctoral education on new topics. Faculty sabbaticals spent in industry or non-academic environments afford the research community yet another opportunity to connect on real-world challenges. Finally, statistics departments increasingly engage industry and non-academic members on their program advisory boards, such as for a professional master's degree. Non-academic professionals can offer insights into curricula or program directions that help graduates improve their employability and contributions to the science and engineering ecosystem. The impact from these examples increases if departments cultivate a collaborative environment in which students work together and discuss experiences, and faculty contributions are recognized.

It is worth articulating a few success stories where students benefitted from strong opportunities for experiential learning in collaboration with the public sector. The first program highlighted is the National Data Mine effort led by Dr. Mark Ward from Purdue University. This program provides academic-year stipends and course credit for undergraduate students pursuing data science education, and graduate student mentors, to interact with a vast array of industrial, governmental, and nongovernmental agency partners. Strongly supported by Purdue University and the National Science Foundation (NSF), this successful program for Purdue undergraduates

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was expanded to a national footprint in 2022. The National Data Mine Network is managed by ASA in partnership with Purdue and supported by NSF (Eikenberry and Samara n.d.). The Purdue Data Mine has approximately 100 partnerships each year, drawing students from approximately 60 colleges and universities. During the past 3 years, as the Data Mine launched the National Data Mine Network, this has included students from more than 30 historically Black colleges and universities, Hispanic-serving institutions, and minority-serving institutions each year. The Purdue Data Mine has engaged more than 6,000 students since 2018, including more than 2,000 graduate and undergraduate students participating in 2025 (Betz et al. 2024). This longstanding visionary program is an exemplar of innovative implementation that brings statistics and data science education and training to real-world challenges at a national scale.

Similar programs exist at other universities to serve their students (Allen 2021). An excellent example of a modernized partnership structure is the Department of Statistics and Data Science at Carnegie Mellon University and Google Public Sector. As part of this relationship, the school has received a large cloud-based graphical processing unit cluster, positioning itself as a leader in university cloud-based research computing (CMU 2025). Washington State University's recently established PhD program in statistical science encourages students to create industry connections and to pursue internships (WSU n.d.). The Rice University Department of Statistics hosts the Center for Computational Finance and Economic Systems (CoFES), which actively promotes internships across all degree programs in the area of quantitative finance. Recently Rice and CoFES have experimented with externships, where students spend academic time with approved industry partners and are required to write an in-depth project report. Another example of experiential learning includes the Data to Knowledge or D2K program at Rice University. In addition to university-led programs, professional societies provide learning opportunities through data challenges, such as the long running ASA DataFest and the well-known Kaggle competitions. These programs are highly valuable for engaged students and non-academic partners.

Experiential learning can be resource-intensive. However, there are other strategies to bring experiential learning into the classroom. Case studies are exceptional ways to educate across a spectrum of statistics and data science methodologies (Matuk et al. 2022). Some recent advances in classroom-based extensions of experiential learning include the "just-in-time-

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learning" model, or flipped classroom, where students are immediately assessed on experiential learning opportunities (McGee et al. 2016).

The field of statistics has a rich history of advancing undergraduate and graduate education through NSF-, the National Security Agency (NSA)-, and the National Institutes of Health (NIH)-sponsored REUs and similar programs. Participating undergraduates are introduced to the field of statistics and its related career opportunities, while current graduate students in hosting departments further their practical knowledge by assisting with these programs as mentors and actively leading research and experiential learning projects. Such programs bring together students from across the country, generally over the summer, for experiential learning and research opportunities. The long-running NSF Division of Mathematical Sciences Vertical Integration of Research and Education in the Mathematical Sciences program offered universities opportunities to enhance the experiential learning culture for their students, providing significant transformations across the statistics educational ecosystem (NRC 2009). Similarly, the Summer Institutes for Training in Biostatistics (SIBS) sponsored by the National Heart, Lung, and Blood Institute also have a long history of recruiting and motivating tomorrow's scholars, researchers, and professionals, with many successful biostatisticians pointing to SIBS as the motivation for their careers (Sullivan et al. 2013). Such programs represent a low-cost, high-reward investment in recruiting the next generation of individuals working at the frontiers of statistics in science and engineering.

Conclusion 6-5: Investments from agencies such as NSF, NIH, and NSA in undergraduate and graduate training in statistics, biostatistics, and data science have had tangible impacts on U.S. competitiveness in science, engineering, and medicine.

The number of internship and externship opportunities for statistics students across industries as well as national laboratories is growing. A few notable examples include the following:

- The Artificial Intelligence (AI) Residency Program at Meta is a 1-year research training position designed to provide hands-on experience with AI research while working in AI at Meta.

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- The Predoctoral Research Assistant Program at Microsoft Research (MSR) is a program at MSR New England and New York City designed to offer recent graduates research experience prior to pursuing a PhD.
- The National Aeronautics and Space Administration's Jet Propulsion Laboratory (JPL) offers a 10-week summer internship for full-time students at all levels across science, technology, engineering, or mathematics. Students are partnered with JPL scientists, and statisticians are actively recruited.
- The Google AI Residency Program is a 12-month research training role designed to jump-start or advance careers in machine learning research.
- The Statistics and Quantitative Sciences Internship Program at Takeda Pharmaceuticals is a 10–12-week summer opportunity where applicants get the opportunity to develop predictive models, perform high-dimensional data analysis, design efficient trials, and contribute to clinical development strategies happening across the company.
- The Statistics and AI/ML [Machine Learning] Summer Internship Program jointly managed by Pfizer and Atrium Works offers current graduate students the opportunity to work on impactful biomedical projects during the summer while gaining valuable exposure to the pharmaceutical industry.
- The Intramural Research Program at NIH offers college, graduate, and professional students interested in exploring careers in research and health care the opportunity to work directly with principal investigators conducting basic, translational, and clinical research. Students interested in both biostatistics and statistics are encouraged to apply.
- Most financial companies offer summer internships for students at all levels. Statisticians interested in quantitative financial markets and insurance are actively sought for such opportunities.
- Biopharmaceutical companies and NIH often offer summer internship opportunities for students who are interested in pursuing biological and health sciences.

Looking beyond academia, ASA has strong membership of professional statisticians from academia, industry, and government. The Biopharmaceutical Section of ASA has a long history

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of cross-fertilization between academia and industry. The Institute of Mathematical Statistics boasts substantive collaborations with ACM. The Eastern North American Region and Western North American Region of the International Biometric Society bring together thought leaders from all sectors around biostatistics, bioinformatics, epidemiology, and public health. The annual Joint Statistical Meetings of global statisticians—which includes the professional societies listed above—is the largest gathering of professional statisticians in North America. The associations maintain consistent and effective outreach and mentoring to students and young professionals and provide extensive opportunities for ongoing professional development. This culture of cross-fertilization across sectors supports the transition of statistical innovation between academia and the public sector in science and engineering.

Conclusion 6-6: The collaborative nature of statistics highlights the importance of ongoing interaction between researchers in science, engineering, medicine, and statistics, especially at the interface of method and application. Training for such an interactive research environment is best encountered through experiential learning and hands-on collaborations. The statistics education community highlights many successful examples of such programs.

FRONTIERS OF STATISTICS EDUCATION: WHAT AND HOW

As discussed above, statistics education will continue to mature in what we teach, how we teach it, and how we build the necessary ecosystems and partnerships necessary to sustain the discipline's leadership and contributions in today's science and engineering environment.

Modernizing Core Statistics Education

To meet the demands of a rapidly evolving landscape, core statistics education must undergo meaningful modernization, which can be drawn from recent important recommendations regarding data science training (Bargagliotti et al. 2021; CSAB 2021; NASEM 2018, 2022).

Many traditional curricula, especially at the doctoral level, require students to progress through multiple series of foundational courses, which can discourage exploration and delay exposure to and engagement with the frontiers of theoretical, methodological, and applied

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collaborative research. While each department is different and must rely on its strengths to serve its community effectively, some departments are defining a smaller and essential core set of courses to build fluency in fundamental concepts of probability, inference, and modeling (Carver et al. 2016; He et al. 2019). The smaller core focuses on key foundational mathematical statistical methodological principles that are necessary building blocks for developing statistical thinking. Such an approach grants students flexibility to pursue specialized pathways earlier in their education, driven by their research interests. Statistics departments should continually reflect on the size and structure of their core curriculum to ensure it addresses both the needs of the future workforce and the changing nature of the field, particularly as the field adapts to new demands, theoretical developments, data sources and types, and technological advances.

Educators and researchers also are increasingly recognizing the importance of integrating computing into statistics curricula (Horton and Hardin 2021), including integration with high-performance computing (Abdulah et al. 2025; NASEM 2025e) and cloud-based platforms. With the availability of user-friendly computational ecosystems that utilize AI programming agents, the technical barriers to developing methods and tools targeting complicated data products have been reduced significantly. Incorporating software environments such as those used in industry or government laboratories also shortens the distance between training and practice. Curricula might include large-scale applications and courses on deep learning, statistical machine learning, and causal inference while linking the rigor of statistical thinking for complicated data collection and adaptive experimental design methodologies. Moreover, coursework will ideally include programming best practices: producing quality code as well as engaging students in code testing, reproducibility, version control, collaborative code review, and project management. The intention is not to convert statistics students into software engineers but rather for them to develop programming agility and a certain level of computing expertise. Statistics programs increasingly include computation-intensive, hands-on learning experiences throughout the curricula, as noted above.

Conclusion 6-7: Large-scale computational training for future statisticians is vital yet remains a gap in the educational curriculum.

Ethics concerning data and algorithms have a natural place in statistical core curricula and are increasingly taught not as an abstract add-on but as an integral part of any statistics and data science workflow and associated decisions. Students are increasingly learning to examine

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these decisions as part of data collection, problem setup, model assumptions, algorithm design, and visualization choices and interpretation, for instance (Baumer et al. 2022; Ellenberg 1983; Hurwitz and Gardenier 2012; Raman et al. 2023).

Statisticians contribute strongly to the science behind the use of machine learning. For example, the "population, question of interest, representativeness of training data, and scrutiny of results" framework spearheaded by Bin Yu and Karl Kumbier offers such a framework, where ethics, or the responsibility of a statistician when producing data-driven results, can be practically scaffolded into the data science curriculum through the core principles of predictability, computability, and stability (PCS) (Yu and Kumbier 2018). At each stage of the data science life cycle, decisions, or "judgment calls," can introduce uncertainty of ethical importance. A focus on scrutiny of results requires that students not only perform these decisions but also run PCS-guided evaluative checks. In particular, predictability evaluation concerns whether the results generalize to new data scenarios in alignment with domain expectations, and reflect robust patterns, not artifacts in the training data or algorithmic overfitting. Stability evaluation invites students to stress-test their entire workflow under reasonable perturbations such as alternative cleaning methods, variations in feature selection, or modeling choices. When conclusions shift under such perturbations, the deployment of such data-driven results must be carefully considered. Ensuring that every stage of the workflow is fully documented, reproducible, and transparent via both code and narrative sets a foundation of accountability and trust.

The statistics education community can expand its contributions to the co-development of curricula with professional disciplines. To scale such efforts, this community needs to support ongoing curricular evolution to continually update, modularize, and standardize core statistics education building blocks for the AI era (Gibbs and Taback 2021), to apply across disciplines and levels of expertise. These blocks integrate into professional education programs, as part of a tailored training adapting to the specific needs of future scientists, engineers, doctors, lawyers, policy makers, and business leaders.

Scaling such efforts requires expanded collaboration across all the data, computation, and machine learning disciplines, including but not limited to computer science, data science, optimization, information theory, and systems science. Statistics educators increasingly lead and

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participate in organized and cross-disciplinary partnerships for institutional educational infrastructure, faculty development, and innovative curriculum designs.

Conclusion 6-8: There are opportunities for cohesive discussion around updating graduate training to keep pace with technological progress and attract the next generation of the statistical workforce.

Professional societies could serve an important role in facilitating these conversations.

Conclusion 6-9: Statistics curricula and training need continual modernization to keep pace with the evolution of the field. As part of this update, statistics education would benefit from a formal infrastructure for educational innovation and professional development, as well as large-scale partnerships between disciplines and sectors. Such an effort could be modeled after the successful Vertical Integration of Research Education in the Mathematical Sciences and Research Training Group programs.

Conclusion 6-10: Academic statistics departments and departments that offer statistics degrees have started rethinking and adapting their programs to embrace new opportunities. To ensure long-term success, these departments will have to demonstrate agility to rapid technological change and the growing demands of scientific disciplines for statistical innovation. It is vital that statistics sustains forward-looking disciplinary renewal to meet the challenges of today's research and technology advances. Equally important is active participation in efforts to develop a workforce prepared for the responsible and ethical development and deployment of data-driven solutions.

Leading in AI Education Efforts

As society becomes progressively driven by data and algorithms, opportunities abound for the statistical community to increase its leadership roles in educating individuals across all stages of learning, especially including support of AI education broadly, from reskilling the current workforce through online education, to collaborating with K–12 educators, to contributing to national and local AI curricula discussions on data literacy, ethics, and algorithmic bias. Statisticians and statistics departments can, should, and will continue to provide expertise on uncertainty, bias, and rigorous inference through partnership and co-design efforts. Such efforts should be considered core to statistics education.

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Building Infrastructure, Partnerships, and Ecosystems for Sustainable Leadership

Expanding the frontiers of statistics education to meet today's societal demands requires more than individual local initiatives and experimentation. It calls for coordinated, well-resourced, and community-driven efforts across institutions and sectors. At the foundation of these efforts is the need to support discipline-level initiatives to continually reimagine the statistical core curriculum through reduction, standardization, and modularization. Professional societies and funding agencies can catalyze and enhance such initiatives by supporting broad buy-in from academic departments. Maintaining discipline-level support enables coordination moving beyond siloed departmental reforms and drives ongoing rapid and broad progress across theory and application. A focused consortium could organize this national effort, with early commitments from participating departments and institutional leadership. Seed funding can support initiatives in departments to implement curricular reform locally in alignment with the shared consortium goals. NSF's support for ASA to advance the National Data Mine Network in partnership with Purdue University is an example of how such a consortium could work and an illustration of the successes enabled by such efforts.

To sustain the continuing evolution of statistical curricula in response to the drivers of statistical frontiers outlined in previous chapters, establishing a community of practice for ongoing curricular renewal can actively integrate statistics education with current scientific research and industrial applications. Instead of simply expanding existing curricula through additional statistical courses, such integration more effectively focuses on flexible redesign and substitution. For example, increasing intentional opportunities for experiential learning (i.e., "learning by doing") within existing courses can weave research-inspired data challenges into ongoing coursework. Open curriculum initiatives such as the Open Climate Curriculum[14] allow the community to share syllabi and reusable course materials, accelerate everyone's progress, and reduce redundant development efforts. Professional societies (e.g., ASA's Section on Statistics and Data Science Education) can support such a community in multiple ways, including recruitment of participants and dissemination of effective practices.

[14] The website for the Open Climate Curriculum is https://openclimatecurriculum.org, accessed January 16, 2026.

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Key Finding 6-1: Professional societies play a critical role in supporting and amplifying these efforts by energizing and mobilizing the community through opinion pieces, online forums, policy monitoring, and conferences.

To institutionalize and support such change, mechanisms for sustaining public sector engagement and routine curriculum updates are key (Chamely-Wiik et al. 2019; Weeks and Harbor 2014, 2). Universities and departments can extend existing relationships, linking collaborators and alumni who are successful professionals and entrepreneurs in technology, finance, health, government, and so on. Industry advisory boards focusing on curricular design can help departments update curricula effectively and can provide support, resources, and project mentors while benefiting from graduates trained in skills they can put to use immediately upon graduation. Departmental incentive structures and academic policies should explicitly recognize the importance of faculty engagement in this area, which has traditionally been undervalued. Departments should integrate participation in such activities and initiatives into core faculty responsibilities and funding decisions—for example, by rotating service assignments, funded fellowships for both faculty and graduate students, or well-resourced education leadership roles. Moreover, success in these areas should be meaningfully incorporated into promotion, tenure, and annual review processes as evidence of educational innovation, community engagement, and real-world impact. Without adequate incentives and supportive infrastructure, even the most promising and exciting reform could quickly lose momentum.

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7
Recommendations

This chapter introduces the report's recommendations and includes the supporting conclusions and key findings from throughout the report. Given the scope of important and competing initiatives that agencies and institutions must balance, and in recognition of a dynamic funding landscape, the committee offers the roadmap below for recommendation implementation. This prioritization is based both on urgency and need as well as ease of adoption (i.e., which of the below should be prioritized regardless of—or despite—fluctuations in funding).

1. Recommendation 7-3 (artificial intelligence [AI]) National Science Foundation (NSF) and all agencies
2. Recommendation 7-1 (statistical foundations and innovation) NSF
3. Recommendation 7-2 (statistical foundations and innovation) National Institutes of Health (NIH)
4. Recommendation 7-7 (public–private partnerships) NSF, NIH, and all agencies
5. Recommendation 7-8 (academic reform) Academic departments and journals
6. Recommendation 7-4 (hybrid models) All agencies
7. Recommendation 7-5 (adaptive design) NSF, NIH, and all agencies
8. Recommendation 7-6 (Research Training Group) NSF and NIH

FUTURE OF STATISTICS

Key Finding 2-5: The foundations of statistics consist of theory, methodology, applications, training, and education, with the practice of statistics serving as a central driver of its development and evolution. Statistical ideas emerge, advance, and mature

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through use, interaction with real-world complexity, and iterative refinement informed by both theoretical insights and application.

Conclusion 2-2: Breakthroughs today stem from decades of investments in foundational research within statistics. Theoretical innovations may take years to develop before statisticians and their collaborators can put them into practice. This theoretical research lays the groundwork for the fundamental science that drives advancements at the forefront of the discipline.

Conclusion 2-3: Investments in robust statistics and extreme value theory, for example, have helped experts predict and track natural disasters and thus informed disaster resilience.

Conclusion 2-1: Statistical scientists drive the statistical innovation life cycle through the adoption and rigorous evaluation of emerging technologies and applications. Sustaining the agility and impact of this innovation cycle requires increased funding and prioritization of research contributions that are practice-informed and application-driven to advance both methodological and applied work across the discipline.

Conclusion 2-4: Investing in research in statistical foundations is critical to ensure a careful and measured use of the tools developed in statistics, machine learning, and AI. Without this investment and support for statistical rigor, these tools could misguide decision making and even cause harm.

Conclusion 2-8: Statistics has been supported broadly across multiple directorates and divisions at NSF, speaking to its critical role throughout science and engineering.

Conclusion 3-1: Statistical research enhances data utility and application. As new data types emerge and as complexity increases, statistics and analytic tools will need to evolve to keep pace with the changing landscape.

Conclusion 5-7: Research to assess the validity, accuracy, and reproducibility of models leads to higher-quality results, reduces bias, and decreases the risk of misinterpretation, and thus is critical for responsible decision making. Topics of particular interest include

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the development and evaluation of improved and robust measures of risk, fairness, trust, transparency, and accountability.

Conclusion 5-5: As new decentralized technologies—such as the Internet of Things, sensing devices, autonomous vehicles, advanced biotechnology, and blockchain—continue to evolve, the availability and nature of data will continue to change rapidly, catalyzing a new frontier in distributed statistical inference. Traditional methods supporting data privacy also rely on decentralized data collection, storage, and analysis. These interacting motivations and constraints, among others, define an ongoing and evolving challenge to develop statistical methods appropriate for a distributed data setting.

Conclusion 2-5: High-dimensional data include both extrinsic and intrinsic geometric structures. Statistical methods and theory are required to enhance discoverability of hidden intrinsic geometric structures while accounting for known extrinsic geometries in the observed or generated data elements. Development of generalized approaches to account for such geometric "lenses" will enhance model accuracy, reliability, generalizability, and analytic efficiency, as well as bridge the gap between AI-driven scientific research and scientific reasoning.

Conclusion 2-7: Professional associations for statistics offer robust publishing opportunities for high-quality, rigorous, and reproducible research. New journals supporting statistical science have been established to acknowledge the importance of innovative methods and applications in the field, while still upholding the strict review standards maintained by the statistics community.

Conclusion 6-8: There are opportunities for cohesive discussion around updating graduate training to keep pace with technological progress and attract the next generation of the statistical workforce.

Recommendation 7-1: Prioritized investment in the field's foundations is essential to accelerate trustworthy, data-driven scientific discovery; keep pace with the modern technological evolution; and maximize positive societal impact by enabling future breakthroughs at the frontiers of statistics. The National Science Foundation (NSF)

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should bolster its ongoing support for statistics, spanning methods, applications, theory, training, and education, and embed statistical science across the agency.

A successful program (or suite of programs) would advance statistical foundations, methods, applications, and training across NSF's scientific and engineering portfolio and include the following:

a. **<u>Foundational Statistical Methods and Applications.</u> Targeted cross-agency funding opportunities should highlight the importance of these translational contributions and the leadership by statisticians.**
    i. **Statistical methods and applications driven by domain scientists with statisticians playing a co-leadership role in advancing the emerging science and technology.**
    ii. **New frameworks and optimality criteria to enable analysis and inference in highly distributed environments, as well as statistical methods for the integrative analysis of observed and model-generated synthetic data.**
    iii. **Statistical research explicitly targeting decision making in areas including precision medicine, precision agriculture, resilient infrastructure, robotics, autonomous driving, and digital twins.**[15]
    iv. **Methods that integrate quantum computing into scientific inquiry, including statistical algorithms that exploit quantum computing and statistical tools to quantify uncertainty and optimize deployment of quantum sensing devices.**
    v. **Statistical research to investigate and improve the quality, scalability, and representativeness of novel data streams and data types, including those that involve human feedback and generative artificial intelligence.**

[15] This report uses the National Academies' definition: "A digital twin is a set of virtual information constructs that mimics the structure, context, and behavior of a natural, engineered, or social system (or system-of-systems), is dynamically updated with data from its physical twin, has a predictive capability, and informs decisions that realize value. The bidirectional interaction between the virtual and the physical is central to the digital twin" (NASEM 2024b, 1).

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b. Foundational Statistical Research. Some practice-informed needs for investment in theoretical research include the following:

i. Statistical inference in highly distributed settings, where as much computation as feasible occurs at or as close to the sensor as possible. As data collection becomes embedded in physical and social infrastructure, "edge" computing and federated learning will become requirements. Privacy- and confidentiality-preserving data analysis methods, particularly in distributed settings.

ii. Statistical theory to improve inference and understand intrinsic or extrinsic structure in high-dimensional, complex observed, or model-generated synthetic data.

iii. Statistical research to investigate and improve the quality, scalability, and representativeness of all data types.

iv. Statistical theory to assess validity, accuracy, and reproducibility of models, methods, and data to support decision making under uncertainty in complex systems. These programs should include support for advances in understanding the intersection of heterogeneous data with causal inference methods.

c. Foundational Statistical Education. Collaboration with other agencies to establish a nationally coordinated program supporting ongoing evaluation and reform of graduate statistical education, particularly in light of the rapid pace of technological evolution. The coordinated effort should develop strategies to strengthen the domestic pipeline for graduate students in statistics.

Key Finding 2-4: It is of paramount importance for the statistical community to position itself to take advantage of existing opportunities and emerging application areas by pursuing relevant research priorities and funding calls broadly. Doing so requires strengthening community-level mechanisms and infrastructure that enable statisticians to engage early, collectively, and strategically. Examples include offering leadership and team-science training workshops, sponsoring task forces to develop shared methodological and computational resources, creating grand community research challenges (such as KDD

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Cup and DREAM challenges), and organizing informal proposers' days in response to large interdisciplinary funding opportunities.

Conclusion 4-1: The foundations of statistics—across theory, methods, and applications—have provided the scientific infrastructure for major advances in medicine, public health, neuroscience, genomics, and, more broadly, the biomedical sciences. Historically, NIH has provided critical support for groundbreaking statistical development supporting these advances. Continued NIH investment in foundational statistics research and training is needed to sustain and to advance the frontiers of both biomedical science and the field of biostatistics itself. Such joint investment in biomedical science and biostatistics is essential for continued development of accurate, reproducible, and actionable insights from modern, complex data sources. Expansion of this support structure would encourage ongoing innovations in the foundations of statistical research focused on health and medicine.

Recommendation 7-2: The National Institutes of Health (NIH) should expand its support for the foundations of statistics in health science across the biomedical sciences, spanning methods, applications, theory, training, and education.

a. **Continue strengthening NIH support for cutting-edge statistical methodological research and applications, which are essential for accelerating discovery from existing and emerging biomedical data.**
b. **Evaluate and ensure that NIH study sections explicitly recognize statistical methodological research as a priority by assigning proposals to panels with appropriate statistical expertise and by applying review criteria that reflect the nature and impact of methodological contributions.**
c. **Evaluate and ensure that training (T, F) and career development (K) award programs are structured and reviewed in ways that prioritize training of future biostatisticians, apply evaluation criteria appropriate to methodological and interdisciplinary preparation, and involve reviewers with relevant statistical expertise.**

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Conclusion 4-2: Areas of investment that NIH might prioritize include statistical and machine learning methods for the analysis of multimodal genomic data, biobanks, and electronic health records; causal inference for complex biomedical data; adaptive designs and real-time intervention; federated and privacy-preserving statistical learning from health care data; and integrative statistics and AI methods for biomedical science and public health.

STATISTICS FOR ARTIFICIAL INTELLIGENCE

Conclusion 5-1: The rapid advancement and deployment of AI technologies call for continued and increasing contributions from the field of statistics. Immediate need and opportunity exist to invest in development and deployment of innovative statistical theory and methods to ensure accurate, reliable, and reproducible conclusions drawing from and using AI technologies as well as to accelerate trustworthy scientific discovery by integrating AI and rigorous statistical principles. Statisticians bring unique expertise to evaluate and enhance the rigor, interpretability, and trustworthiness of AI models and AI-based research and application. Conclusion 6-3: The need for embedding statistical thinking into AI education is both urgent and rapidly increasing, especially as AI tools are adapted and deployed within data-driven workflows.

Conclusion 2-9: NSF invests heavily in AI research and education across multiple directorates. Incorporating the depth and breadth of perspectives from the statistics community into these investments increases their value and impact for U.S. competitiveness.

Conclusion 2-10: NSF would benefit from an overarching survey of statistical progress within the National AI Research Institutes. Increasing the footprint of the field of statistics in the NSF-supported AI space and providing coordination across statisticians working at this interface are paramount. Professional societies may be well positioned to support this effort.

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Conclusion 5-8: The statistics community needs to recognize, support, and coordinate efforts focused on reproducibility and high-caliber statistical research. Recognition for these efforts includes publications, promotions, tenure, and scholarly awards and fellowships. The U.S. statistics community has led efforts to promote rigor and reproducibility in research and peer-reviewed publications, which becomes increasingly imperative with the proliferation of AI. The Journal of the American Statistical Association *implemented a formal reproducibility review of all accepted submissions in 2016, guided by a dedicated reproducibility editorial team (Wrobel et al. 2024). Reproducibility has become a standard expectation across the spectrum of professional statistics journals. Leading scientific journals have adopted similar initiatives; for example,* Nature *and* Science *each have statistical review boards, and rigor and reproducibility are central to all NIH-funded research. Statisticians contribute to the broad corpus of research on advancing and using AI, while maintaining the fundamental, non-negotiable constructs of sound statistical practices.*

Recommendation 7-3: Federal funding agencies should prioritize funding to expand and accelerate rigorous ongoing statistical thinking in artificial intelligence (AI) research and education initiatives. Successful initiatives would embed statistics into AI programs in the following ways:

a. **<u>AI-Based Research</u>. Specifically, federal funding agencies such as the National Science Foundation (NSF), National Institutes of Health (NIH), Department of Defense, Department of Energy, National Aeronautics and Space Administration, and National Security Agency should do the following:**
    i. **Support the identification of crosscutting opportunities to integrate statistical thinking into AI and AI-based research through centers of excellence or major collaborations that build on the success of the National AI Research Institutes and broaden statistical research to ensure the accurate, reliable, and reproducible use of AI for improved research results.**
    ii. **Elevate the role of statistical rigor in data-driven and AI research by incorporating explicit guidance in calls for proposals and proposal**

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review criteria and including reviewers with statistical expertise. Language should include guidance regarding statistical validity of inference and accuracy of uncertainty quantification.

iii. Support a comprehensive survey that identifies the footprint of the field of statistics, the statisticians involved, and the foci of their research, throughout the network of National AI Research Institutes.

b. AI Education. Future NSF, NIH, and other federal agencies' AI-related training funding, such as to accelerate AI education programs, should include statistical thinking and rigor as a major pillar. Calls should explicitly seek collaboration with the statistics community.

INTERFACES

Conclusion 5-4: Transdisciplinary collaborations allow hybrid development of data-generative models, analytic models, and predictive models, along with inferential systems for increasing knowledge, supporting decisions, and expanding understanding. Hybrid statistical–scientific models leverage available and synthetic data and known scientific principles to better understand complex phenomena.

Conclusion 5-6: Hybrid formulations are often the foundation of uncertainty quantification. Statisticians working at this intersection think carefully about the transparency in assumptions, computational cost, and complexity of implementation—often addressing all frontiers with their contributions. A great opportunity exists to expand support for rigorous collaborations at this interface, with statisticians playing a leading or co-leading scientific role. The benefits span the breadth of science and engineering discovery.

Recommendation 7-4: Funding agencies should collaboratively identify crosscutting, translational issues to inform future programs supporting the emergence of hybrid statistical–scientific models. Such programs should foster collaborative advances in statistical, scientific, and engineering theory, analytic methods, and applications.

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Conclusion 5-2: Interdisciplinary effort is greatly needed to build a more comprehensive understanding of principled methodologies for adaptive statistical design problems and their statistical performance at scale, with statisticians playing a critical role in these endeavors.

Conclusion 5-3: Adaptive and autonomous statistical design—methods designed to change sampling and simulation strategies in real time and in response to insights obtained through sequential learning—will be an important area for cross-agency collaboration. Scientific, engineering, industrial, and other domain-specific substantive problems will provide use cases to guide this research, and ultimately such capabilities can be expected to be implemented through AI agents. Advanced clinical trials and advances in the design of materials provide key examples of statistical innovation in this area.

Recommendation 7-5: In the near term, the National Institutes of Health (NIH) and National Science Foundation (NSF) should establish an interdisciplinary and translational cross-agency initiative incorporating innovative adaptive and autonomous statistical design, in which statisticians play a critical role in funded teams. Although NIH and NSF may serve as the leaders of this cross-agency initiative, the National Security Agency, Department of Energy, National Aeronautics and Space Administration, national laboratories, U.S. Department of Agriculture, and other federal research agencies are also in a position to benefit directly from U.S. leadership in the space of adaptive and autonomous statistical design.

EDUCATION

Conclusion 6-9: Statistics curricula and training need continual modernization to keep pace with the evolution of the field. As part of this update, statistics education would benefit from a formal infrastructure for educational innovation and professional development, as well as large-scale partnerships between disciplines and sectors. Such an effort could be modeled after the successful Vertical Integration of Research Education in the Mathematical Sciences and Research Training Group programs.

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Conclusion 6-4: Recent advances by ABET and CSAB, of which the American Statistical Association is a member, established accreditation criteria for undergraduate data science programs, which went into practice in the 2022-2023 cycle. These criteria recognize the importance of contributions across computing, statistics, and applied mathematics. ABET—through CSAB—is also pursuing criteria for master's programs in data science as well as a breadth of AI-focused programs. The criteria for these programs clearly articulate the important role of statistics and statistical thinking.

Conclusion 6-6: The collaborative nature of statistics highlights the importance of ongoing interaction between researchers in science, engineering, medicine, and statistics, especially at the interface of method and application. Training for such an interactive research environment is best encountered through experiential learning and hands-on collaborations. The statistics education community highlights many successful examples of such programs.

Conclusion 6-7: Large-scale computational training for future statisticians is vital yet remains a gap in the educational curriculum.

Conclusion 6-5: Investments from agencies such as NSF, NIH, and the National Security Agency in undergraduate and graduate training in statistics, biostatistics, and data science have had tangible impacts on U.S. competitiveness in science, engineering, and medicine.

Recommendation 7-6: The National Science Foundation, National Institutes of Health, and National Security Agency should invest in undergraduate and graduate training in statistics, biostatistics, and data science. Mechanisms include graduate fellowships, research experiences for undergraduates, and innovative programs that vertically and cross-sectionally integrate students, postdoctoral graduates, faculty, and industry researchers in statistics. Strategic priorities for such support should focus on cross-disciplinary, application-driven training; interuniversity collaboration; partnerships with industry and government; and computational training for the next generation of statisticians capable of leveraging the advanced large-scale computing infrastructure developed in industry today.

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PUBLIC–PRIVATE PARTNERSHIPS

Conclusion 4-3: Public–private partnerships increase the vitality of collaborative statistics research for both industry and academia, particularly when academic researchers are given the latitude to focus on foundational research.

Conclusion 4-4: Such historical statistical investments have led to breakthroughs in medicine, health care, agriculture, energy, the discovery of new materials, the financial sector, insurance, weather prediction, and resilient infrastructure and communities, and will lead to significant advancements in the future, ensuring sustained U.S. competitiveness.

Recommendation 7-7: Both the National Science Foundation and National Institutes of Health, as well as other funding agencies, should expand opportunities to advance science through the inclusion of statistics with current and emerging public–private partnerships. Such historical investments have led to breakthroughs in medicine, health care, agriculture, energy, the discovery of new materials, the financial sector, insurance, weather prediction, and resilient infrastructure and communities, and will lead to significant advancements in the future, ensuring sustained U.S. competitiveness.

ACADEMIC HIRING AND PROMOTION

Conclusion 2-11: The statistical innovation life cycle operates as a dynamic continuum between scientific inquiry and innovation in statistical theory, methods, and applications. The broader statistical community would benefit from recognizing the many ways in which researchers and practitioners advance the field throughout this life cycle. The community could support statistical innovation through transdisciplinary collaboration by recognizing high-risk, high-reward foundational research and enabling statisticians to contribute as leaders and collaborators across disciplines. Accelerating this cycle is critical for addressing today's data-driven challenges, and existing infrastructure and

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incentives for statistical research and education need to balance recognition across the full spectrum of research.

Recommendation 7-8: Universities should strengthen support for and champion the contributions of statistical departments, and statistical departments should prioritize hiring, tenure, and promotion practices that reward innovation, deep and ongoing collaboration, and high-impact contributions, including contributions in statistical software, analytic pipelines, and curated data sets. Journals should create environments that encourage and reward high-risk, high-reward research and education initiatives, including novel methodological advances, interdisciplinary applications, and collaborations. Funding agencies and industrial partners should create opportunities for statisticians to serve not only as collaborators but also as leaders in interdisciplinary research teams and education efforts, reinforcing a culture that values their central role in scientific progress.

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Appendix A
Statement of Task

The National Academies will assemble an interdisciplinary ad hoc committee to produce a forward-looking assessment of the current state of the statistical sciences and emerging opportunities for the discipline and its stakeholders as they look ahead to 2035 and beyond. The study may make recommendations to funding agencies on how to adjust and expand their portfolios of activities and partnerships to reflect the evolution of the field and strengthen the impact of the discipline.

Specifically, the study will assess:

- **Statistical research.** The state of research in the statistical sciences, examining aspects such as the significance of recent developments and highlighting current and emerging trends, challenges, and directions;
- **Statistical research in allied fields.** Interactions between the statistical sciences and the mathematical, computational, engineering, materials science, and related fields such as biostatistics and probability; the statistical foundations that underpin and motivate machine learning, artificial intelligence, the internet, and data science; the growth of data-intensive research arising in fields that incorporate statistical and data science thinking; and the role of statistics as a bridge to translational and use-inspired research;
- **Statistical needs to support scientific and technological advances.** Urgent needs for statistical innovation in areas such as verification, validation, and uncertainty quantification; privacy and security; and working with heterogeneous and complex data, among others, to enable responsible use of nascent scientific and technological developments like large language models; complex and dynamic systems (e.g. digital twins); hybrid models and suites of models that rely on both physics-based and data-driven approaches; broad data gathering and processing efforts; and policies around open data sharing;

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- **Importance of statistical sciences and strength of interdisciplinary collaborations.** The role of statistical sciences in key research domains for American competitiveness, including manufacturing, materials science, finance and blockchain, biomedical and biological sciences, public health, medicine, geosciences, environmental health and science, and energy applications; and
- **Statistical education, training, and workforce development.** Approaches toward statistics education and training to enhance workforce development across science and engineering domains (both within statistics and in fields that incorporate statistical thinking, such as artificial intelligence and data science) for a myriad of data-rich applications relevant to industry and technology, innovation and economic competitiveness, health and well-being, national security, and other areas of national interest.

MAPPING OF REPORT SECTIONS TO STATEMENT OF TASK

Statement of Task Bullet	Corresponding Report Chapter
Statistical research	Chapters 1, 2, 3, 4, 5
Statistical research in allied fields	Chapters 2, 4, 5
Statistical needs to support scientific and technological advances	Chapters 2, 5
Importance of statistical sciences and strength of interdisciplinary collaborations	Chapters 1, 2, 4, 5
Statistical education, training, and workforce development	Chapter 6

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Appendix B
Call for Perspectives and Public Meeting Agendas

CALL FOR PERSPECTIVES

Form Open on Website During Information-Gathering Period of Study

The National Academies has assembled an interdisciplinary ad hoc committee to produce a decadal-like assessment of the current state of the statistical sciences and emerging opportunities for the discipline and its stakeholders as they look ahead to 2035 and beyond (please navigate to the webpage to read the full statement of task). The study may make recommendations to funding agencies on how to adjust and expand their portfolios of activities and partnerships to reflect the evolution of the field and strengthen the impact of the discipline. We invite you to share your thoughts about how statistics is operating and will evolve at the frontiers. Your perspectives are invaluable and help ensure we have a holistic understanding of the statistical sciences and their broad implications for science and engineering.

1. What are some key developments and trends that will inform the state of the statistical sciences over the next decade?
2. How are data evolving to meet the demands of science and engineering? How do you think data (and our notions of data) will continue to evolve? How is the field of statistics evolving to meet the changing nature of data?
3. In what ways do the statistical sciences foster cross-fertilization between fields of science and engineering and support the advancement of other domains? In what ways do other domains further the field of statistics?
4. What role does the scientific discipline of statistics play in enabling and supporting technological advances?

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5. How will statistical training and education evolve over the next decade?
6. Do you have favorite case studies that highlight the contributions of statistics to science and engineering?

PUBLIC MEETING AGENDAS

November 18, 2024

3:00 p.m. **Welcome and Sponsor Introductions**

Kathy Ensor

- NSA
  - Mathematical Sciences Program

    Jay Servidea
- NSF
  - Directorate for Mathematical and Physical Sciences
    - Division of Mathematical Sciences

      David Manderscheid, Tapabrata Maiti, Yong Zeng, Yulia Gel, Jun Zhu, John Kolassa
    - Division of Materials Research

      John Schlueter
  - Directorate for Computer and Information Science and Engineering
    - Division of Computer and Network Systems

      Deepankar Medhi
    - Division of Computing and Communication Foundations

      Al Hero
  - Directorate for Biological Sciences
    - Division of Environmental Biology

      Kendra McLauchlan
  - Directorate for Engineering
    - Division of Electrical, Communications and Cyber Systems

      Anthony Kuh
  - Directorate for Geosciences

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        - Division of Atmospheric and Geospace Sciences
          Eric DeWeaver
    - Directorate for STEM Education
        - Division of Research on Learning in Formal and Informal Settings
          Margret Hjalmarson
        - Division of Undergraduate Education
          Ellen Carpenter
- NIH
    - National Cancer Institute
      Dana Wolff-Hughes
    - Office of Data Science Strategy
      Mohd Anwar

3:15 p.m. **Q&A with Sponsors**
Kathy Ensor

4:00 p.m. **ADJOURN**

January 13, 2025

11:15 a.m. **Causal Inference: Frontiers and Opportunities**
Moderator: Xihong Lin, *Committee Member (Harvard)*
Panelists: Hongzhe Li *(U Penn)*, Peng Ding *(Berkeley)*, Michael Kosorok *(UNC)*, Betsy Ogburn *(JHU)*

12:15 p.m. **MOVE TO CLOSED SESSION**

February 4, 2025

11:15 a.m. **Journal Editors' Perspectives on Trends in Statistical Sciences**
Moderator: Daniela Witten, *Committee Member (U of Washington)*
Panelists: Hani Doss *(U of Florida)*, Lexin Li *(UC Berkeley)*, Annie Qu *(UC Irvine)*, Dylan Small *(U of Pennsylvania)*, Michael Stein *(Rutgers)*, Lan Wang *(U of Miami)*

12:45 p.m. **MOVE TO CLOSED SESSION**

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February 20, 2025

1:30 p.m. Statistics at the National Labs

Moderators: Amy Braverman, *Committee Member (Jet Propulsion Laboratory, California Institute of Technology)*, Stephan R. Sain, *Committee Member (Jupiter Intelligence)*

Panelists: Carlo Graziani *(Argonne National Lab)*, David Schimel *(Jet Propulsion Laboratory, California Institute of Technology)*, Lyndsay Shand *(Sandia National Labs)*, Scott Vander Wiel *(Los Alamos National Lab)*

3:00 p.m. MOVE TO CLOSED SESSION

March 4, 2025

11:15 a.m. Statistical Privacy Panel

Moderator: Rina Barber, *Committee Member (U of Chicago)*

Panelists: Bonnie Berger *(MIT)*, John Duchi *(Stanford)*, Weijie Su *(U Penn)*

12:45 p.m. MOVE TO CLOSED SESSION

April 1, 2025

11:15 a.m. Statistics in Industry

Moderator: Lorin Crawford, *Committee Member (Microsoft Research)*

Panelists: Sivan Aldor-Noiman *(Banff Advisors)*, Bonnie Ghosh-Dastidar *(RAND)*, Amarjot Kaur *(Merck)*, Lester Mackey *(Microsoft Research)*, Vijay Nair *(Wells Fargo)*

12:45 p.m. MOVE TO CLOSED SESSION

April 15, 2025

11:15 a.m. Statistics in Sustainable and Renewable Agriculture

Moderator: Kathy Ensor, *Committee Chair (Rice University)*

Panelists: Trevor Hefley *(Kansas State University)*, Eunchun Park *(U of Arkansas)*, Frederi Viens *(Rice University)*, Linda Young *(USDA National Agricultural Statistics Service)*

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12:45 p.m. **MOVE TO CLOSED SESSION**

April 23, 2025

Panel at the University of Chicago Institute for Mathematical and Statistical Innovation's Workshop on Uncertainty Quantification for Material Science and Engineering Workshop

2:00 p.m. **Panel on the Frontiers of Statistics**

Moderator: *Kathy Ensor*

Panelists: *Peter Chien, Ralph Smith, Wei Chen, Roshan Joseph, Anh Tran*

3:00 p.m. **ADJOURN**

July 10, 2025

1:00 p.m. **Panel with the ASA's Caucus of Academic Representatives**

Moderator: *Kathy Ensor*

Panelists: *Veera Baladandayuthapani, Catherine Calder, Nicole Alana Lazar, Shari Messinger, Jeffrey Morris, Dan Nettleton, Robert Perera, Abel Rodriguez, Christopher Saunders, Ming Tony Tan*

2:30 p.m. **ADJOURN**

September 9, 2025

11:00 a.m. Blockchain discussion with *Yulia Gel and Ricky Rambharat*

12:00 p.m. Statistics Education discussion with Donna LaLonde *(ASA)*, Dave Hunter *(Penn State)*, Beth Chance *(Cal Poly)*, Monnie McGee *(Southern Methodist University)*, Mark Ward *(Purdue)*, Jan Dasgupta *(Washington State)*

1:00 p.m. **ADJOURN**

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Appendix C
Committee Member Biographical Information

KATHERINE BENNETT ENSOR (*Chair*) is the Noah G. Harding Professor of Statistics in the George R. Brown School of Engineering at Rice University where she also serves as the director of the Center for Computational Finance and Economic Systems. Currently, Professor Ensor also oversees the development of the Kinder Institute Urban Data Platform, a resource for the greater Houston area. She is an expert in many areas of modern statistics and develops innovative statistical techniques to answer important questions in science, engineering, and business with a focus on the environment, energy, and finance. Her research interests include methods for dependent data including time series and spatial and spatiotemporal, unique applications of Bayesian hierarchical modeling and approximate Bayesian computation, and stochastic process modeling and information integration. Professor Ensor is a fellow of the American Statistical Association (ASA) and the American Association for the Advancement of Science, and she has been recognized for her leadership, scholarship, and mentoring. She serves as vice president of ASA and as a member of the National Academies of Sciences, Engineering, and Medicine's Committee on Applied and Theoretical Statistics. She has a BSE and an MS in mathematics from Arkansas State University and a PhD in statistics from Texas A&M University.

LANCE A. WALLER (*Vice Chair*) is a professor and the former chair in the Department of Biostatistics and Bioinformatics in the Rollins School of Public Health at Emory University. His research involves the development of statistical methods for geographic data including applications in environmental justice, epidemiology, disease surveillance, and disease ecology. His research appears in biostatistical, statistical, environmental health, and ecology journals and in the textbook *Applied Spatial Statistics for Public Health Data* (2004, Wiley). Dr. Waller earned his bachelor's degree in mathematical sciences from New Mexico State University in

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1986 and his doctorate in operations research from Cornell University in 1991. Dr. Waller currently serves as the co-chair of the National Academies' Committee on Applied and Theoretical Statistics, is a former member of the Board on Mathematical Sciences and Analytics, and a participant in multiple National Academies' workshops and consensus studies. Dr. Waller also serves as a member of the U.S. Census Scientific Advisory Committee. Dr. Waller is a fellow of ASA and a fellow of the Royal Geographical Society.

RINA FOYGEL BARBER is the Louis Block Professor of Statistics at The University of Chicago, where she has been faculty since January 2014. Barber's research focuses on developing theory and methodology for statistical problems in challenging modern settings, including distribution-free inference, high-dimensional multiple testing, and sparse and low-rank estimation, as well as nonconvex optimization with applications in medical imaging. Her research has been recognized by awards including the Committee of Presidents of Statistical Societies (COPSS) Presidents' Award (2020), the Peter Gavin Hall Institute of Mathematical Statistics (IMS) Early Career Prize (2020), the IMS Medallion Lecture and Award (2022), and a MacArthur Fellowship (2023). She was elected as a fellow of IMS in 2023. Prior to joining the faculty at The University of Chicago, she was a National Science Foundation (NSF) postdoctoral fellow at Stanford University advised by Emmanuel Candès, and she received a PhD in statistics at The University of Chicago in 2012 advised by Mathias Drton and Nathan Srebro, an MS in mathematics at The University of Chicago in 2009, and an ScB in mathematics from Brown University in 2005.

AMY BRAVERMAN is a senior research scientist at the Jet Propulsion Laboratory, California Institute of Technology. She is the technical group lead for statistical methods and applications in the Uncertainty Quantification and Statistical Analysis Group. Her primary expertise is in statistical methods and uncertainty quantification for remote sensing data, especially for massive data sets obtained from Earth-orbiting satellites. Dr. Braverman actively collaborates with academic colleagues to engage them and their graduate students in research at the forefront of statistical problems relevant to NASA and climate and environmental sciences. She is the recipient of the NASA Exceptional Public Service Medal, the University of California, Los Angeles (UCLA) Physical Sciences Luminary Award in Statistics, and ASA's Section on the

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Environment Distinguished Achievement Award, and she is a fellow of ASA. She is a member of the Society for Industrial and Applied Mathematics (SIAM), currently serving as the Uncertainty Quantification Activity Group Chair, and a member of the American Geophysical Union. Dr. Braverman holds a PhD in statistics from UCLA (1999), an MA in mathematics from UCLA (1992), and a BA in economics from Swarthmore College (1982). Dr. Braverman served on the National Academies' Committee on Applied and Theoretical Statistics in the early 2000s.

LORIN CRAWFORD is a principal researcher at Microsoft Research. The central aim of his research program is to build machine learning algorithms and statistical tools that aid in the understanding of how genetic and gene-by-environmental interactions contribute to the architecture of complex traits and disease progression. His work has landed him a place on *Forbes*' 30 Under 30 list and recognition as a member of *The Root*'s 100 Most Influential African Americans. He has also been awarded an Alfred P. Sloan Research Fellowship, a David & Lucile Packard Foundation Fellowship for Science and Engineering, a COPSS Emerging Leader Award, and the Annie T. Randall Innovator Award from the Biometrics Section of ASA. Dr. Crawford is an associate editor at the journal *Biostatistics*, an associate editor of reproducibility at the *Journal of the American Statistical Association*, and a regional committee member of the International Biometric Society Eastern North American Region. Dr. Crawford received his PhD from the Department of Statistical Science at Duke University where his dissertation was awarded a Leonard J. Savage Award in Applied Methodology. He received his BS in mathematics from Clark Atlanta University.

DAVID B. DUNSON is currently the Arts and Sciences Distinguished Professor of Statistical Science and Mathematics at Duke University. He is an internationally leading expert in Bayesian statistics and methods for modeling and analysis of complex and high-dimensional data, with an emphasis on applied-driven methodology with theoretical support. He is heavily engaged in collaborations in biodiversity and ecology research, environmental health, and neurosciences. His work has been widely cited with an H-index of 99, and he has won numerous prestigious awards, including the COPSS President's Award, the Mortimer Spiegelman Award, a gold medal from the Environmental Protection Agency, the Mitchell Prize (three times) for the best Bayesian applications publication, and the COPSS George W. Snedecor Award. He has an exceptional

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track record as a mentor with more than 100 PhD students and more than 100 postdoctoral associates.

OMAR GHATTAS is a professor of geological sciences and mechanical engineering at The University of Texas (UT) at Austin. He is also the director of the Center for Computational Geosciences and Optimization in the Oden Institute for Computational Engineering and Sciences and holds the John A. and Katherine G. Jackson Chair in Computational Geosciences. He is a member of the faculty in the computational science, engineering, and mathematics interdisciplinary PhD program in the Oden Institute and holds courtesy appointments in computer science and biomedical engineering. Before moving to UT Austin in 2005, he spent 16 years on the faculty of Carnegie Mellon University (CMU). He holds a BSE (civil and environmental engineering) and an MS and a PhD (computational mechanics) from Duke University. With collaborators, he received the Association for Computing Machinery Gordon Bell Prize in 2003 (for Special Achievement) and again in 2015 (for Scalability), and was a finalist for the 2008, 2010, and 2012 Bell Prizes. He received the 2019 SIAM Computational Science & Engineering Best Paper Prize and the 2019 SIAM Geosciences Career Prize. He is a fellow of SIAM.

FRAUKE KREUTER is the director of the Joint Program in Survey Methodology (JPSM) at the University of Maryland, College Park; a professor of statistics and methodology at the University of Mannheim; and the head of the Statistical Methods Research Department at the Institute for Employment Research in Nürnberg, Germany. Before joining the University of Maryland, she was a postdoc at the UCLA Statistics Department. Dr. Kreuter's research focuses on sampling and measurement errors in complex surveys. In her work at JPSM, she maintains strong ties to the federal statistical system and serves in advisor roles for the National Center for Educational Statistics and the Bureau of Labor Statistics. She has served as a member on the Panel on Improving Federal Statistics for Policy and Social Science Research Using Multiple Data Sources and State-of-the-Art Estimation Methods at the National Academies. Dr. Kreuter is the author or co-author of several books, including *Data Analysis Using Stata* and *Practical Tools for Designing and Weighting Survey Samples*. She has an MA in sociology from the

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University of Mannheim, Germany, and a PhD in survey methodology from the University of Konstanz.

XIHONG LIN is a professor and the former chair of biostatistics, the coordinating director of the Program in Quantitative Genomics at the Harvard T.H. Chan School of Public Health, and a professor of statistics of Harvard University. Dr. Lin's research interests lie in development and application of scalable statistical and machine learning methods for analysis of massive and complex genomics, epidemiological, biobank, and health data. Her research has been supported by the Outstanding Investigator Award (R35) (2015–2029) from the National Cancer Institute. Dr. Lin is an elected member of the National Academy of Sciences and the National Academy of Medicine. She received the 2002 Mortimer Spiegelman Award from the American Public Health Association, the 2006 Presidents' Award and the 2017 Florence Nightingale David Award of the COPSS, the 2022 National Institute of Statistical Sciences Jerome Sacks Award for Outstanding Cross-Disciplinary Research, and the 2022 Marvin Zelen Leadership in Statistical Science Award. She is an elected fellow of ASA, IMS, and the International Statistical Institute. Dr. Lin is the former chair of the COPSS (2010–2012) and a former member of the Committee on Applied and Theoretical Statistics of the National Academies. She is a member of the National Academies' study on International Talent Programs in the Changing Global Environment. She is the former coordinating editor of *Biometrics* and the founding co-editor of *Statistics in Biosciences*.

BRIAN J. REICH is the Gertrude M. Cox Distinguished Professor of Statistics at North Carolina State University. His research interests include Bayesian methods, spatial statistics, extreme value analysis, and machine learning applied to ecology, epidemiology, meteorology, and climate. He has published more than 200 papers in leading statistics and subject-matter journals (including *Nature* and the *Proceedings of the National Academy of Sciences*) and the textbook titled *Bayesian Statistical Methods*. He has also supervised more than 50 doctoral students and postdoctoral fellows; served as editor-in-chief of the *Journal of Agricultural, Biological and Environmental Statistics*; and led NSF training grants to support undergraduate through postdoctoral research. For this body of work, he was given the ASA Outstanding Statistical Application Award and ASA's Section of Statistics and the Environment Early

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Investigator Award, and he was named a fellow of ASA. Prior to joining the faculty at North Carolina State University, he completed his PhD in biostatistics from the University of Minnesota and a postdoctoral fellowship at North Carolina State University.

STEPHAN R. SAIN is a senior principal data scientist and senior director at Jupiter Intelligence, where he heads the data sciences team and is responsible for operational aspects of the broader Jupiter science organization. Jupiter provides data and analytics services to better predict and manage risks from weather and sea level rise, storm intensification, and changing temperatures caused by medium- to long-term climate change. Sain is an experienced data science leader and applied statistician who has long worked at the intersection of climate research, applied statistics, and machine learning, including a focus on spatial methods for large data sets, extremes, uncertainty quantification, and climate risk analytics. From 2006 to 2014, he was the head of the Geophysical Statistics Project and a scientist in the Institute for Mathematics Applied to Geosciences at the National Center for Atmospheric Research in Boulder, Colorado. Sain is a fellow of ASA and is a past recipient of the Distinguished Achievement Award from ASA's Section on Statistics and the Environment. Sain also serves on the steering committee for ASA's newly formed Caucus of Industry Representatives, is a past member of ASA's Advisory Committee on Climate Change Policy, is an affiliate faculty member in the University of Colorado's Department of Applied Mathematics, and is a member of the advisory board for the Institute for Mathematical and Statistical Innovation at The University of Chicago.

AARTI SINGH is the FORE Systems Professor of Machine Learning at CMU and the director of the NSF AI Institute of Societal Decision Making. Her research focuses on interactive algorithms that can guide learning and decision making, with applications to enabling social and scientific discoveries. Her work is recognized by an NSF CAREER Award, a U.S. Air Force Young Investigator Award, an A. Nico Habermann Faculty Chair Award, a Harold A. Peterson Best Dissertation Award, and multiple paper awards. She has served on the National Academies' Committee on Applied and Theoretical Statistics, National Academies' Board on Mathematical Sciences and Analytics, and World Economic Forum expert network. She has also served as a lead expert on multiple National Academies' and Office of Naval Research/National Institute of Standards and Technology study committees, general and program chair for the International

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Conference on Machine Learning 2025 and 2020, program chair for the Artificial Intelligence and Statistics 2017 conference, and action editor for *IEEE Transactions on Information Theory* and *Journal of Machine Learning Research*. She received her PhD in electrical and computer engineering from the University of Wisconsin–Madison in 2008 and was a postdoctoral research associate at the Program in Applied and Computational Mathematics at Princeton University from 2008 to 2009, before joining CMU.

DANIELA M. WITTEN is a professor of statistics and biostatistics at the University of Washington and the Dorothy Gilford Endowed Chair in Mathematical Statistics. She develops statistical machine learning methods for high-dimensional data, with a focus on unsupervised learning. She has received several awards for her research in statistical machine learning, most notably the Spiegelman Award from the American Public Health Association for a (bio)statistician under age 40 and the Presidents' Award from the COPSS for a statistician under age 41. She is also a fellow of ASA and IMS. Dr. Witten is a co-author of the textbook *Introduction to Statistical Learning* and since 2023 serves as the joint editor of *Journal of the Royal Statistical Society, Series B*. She received a BS in mathematics and biological sciences in 2005 and an MS and a PhD in statistics in 2006 and 2010 from Stanford University.

TIAN ZHENG is a professor of statistics at Columbia University. Her research focuses on developing statistical and machine learning methods for exploring and understanding patterns in complex and high-dimensional data across application domains including biology, psychology, and climate modeling. Professor Zheng is the co–principal investigator and deputy director of the NSF Science and Technology Center for Learning the Earth with Artificial Intelligence and Physics, where she contributes to research integrating statistical foundations, AI, and climate science. She previously served as the chair of the Department of Statistics at Columbia University from 2019 to 2025 and as the associate director for education at the Columbia Data Science Institute from 2017 to 2020, where she led the development of multiple data science education initiatives. She is currently the inaugural chair of Stats Up AI, an interest group of ASA that aims to empower statisticians to contribute to research and applications in AI. Her research and teaching have been recognized by the Outstanding Statistical Application Award from ASA, the Mitchell Prize from the International Society for Bayesian Analysis, a Google

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Research Award, the 2017 Columbia Presidential Award for Outstanding Teaching, and the 2021 Lenfest Distinguished Columbia Faculty Award for excellence in teaching and mentoring. In 2025, she received a Google gift, as part of the Empire AI consortium, to establish the AIX Faculty Fellowship program supporting cross-disciplinary AI education initiatives at Columbia. Professor Zheng is a fellow of ASA, IMS, the American Association for the Advancement of Science, and the Asian American Scholar Forum. She received a BS in applied mathematics with a minor in computer science from Tsinghua University and a PhD in statistics from Columbia University.

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